

MA2213 Numerical Analysis I

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AY25/26 Sem 2

This is a first course on the theory and applications of numerical approximation techniques. Through the study of this course, the students will gain an understanding of how in practice mathematically formulated problems are solved using computers, and how computational errors are analysed and tackled. The students will be equipped with a number of commonly used numerical algorithms and knowledge and skill in performing numerical computation using MATLAB. The course is intended for mathematics majors and students from engineering and physical sciences. It will provide a firm basis for future study of numerical analysis and scientific computing. Major topics: Computational errors, direct method for systems of linear equations, interpolation and approximation, numerical integration, use of MATLAB software.

Prerequisite

If undertaking an Undergraduate Degree THEN((must have completed 1 of MA1312 / MA1505 / MA1511 / MA1521 / MA2002 at a grade of at least D)AND(must have completed 1 of MA1508E / MA1513 / MA1522 / MA2001 at a grade of at least D))

Preclusion

If undertaking an Undergraduate Degree THEN((must not be undertaking 1 of 1006DSAHON Bachelor of Science - Data Science and Analytics (Hons), 1006DSEXDP Bachelor of Science - Data Sci & Econs (Hons) XDP)AND(must not have completed 1 of DSA2102/ME3291 at a grade of at least D))

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1 Computer Arithmetic and Algorithms

1.1 Binary machine numbers

Due to the finite length of the memory, only finite real numbers can be represented exactly. Computers use **binary numbers** instead of decimal numbers:

$$(a_m a_{m-1} \dots a_1 a_0 . b_1 \dots b_n)_2$$

where $a_k, b_k \in \{0, 1\}$. We can convert it to a decimal number as such:

$$a_m \times 2^m + \dots + a_1 \times 2^1 + a_0 + 0 + b_1 \times 2^{-1} + \dots + b_n \times 2^{-n}.$$

In general, to convert a number of arbitrary base $(a_m a_{m-1} \dots a_1 a_0 . b_1 \dots b_n)_{\text{base}}$ into decimal number,

$$\sum_{i=0}^m a_i \times (\text{base})^i + \sum_{j=1}^n b_j \times (\text{base})^{-j}.$$

In scientific notation, we write

$$(\pm 1.b_1 b_2 \dots b_n \times 10^m)_2 = \pm (1.b_1 b_2 \dots b_n) \times 2^m.$$

In the computer memory, every binary digit is called a **bit**.

Example.

$$\begin{aligned} (10101)_2 &= (1 \times 2^4 + 0 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1)_2 \\ &= (1 \times 2^4 + 1 \times 2^2 + 1)_2 \\ &= (16 + 4 + 1)_{10} = (21)_{10} \end{aligned}$$

Example. $(1.0101)_2 = (1 + 1 \times 2^{-2} + 1 \times 2^{-4})_{10} = (1.3125)_{10}$

Example. $(101.101)_2 = (2^2 + 1 + 2^{-1} + 2^{-3})_{10} = (5.625)_{10}$

Addition has the following rules:

$$\begin{aligned} (0)_2 + (0)_2 &= (0)_2 \\ (0)_2 + (1)_2 &= (1)_2 \\ (1)_2 + (0)_2 &= (1)_2 \\ (1)_2 + (1)_2 &= (10)_2 \\ (1)_2 + (1)_2 + (1)_2 &= (11)_2 \end{aligned}$$

Use the column method, line up the binary points

$$\begin{array}{r} 1111 \\ 11011 \\ + 10111 \\ \hline 110010 \end{array}$$

$$\begin{array}{r} 11 \\ 10.011 \\ + 1.0111 \\ \hline 11.1101 \end{array}$$

Subtraction works similarly.

In scientific notation, we write

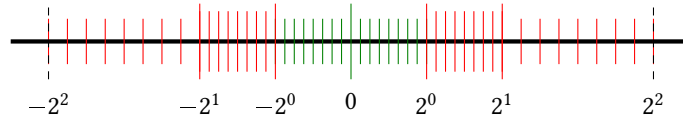
$$(\pm 1.b_1b_2 \dots b_n \times 10^m)_2 = \pm (1.b_1b_2 \dots b_n)_2 \times 2^m.$$

Definition 1.1. Any number p provided to the computer is approximated to the closest representable number $fl(p)$. The error of this approximation can be measured in the following ways:

- The **absolute error** is $|p - fl(p)|$.
- The **relative error** is $\frac{|p - fl(p)|}{|p|}$, provided that $p \neq 0$.

Example. If $p = \sqrt{2} \approx 1.414$, then $fl(p) = 1.375$.

- Absolute error: $|p - fl(p)| \approx 0.0392$
- Relative error: $|p - fl(p)|/|p| \approx 0.0277$



Proposition 1.2. If $|p| \in [1, 3.75]$, the relative error $\leq 2^{-4}$. If $|p| < 1$, then the relative error may be $> 2^{-4}$.

Proof. For each p , we want to find some constant c independent of p such that $\frac{|p - fl(p)|}{|p|} \leq c$. For normal numbers, consider two cases:

Case 1: $1 \leq |p| \leq 2$. Then there exists k such that

$$p \in [2^0 + k \cdot 2^{-3}, 2^0 + (k+1)2^{-3}).$$

Then

$$\frac{|p - fl(p)|}{|p|} \leq \frac{|p - fl(p)|}{1} \leq \frac{1}{2} \cdot 2^{-3} = 2^{-4}.$$

Case 2: $2 \leq |p| \leq 3.75$.

$$\frac{|p - fl(p)|}{|p|} \leq \frac{|p - fl(p)|}{2} \leq \frac{\frac{1}{2} \cdot \text{length of subinterval}}{2} = 2^{-4}.$$

For denormal numbers where $|p| < 1$, the relative error may be larger than 2^{-4} . For example, when $p = 2/9$, $fl(p) = 1/4$. The relative error is $\frac{|p - fl(p)|}{|p|} = \frac{1}{8} > 2^{-4}$. $\square$

A general model to represent real numbers is as follows:

s	e_1	e_2	$\dots$	e_m	b_1	b_2	$\dots$	b_n
-----	-------	-------	---------	-------	-------	-------	---------	-------

(1)

- 1 bit to store the sign s
- m bits to store the exponent, where the base for the exponent is 2
- n bits to store significant digits

The binary number (1) is interpreted as follows:

Case 1: Normal numbers. If $(e_1, \dots, e_m) \neq (0, \dots, 0)$ and $(e_1, \dots, e_m) \neq (1, \dots, 1)$,

$$\left((-1)^s \times 1.b_1 \dots b_n \times 10^{e_1 \dots e_m - \overbrace{11 \dots 1}^{m-1 \text{ ones}}} \right)_2$$

where we shift the exponent, so that the exponent can take on negative values.

Relative error $\leq 2^{-(n+1)}$ - exercise to prove this

Case 2: Smallest exponent. If $(e_1, \dots, e_m) = (0, \dots, 0)$, we allow $a_0 = 0$:

$$\left((-1)^s \times 0.b_1 \dots b_n \times 10^{1 - \overbrace{11 \dots 1}^{m-1 \text{ ones}}} \right)_2$$

The effect of this modification is redistributing the numbers, since there is a gap between zero and the first positive number.

Denormal numbers have $a_0 = 0$, **normal numbers** have $a_0 \neq 0$.

Case 3: Largest exponent. If $(e_1, \dots, e_m) = (1, \dots, 1)$, we reserve the binary sequences to represent infinity and NaN (not-a-number).

- The sequence where exponents all 1, significant digits all 0, sign positive means $+\infty$:

0	1	...	1	0	...	0
---	---	-----	---	---	-----	---

- The sequence if sign is negative means $-\infty$:

1	1	...	1	0	...	0
---	---	-----	---	---	-----	---

- If any $b_k \neq 0$, then the sequence means NaN:

0	1	...	1	b_1	...	b_k
---	---	-----	---	-------	-----	-------

- If negative, then the sequence means $-\text{NaN}$:

1	1	...	1	b_1	...	b_k
---	---	-----	---	-------	-----	-------

Example. $\sqrt{-1} = \text{NaN}$, $\arcsin 1.1 = \text{NaN}$. $\frac{1}{+0} = +\infty$, $\log(+0) = -\infty$.

There are two representations of zero:

0	0	...	0	0	...	0	means +0
1	0	...	0	0	...	0	means -0

The **single-precision floating-point format** has a size of 32 bits (4 bytes), where $m = 8$, $n = 23$.

s	e_1	e_2	...	e_8	b_1	b_2	...	b_{23}
-----	-------	-------	-----	-------	-------	-------	-----	----------

The minimum positive number is

$$(0.00 \dots 01 \times 10^{1-1111111})_2 = 2^{-149} \approx 1.40 \times 10^{-45}$$

and the maximum positive number is

$$(1.11 \cdots 11 \times 10^{11111110-1111111})_2 = 2^{127} \times (2 - 2^{-23}) \approx 3.40 \times 10^{38}.$$

For normal numbers, the relative error is less than $2^{-24} = 5.96 \times 10^{-8}$. Thus, normal numbers have 7 or 8 decimal significant digits.

The **double-precision floating-point format** has a size of 64 bits (8 bytes), where $m = 11$, $n = 52$.

s	e_1	e_2	$\cdots$	e_{11}	b_1	b_2	$\cdots$	b_{52}
-----	-------	-------	----------	----------	-------	-------	----------	----------

1.2 Finite-digit arithmetic

In addition to inaccurate representation of numbers, the arithmetic performed in a computer is not exact. Suppose the single-precision floating-point format is used. For $x = 5/7$ and $y = 1/3$, when computers receive the instruction to compute $x + y$, the values of x and y are changed to the **truncations** $fl(x)$ and $fl(y)$:

$$fl(x) = (0.1011011011011011011011)_2, \quad fl(y) = (0.0101010101010101010101)_2.$$

Then

$$fl(x) + fl(y) = (1.0000110000110000110000111)_2.$$

However, $fl(x) + fl(y)$ cannot be represented exactly in single-precision floating-point format, so another **truncation** is applied:

$$fl(fl(x) + fl(y)) = (1.00001100001100001100010)_2.$$

Suppose the floating-point representations $fl(x)$ and $fl(y)$ are given for the real numbers x and y .

Definition 1.3. Let $\oplus, \ominus, \otimes, \oslash$ denote machine addition, subtraction, multiplication, and division operations, respectively. Define **finite-digit arithmetic** by

$$\begin{aligned} x \oplus y &:= fl(fl(x) + fl(y)), \\ x \ominus y &:= fl(fl(x) - fl(y)), \\ x \otimes y &:= fl(fl(x) \times fl(y)), \\ x \oslash y &:= fl(fl(x) / fl(y)). \end{aligned}$$

This arithmetic corresponds to performing exact arithmetic on the floating-point representations of x and y , then converting the exact result to its finite-digit floating-point representation.

Using the definition above, one can show various properties of finite-digit arithmetic, such as:

- addition is commutative: $x \oplus y = y \oplus x$
- $x \ominus y = -(y \ominus x)$
- multiplication is commutative: $x \otimes y = y \otimes x$

However, in general,

- Addition is not associative: $(x \oplus y) \oplus z \neq x \oplus (y \oplus z)$
- Not distributive: $(x \oplus y) \otimes z \neq (x \otimes z) \oplus (y \otimes z)$

Example. For $x = 3.14159$, $y = 2.71828$, $z = 1000000$, the results of $(x \oplus y) \oplus z$ and $x \oplus (y \oplus z)$ are different:

$$\begin{aligned} (x \oplus y) \oplus z &= 1000005.875 \\ x \oplus (y \oplus z) &= 1000005.8125 \end{aligned}$$

Some operations may introduce significant numerical error:

- Sum of two numbers with significantly different magnitudes:

$$10^6 \oplus 12.3456 = 1.000012375 \times 10^6$$

- Subtracting two numbers very close to each other:

$$8381.02 \ominus 123.45 \otimes 67.89 = -9.765625 \times 10^{-4}$$

- Product of two small numbers:

$$F = G \frac{mM}{r^2}$$

where $G = 6.674302 \times 10^{-11}$, $m = 9.109384 \times 10^{-31}$, $M = 2.18732 \times 10^{31}$, $r = 3.248678 \times 10^4$. Then $G \otimes (m \otimes M) \ominus (r \otimes r)$ is more accurate than $(G \otimes m) \otimes M \ominus (r \otimes r)$, since the latter involves a denormal number $G \otimes m$ which is less accurate.

Three don'ts:

- $a + b$ for $|a| \gg |b| > 0$
- $a - b$ for $a \approx b$
- $a \times b$ for $0 < |a|, |b| \ll 1$

1.3 Algorithms and pseudocodes

An **algorithm** is a finite sequence of unambiguous computer-implementable instructions that solves a class of problems.

Given $x_1, \dots, x_n$, to compute the sum $x_1 + x_2 + \dots + x_n$, a naive algorithm is:

Define a variable s , and set its value to be zero. Add s by x_1 . Add s by x_2 Add s by x_n . The value of s is the sum.

Computation of $x_1 + x_2 + \dots + x_n$

```

1:  $sum \leftarrow 0$ 
2: for  $i = 1, \dots, n$  do
3:    $sum \leftarrow sum + x_i$ 
4: return  $sum$ 

```

The return value of the above algorithm is

$$x_1 \oplus \dots \oplus x_n$$

whose absolute error is

$$|(x_1 + \dots + x_n) - (x_1 \oplus \dots \oplus x_n)|.$$

The error produced by the difference between the exact arithmetic and computer arithmetic is known as the **round-off error**, due to truncation as a result of finite-digit representation.

Our numerical experiment shows that

$$\underbrace{7 \oplus \dots \oplus 7}_{12345678 \text{ sevens}} = 96368672 \quad \text{but} \quad \underbrace{7 + \dots + 7}_{12345678 \text{ sevens}} = 86419746.$$

The reason for this is round-off error: since there are 24 binary digits in floating point representation, any number

that exceeds that is truncated.

$$\begin{aligned}
 16777236 \oplus 7 &:= fl((1000000000000000000010100)_2 + (111)_2) \\
 &= fl((100000000000000000001101\mathbf{1})_2) \\
 &= (100000000000000000001110\mathbf{0})_2 \\
 &= 16777244 = 16777236 + 8
 \end{aligned}$$

Similarly,

$$\begin{aligned}
 16777244 \oplus 7 &:= fl((1000000000000000000011100)_2 + (111)_2) \\
 &= fl((100000000000000000001000\mathbf{1})_2) \\
 &= (1000000000000000000010010\mathbf{0})_2 \\
 &= 16777252 = 16777244 + 8
 \end{aligned}$$

and

$$\begin{aligned}
 16777252 \oplus 7 &:= fl((10000000000000000000100100)_2 + (111)_2) \\
 &= fl((100000000000000000001010\mathbf{1})_2) \\
 &= (1000000000000000000010110\mathbf{0})_2 \\
 &= 16777260 = 16777252 + 8
 \end{aligned}$$

Kahan's algorithm minimises round-off errors.

Algorithm 1 Kahan's algorithm: Computation of $x_1 + x_2 + \dots + x_n$

```

1:  $sum \leftarrow 0$   $\triangleright$  Prepare the accumulator
2:  $c \leftarrow 0$   $\triangleright$  A running compensation for lost low-order bits
3: for  $i = 1, \dots, n$  do
4:    $y \leftarrow x_i - c$   $\triangleright$   $c$  is zero the first time around
5:    $t \leftarrow sum + y$   $\triangleright$  Alas,  $sum$  is big,  $y$  small, so low-order digits of  $y$  are lost
6:    $c \leftarrow (t - sum) - y$   $\triangleright$   $t - sum$  cancels the high-order part of  $y$ ; subtracting  $y$  recovers negative (low part of  $y$ )
7:    $sum \leftarrow t$ 
8: return  $sum$   $\triangleright$  Next time around, the lost low part will be added to  $y$  in a fresh attempt

```

Consider the function

$$g(x) = \int_x^\infty t^{-1/2} e^{-t} dx.$$

We can rewrite $g(x)$ as

$$\begin{aligned}
 g(x) &= \int_x^\infty t^{-1/2} e^{-t} dx \\
 &\stackrel{t=s^2}{=} \int_{\sqrt{x}}^\infty s^{-1} e^{-s^2} 2s ds \\
 &= 2 \int_{\sqrt{x}}^\infty e^{-s^2} ds = 2 \left(\int_0^\infty e^{-s^2} ds - \int_0^{\sqrt{x}} e^{-s^2} ds \right) \\
 &= \int_{-\infty}^\infty e^{-s^2} ds - 2 \int_0^{\sqrt{x}} e^{-s^2} ds = \sqrt{\pi} - 2 \int_0^{\sqrt{x}} e^{-s^2} ds.
 \end{aligned}$$

examples
of kahan
algo

By the Taylor expansion $e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}$, we write

$$\int_0^{\sqrt{x}} e^{-s^2} ds = \int_0^{\sqrt{x}} \sum_{k=0}^{\infty} \frac{(-s^2)^k}{k!} ds = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!} \cdot \left[\frac{s^{2k+1}}{2k+1} \right]_0^{\sqrt{x}} = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!} \frac{x^{k+1/2}}{2k+1}.$$

Hence, the integral can be expressed as an infinite series

$$g(x) = \sqrt{\pi} - \sum_{k=0}^{\infty} \frac{(-1)^k x^{k+1/2}}{k!(k+1/2)}.$$

Therefore we can use the approximation

$$g(x) \approx \sqrt{\pi} - \sum_{k=0}^N \frac{(-1)^k x^{k+1/2}}{k!(k+1/2)}$$

for a sufficiently large N . The error introduced by truncation of the series is called **truncation error**. In this case, the truncation error is

$$e_N(x) := \left| g(x) - \left(\sqrt{\pi} - \sum_{k=0}^N \frac{(-1)^k x^{k+1/2}}{k!(k+1/2)} \right) \right| = \sum_{k=N+1}^{\infty} \frac{(-1)^k x^{k+1/2}}{k!(k+1/2)}.$$

A simple pseudocode is as follows:

Computation of $g(x)$

```

1:  $g \leftarrow \sqrt{\pi}$ 
2: for  $k = 0, \dots, N$  do
3:    $g \leftarrow g - \frac{(-1)^k x^{k+1/2}}{k!(k+1/2)}$ 
4: return  $g$ 
```

The problems with the above algorithm are that the computer may not know how to compute the factorial $k!$ and exponentiation $x^{k+1/2}$.

The idea is to use a **recurrence relation**, since the $(k+1)$ -th term of the series is related to the k -th term. Let

$$s_k = \frac{(-1)^k x^{k+1/2}}{k!}.$$

Then s_k satisfies the recurrence relation

$$s_{k+1} = -\frac{x}{k+1} s_k, \quad s_0 = \sqrt{x}.$$

Computation of $g(x)$

```

1:  $g \leftarrow \sqrt{\pi}, s \leftarrow \sqrt{x}$ 
2: for  $k = 0, \dots, N$  do
3:    $g \leftarrow g - \frac{s}{k+1/2}$ 
4:    $s \leftarrow -\frac{xs}{k+1}$ 
```

As another example, consider the elliptical integral of second kind¹:

$$E(k) = \int_0^1 \frac{\sqrt{1-k^2t^2}}{\sqrt{1-t^2}} dt, \quad k \in (0, 1)$$

which has the series expansion

$$E(k) = \frac{\pi}{2} \sum_{n=0}^{\infty} \left(\frac{(2n)!}{2^{2n}(n!)^2} \right)^2 \frac{k^{2n}}{1-2n}.$$

We approximate this function by truncating the series:

$$E(k) \approx \frac{\pi}{2} \sum_{n=0}^N \left(\frac{(2n)!}{2^{2n}(n!)^2} \right)^2 \frac{k^{2n}}{1-2n}.$$

To implement it efficiently, as above, let

$$s_n = \frac{\pi}{2} \left(\frac{(2n)!}{2^{2n}(n!)^2} \right)^2 k^{2n}.$$

Then s_n satisfies the recurrence relation

$$s_{n+1} = \left(\frac{n+1/2}{n+1} k \right)^2 s_n, \quad s_0 = \frac{\pi}{2}.$$

Thus, the original series becomes

$$E(k) \approx \sum_{n=0}^{\infty} \frac{s_n}{1-2n}.$$

Computation of $E(k)$

```

1:  $E \leftarrow 0, s \leftarrow \pi/2$ 
2: for  $n = 0, \dots, N$  do
3:    $E \leftarrow E + \frac{s}{1-2n}$ 
4:    $s \leftarrow \left( \frac{n+1/2}{n+1} k \right)^2 s$ 
5: return  $E$ 
```

The truncation error is

$$e_N(k) := \left| E(k) - \frac{\pi}{2} \sum_{n=0}^N \left(\frac{(2n)!}{2^{2n}(n!)^2} \right)^2 \frac{k^{2n}}{1-2n} \right| = \frac{\pi}{2} \sum_{n=N+1}^{\infty} \left(\frac{(2n)!}{2^{2n}(n!)^2} \right)^2 \frac{k^{2n}}{2n-1}.$$

When $n > 0$, distribute the $2n$ 2's into $(n!)^2$:

$$\frac{(2n)!}{2^{2n}(n!)^2} = \frac{1 \cdot 2 \cdot 3 \cdot 4 \cdots (2n-1) \cdot (2n)}{2 \cdot 2 \cdot 4 \cdot 4 \cdots (2n) \cdots (2n)} \leq \frac{1}{2}.$$

Hence, the truncation error can be bounded from above by

$$e_N(k) \leq \frac{\pi}{8} \sum_{n=N+1}^{\infty} \frac{k^{2n}}{2n-1} \leq \frac{\pi}{8} \sum_{n=N+1}^{\infty} \frac{k^{2n}}{2N} = \frac{\pi k^{2(N+1)}}{16N(1-k^2)} \quad (*)$$

by computing the geometric series. We would like to choose (a sufficiently large) N such that the absolute error

¹it is known that the integral does not have an explicit form.

of the approximation

$$E(k) \approx \frac{\pi}{2} \sum_{n=0}^N \left(\frac{(2n)!}{2^{2n}(n!)^2} \right)^2 \frac{k^{2n}}{1-2n}$$

is less than a certain **threshold** $\varepsilon > 0$. By (*), we can choose N such that

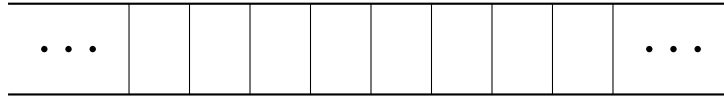
$$\frac{\pi k^{2(N+1)}}{16N(1-k^2)} < \varepsilon.$$

1.4 Matrix multiplication

A **matrix** is of the form

$$A = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix} \quad \text{or} \quad A = (a_{ij})_{m \times n}.$$

In the **linear memory model**, the memory is viewed as a long tape of bytes:



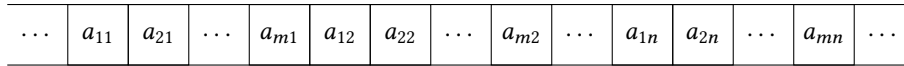
There are two ways in which a matrix $A = (a_{ij})_{m \times n}$ is stored in memory:

- (1) **Row-major order**: store in terms of rows

Location of (i, j) -element is $(i-1)n + j$

- (2) **Column-major order**: store in terms of columns

Location of (i, j) -element is $(j-1)m + i$



Theorem 1.4 (Multiplying two general matrices). *Let $A \in \mathbb{R}^{m \times n}$, $B \in \mathbb{R}^{n \times p}$. Then*

- *Number of multiplications: mnp*
- *Number of additions: $m(n-1)p$*

Proof. Let $C = AB$. Then for all $i = 1, \dots, m$ and $j = 1, \dots, p$,

$$c_{ij} = \sum_{k=1}^n a_{ik} b_{kj}.$$

Multiplication of $A = (a_{ij})_{m \times n}$ and $B = (b_{ij})_{n \times p}$

```

1: for  $i = 1, \dots, m$  do
2:   for  $j = 1, \dots, p$  do
3:      $c_{ij} \leftarrow a_{i1} b_{1j}$ 
4:     for  $k = 2, \dots, n$  do
5:        $c_{ij} \leftarrow c_{ij} + a_{ik} b_{kj}$ 
6:   return  $C = (c_{ij})_{m \times p}$ 

```

Each for loop corresponds to a summation symbol. The total number of multiplications is

$$\sum_{i=1}^m \sum_{j=1}^p \left(1 + \sum_{k=2}^n 1 \right) = \sum_{i=1}^m \sum_{j=1}^p (1 + (n-1)) = \sum_{i=1}^m \sum_{j=1}^p n = \sum_{i=1}^m np = mnp.$$

Similarly, the number of additions is

$$\sum_{i=1}^m \sum_{j=1}^p \sum_{k=2}^n 1 = \sum_{i=1}^m \sum_{j=1}^p (n-1) = \sum_{i=1}^m (n-1)p = m(n-1)p.$$

□

We consider some special matrices.

Definition 1.5. The **identity matrix** $I_n \in \mathbb{R}^{n \times n}$ is

$$I_n = \begin{pmatrix} 1 & & & \\ & 1 & & \\ & & \ddots & \\ & & & 1 \end{pmatrix}$$

Theorem 1.6. For any matrix $A \in \mathbb{R}^{m \times n}$, one has $I_m A = A = A I_n$.

Definition 1.7. A **scalar matrix** $\Lambda \in \mathbb{R}^{n \times n}$ is

$$\Lambda = \begin{pmatrix} \lambda & & & \\ & \lambda & & \\ & & \ddots & \\ & & & \lambda \end{pmatrix} = \lambda I$$

for some $\lambda \in \mathbb{R}$.

Theorem 1.8. For any matrix $A \in \mathbb{R}^{m \times n}$, the scalar matrices $\Lambda_1 = \lambda I_m$ and $\Lambda_2 = \lambda I_n$ satisfy

$$\Lambda_1 A = A \Lambda_2 = \lambda A.$$

Definition 1.9. A **diagonal matrix** $D = \text{diag}\{\lambda_1, \lambda_2, \dots, \lambda_n\} \in \mathbb{R}^{n \times n}$ is

$$D = \begin{pmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_n \end{pmatrix}.$$

Theorem 1.10. For any matrix $A = (a_{ij})_{m \times n}$ and $D = \text{diag}\{\lambda_1, \lambda_2, \dots, \lambda_n\}$, it holds that

$$AD = (a_{ij}\lambda_j)_{m \times n}$$

where we multiply the j -th column by λ_j .

Theorem 1.11. For any matrix $A = (a_{ij})_{m \times n}$ and $D = \text{diag}\{\lambda_1, \lambda_2, \dots, \lambda_m\}$, it holds that

$$DA = (\lambda_i a_{ij})_{m \times n}$$

where we multiply the i -th row by λ_i .

Proof. We can write the matrix D as $(d_{ij})_{m \times m}$, where

$$d_{ij} = \begin{cases} \lambda_i & \text{if } i = j, \\ 0 & \text{if } i \neq j. \end{cases}$$

Thus the (i, j) -entry of DA is

$$\begin{aligned} \sum_{k=1}^m d_{ik} a_{kj} &= d_{i1} a_{1j} + \cdots + d_{ii} a_{ij} + \cdots + d_{im} a_{mj} \\ &= 0 + \cdots + 0 + d_{ii} a_{ij} + 0 \cdots + 0 = \lambda_i a_{ij}. \end{aligned}$$

□

Left-multiply a diagonal matrix $D = \text{diag}\{\lambda_1, \lambda_2, \dots, \lambda_m\}$:

$$DA = \begin{pmatrix} \lambda_1 a_{11} & \lambda_1 a_{12} & \cdots & \lambda_1 a_{1n} \\ \lambda_2 a_{21} & \lambda_2 a_{22} & \cdots & \lambda_2 a_{2n} \\ \vdots & \vdots & & \vdots \\ \lambda_m a_{m1} & \lambda_m a_{m2} & \cdots & \lambda_m a_{mn} \end{pmatrix}$$

Right-multiply a diagonal matrix $D = \text{diag}\{\lambda_1, \lambda_2, \dots, \lambda_n\}$:

$$DA = \begin{pmatrix} a_{11}\lambda_1 & a_{12}\lambda_2 & \cdots & a_{1n}\lambda_n \\ a_{21}\lambda_1 & a_{22}\lambda_2 & \cdots & a_{2n}\lambda_n \\ \vdots & \vdots & & \vdots \\ a_{m1}\lambda_1 & a_{m2}\lambda_2 & \cdots & a_{mn}\lambda_n \end{pmatrix}$$

Definition 1.12. An **upper triangular matrix** $A \in \mathbb{R}^{n \times n}$ is of the form

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ & a_{22} & \cdots & a_{2n} \\ & & \ddots & \vdots \\ 0 & & & a_{nn} \end{pmatrix}$$

where $a_{ij} = 0$ if $i > j$.

Theorem 1.13. Let $A, B \in \mathbb{R}^{n \times n}$ be two upper triangular matrices. Then AB and BA are upper triangular matrices.

Proof. Let $C = AB$. Show that $c_{ij} = \sum_{k=1}^n a_{ik} b_{kj} = 0$ when $i > j$. □

Definition 1.14. A **lower triangular matrix** $A \in \mathbb{R}^{n \times n}$ is of the form

$$A = \begin{pmatrix} a_{11} & & & \\ a_{21} & a_{22} & & \\ \vdots & \ddots & \ddots & \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix}$$

where $a_{ij} = 0$ if $i < j$.

Theorem 1.15. Let $A, B \in \mathbb{R}^{n \times n}$ be two lower triangular matrices. Then AB and BA are lower triangular matrices.

Proof. transpose case of upper triangular. □

Definition 1.16. An **upper Hessenberg matrix** $A \in \mathbb{R}^{n \times n}$ is of the form

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ & a_{32} & a_{33} & \cdots & a_{3n} \\ & & \ddots & \ddots & \vdots \\ & & & a_{n,n-1} & a_{nn} \end{pmatrix}$$

where $a_{ij} = 0$ if $i > j + 1$.

Notice we add one more subdiagonal to the upper triangular matrix.

Remark. The product of two upper Hessenberg matrices may not be upper Hessenberg.

Theorem 1.17. Let $A \in \mathbb{R}^{n \times n}$ be an upper Hessenberg matrix, and $B \in \mathbb{R}^{n \times n}$ be an upper triangular matrix. Then AB and BA are upper Hessenberg matrices.

Proof. Let $C = AB$. Show $c_{ij} = 0$ when $i > j + 1$ □

Definition 1.18. An **lower Hessenberg matrix** $A \in \mathbb{R}^{n \times n}$ is of the form

$$A = \begin{pmatrix} a_{11} & a_{12} & & & 0 \\ a_{21} & a_{22} & a_{23} & & \\ a_{31} & a_{32} & a_{33} & \ddots & \\ \vdots & \vdots & \vdots & \ddots & a_{n-1,n} \\ a_{n1} & a_{n2} & a_{n3} & \cdots & a_{nn} \end{pmatrix}$$

where $a_{ij} = 0$ if $i < j - 1$.

Theorem 1.19. Let $A \in \mathbb{R}^{n \times n}$ be a lower Hessenberg matrix, and $B \in \mathbb{R}^{n \times n}$ be a lower triangular matrix. Then AB and BA are lower Hessenberg matrices.

Proof. take transpose of upper Hessenberg □

Definition 1.20. A **tridiagonal matrix** $T \in \mathbb{R}^{n \times n}$ is of the form

$$T = \begin{pmatrix} a_1 & b_1 & & & \\ c_1 & a_2 & b_2 & & \\ & c_2 & \ddots & \ddots & \\ & & \ddots & \ddots & b_{n-1} \\ & & & c_{n-1} & a_n \end{pmatrix}.$$

$a_{ij} \neq 0$ only if $|i - j| \leq 1$, so $a_{ij} = 0$ if $|i - j| > 1$.

Notice T is tridiagonal because it contains a diagonal, subdiagonal, and superdiagonal.

T is both lower Hessenberg and upper Hessenberg.

Theorem 1.21. Let $T \in \mathbb{R}^{n \times n}$ be a tridiagonal matrix. Then for any lower triangular matrix A and upper triangular matrix B , the matrices TA and AT are lower Hessenberg matrices, and TB and BT are upper Hessenberg matrices.

Proof. apply previous two theorems □

In what will follow, we will see that multiplying two special matrices requires fewer operations than multiplying two general matrices. To save computational cost, we do the following:

- (1) Identify the zero entries in the matrix product, and skip the computation of these entries.

(2) For non-zero entries, avoid adding zero terms in the sum

$$c_{ij} = \sum_{k=1}^n a_{ik} b_{kj}.$$

Theorem 1.22. *Multiplying a matrix and an upper triangular matrix: Let $A \in \mathbb{R}^{m \times n}$, $B \in \mathbb{R}^{n \times n}$ upper triangular. Then*

- Number of multiplications: $mn(n+1)/2$
- Number of additions: $mn(n-1)/2$

Proof. Let $C = AB$. Then for $i = 1, \dots, m$, $j = 1, \dots, n$,

$$c_{ij} = \sum_{k=1}^n a_{ik} b_{kj}.$$

Since B is upper triangular, we have $b_{kj} = 0$ if $k > j$. Thus, we can simplify the above sum as

$$c_{ij} = \sum_{k=1}^j a_{ik} b_{kj}.$$

Multiplication of $A = (a_{ij})_{m \times n}$ and upper triangular $B = (b_{ij})_{n \times n}$

```

1: for  $i = 1, \dots, m$  do
2:   for  $j = 1, \dots, n$  do
3:      $c_{ij} \leftarrow a_{i1} b_{1j}$ 
4:     for  $k = 2, \dots, j$  do
5:        $c_{ij} \leftarrow c_{ij} + a_{ik} b_{kj}$ 
6: return  $C = (c_{ij})_{m \times n}$ 

```

The number of multiplications is

$$\begin{aligned} \sum_{i=1}^m \sum_{j=1}^n \left(1 + \sum_{k=2}^j 1 \right) &= \sum_{i=1}^m \sum_{j=1}^n (1 + (j-1)) = \sum_{i=1}^m \sum_{j=1}^n j \\ &= \sum_{i=1}^m \frac{n(n+1)}{2} = \frac{mn(n+1)}{2}. \end{aligned}$$

The number of additions is

$$\begin{aligned} \sum_{i=1}^m \sum_{j=1}^n \sum_{k=2}^j 1 &= \sum_{i=1}^m \sum_{j=1}^n (j-1) \\ &= \sum_{i=1}^m \frac{n(n-1)}{2} = \frac{mn(n-1)}{2}. \end{aligned}$$

□

Theorem 1.23. *Multiplying two lower triangular matrices: Let $A, B \in \mathbb{R}^{n \times n}$, both are lower triangular.*

- Number of multiplications: $n(n+1)(n+2)/6$
- Number of additions: $(n-1)n(n+1)/6$

Proof. Let $C = AB$. Then for $i = 1, \dots, m, j = 1, \dots, n$,

$$c_{ij} = \sum_{k=1}^n a_{ik} b_{kj}.$$

Since A and B are lower triangular, we have $a_{ik} = 0$ if $i < k$, and $b_{kj} = 0$ if $k < j$.

Claim: The product of two lower triangular matrices is lower triangular. Thus, $c_{ij} = 0$ if $i < j$, so we only need to compute c_{ij} for $i \geq j$.

Case 1: $k \leq i < j$. In this case, $b_{kj} = 0$ for $k, j \in \{1, \dots, n\}$ with $k < j$.

Case 2: $i > k$. In this case, $a_{ik} = 0$ for $i, k \in \{1, \dots, n\}$ with $i < k$.

Hence, for $i < j$, we have

$$c_{ij} = \sum_{k=1}^i a_{ik} b_{kj} + \sum_{k=i+1}^n a_{ik} b_{kj} = 0 + 0 = 0.$$

Therefore, for $i = 1, \dots, n, j = 1, \dots, i$,

$$c_{ij} = \sum_{k=j}^i a_{ik} b_{kj}.$$

Multiplication of lower triangular $A = (a_{ij})_{n \times n}$ and $B = (b_{ij})_{n \times n}$

```

1: Set  $c_{ij}$  to be zero for all  $1 \leq i < j \leq n$ 
2: for  $i = 1, \dots, n$  do
3:   for  $j = 1, \dots, i$  do
4:      $c_{ij} \leftarrow a_{ij} b_{jj}$ 
5:     for  $k = j + 1, \dots, i$  do
6:        $c_{ij} \leftarrow c_{ij} + a_{ik} b_{kj}$ 
7: return  $C = (c_{ij})_{n \times n}$ 

```

The number of multiplications is

$$\begin{aligned}
\sum_{i=1}^n \sum_{j=1}^i \left(1 + \sum_{k=j+1}^i 1 \right) &= \sum_{i=1}^n \sum_{j=1}^i (1 + (i - j)) = \sum_{i=1}^n (i + (i - 1) + \dots + 1) \\
&= \sum_{i=1}^n \frac{i(i+1)}{2} = \frac{1}{2} \sum_{i=1}^n (i^2 + i) \\
&= \frac{1}{2} \left(\frac{n(n+1)(2n+1)}{6} + \frac{n(n+1)}{2} \right) = \frac{n(n+1)(n+2)}{6}.
\end{aligned}$$

The number of additions is

$$\begin{aligned}
\sum_{i=1}^n \sum_{j=1}^i \sum_{k=j+1}^i 1 &= \sum_{i=1}^n \sum_{j=1}^i (i - j) = \sum_{i=1}^n ((i - 1) + \dots + 1 + 0) \\
&= \sum_{i=1}^n \frac{i(i-1)}{2} = \frac{1}{2} \sum_{i=1}^n (i^2 - i) \\
&= \frac{1}{2} \left(\frac{n(n+1)(2n+1)}{6} - \frac{n(n+1)}{2} \right) = \frac{(n-1)n(n+1)}{6}.
\end{aligned}$$

□

Theorem 1.24. Multiplying a tridiagonal matrix and an upper triangular matrix: Let $A, B \in \mathbb{R}^{n \times n}$, A is tridiagonal, B is upper triangular.

- Number of multiplications: $(3n - 2)(n + 1)/2$

- Number of additions: $n(n-1)$

Proof. From A being tridiagonal and B being upper triangular, we have

$$\begin{aligned} a_{ik} \neq 0 &\implies k \in \{i-1, i, i+1\}, \\ b_{kj} \neq 0 &\implies k \leq j. \end{aligned}$$

Hence the only indices that contribute to

$$c_{ij} = \sum_{k=1}^n a_{ik} b_{kj}$$

are those in the intersection $\{i-1, i, i+1\} \cap \{1, 2, \dots, j\}$. Therefore, $c_{ij} = 0$ whenever this intersection is empty. This happens precisely when $i-1 > j$. Thus, we only need to compute c_{ij} for $j \geq i-1$. Since we don't need to compare j and 0 (when $i = 1$), we only need to compute c_{ij} for $j \geq \max(1, i-1)$.

We need to compute

$$c_{ij} = \sum_{k=1}^n a_{ik} b_{kj} \quad (i = 1, \dots, n, j = \max(1, i-1), \dots, n).$$

Note that

- $a_{ik} \neq 0$ only if $\max(1, i-1) \leq k \leq i+1$,
- $b_{kj} \neq 0$ only if $k \leq j$.

Therefore, for $i = 1, \dots, n, j = \max(1, i-1), \dots, n$,

$$c_{ij} = \sum_{k=\max(1, i-1)}^{\min(i+1, j)} a_{ik} b_{kj}.$$

Multiplication of tridiagonal $A = (a_{ij})_{n \times n}$ and upper triangular $B = (b_{ij})_{n \times n}$

```

1: for  $i = 1, \dots, n$  do
2:   for  $j = \max(1, i-1), \dots, n$  do
3:      $k \leftarrow \max(1, i-1)$ ,  $c_{ij} \leftarrow a_{ik} b_{kj}$ 
4:     for  $k = \max(2, i), \dots, \min(i+1, j)$  do
5:        $c_{ij} \leftarrow c_{ij} + a_{ik} b_{kj}$ 
6: return  $C = (c_{ij})_{n \times n}$ 

```

The number of multiplications is

$$\sum_{i=1}^n \sum_{j=\max(1, i-1)}^n \left(1 + \sum_{k=\max(2, i)}^{\min(i+1, j)} 1 \right) = \sum_{i=1}^n \sum_{j=\max(1, i-1)}^n (\min(i+1, j) - \max(1, i-1) + 1). \quad (*)$$

To deal with the maximum term, split the i -summation into $\sum_{i=1}^1$ and $\sum_{i=2}^n$. For the first summation, $i = 1$ implies $i-1 < 1$ so $\max(1, i-1) = 1$. For the second summation, $i \geq 2$ implies $i-1 \geq 1$ so $\max(1, i-1) = i-1$. Thus,

$$\begin{aligned} (*) &= \sum_{j=1}^n \min(2, j) + \sum_{i=2}^n \sum_{j=i-1}^n (\min(i+1, j) - (i-1) + 1) \\ &= (1 + 2 + \dots + 2) + \sum_{i=2}^n \sum_{j=i-1}^n (\min(i+1, j) - (i-1) + 1) \\ &= (2n-1) + \sum_{i=2}^n \sum_{j=i-1}^n (\min(i+1, j) - (i-1) + 1). \end{aligned}$$

To deal with the minimum term, split the j -summation into $\sum_{j=i-1}^i$ and $\sum_{j=i+1}^n$. For the first summation, $j < i + 1$; for the second summation, $j \geq i + 1$.

$$\begin{aligned}
(*) &= (2n - 1) + \sum_{i=2}^n \left(\sum_{j=i-1}^i + \sum_{j=i+1}^n \right) (\min(i + 1, j) - (i - 1) + 1) \\
&= (2n - 1) + \sum_{i=2}^n \sum_{j=i-1}^i (j - (i - 1) + 1) + \sum_{i=2}^n \sum_{j=i+1}^n ((i + 1) - (i - 1) + 1) \\
&= (2n - 1) + \sum_{i=2}^n (1 + 2) + \sum_{i=2}^n \sum_{j=i+1}^n ((i + 1) - (i - 1) + 1) \\
&= (2n - 1) + 3(n - 1) + 3 \sum_{i=2}^n (n - i) \\
&= (2n - 1) + 3(n - 1) + 3 \cdot \frac{(n - 2)(n - 1)}{2} = \frac{1}{2}(3n - 2)(n + 1).
\end{aligned}$$

Similarly, the number of additions is

$$\begin{aligned}
\sum_{i=1}^n \sum_{j=\max(1, i-1)}^n \sum_{k=\max(2, i)}^{\min(i+1, j)} 1 &= \sum_{i=1}^n \sum_{j=\max(1, i-1)}^n (\min(i + 1, j) - \max(2, i) + 1) \\
&= \sum_{j=1}^n (\min(2, j) - 1) + \sum_{i=2}^n \sum_{j=i-1}^n (\min(i + 1, j) - i + 1) \\
&= (0 + 1 + \cdots + 1) + \sum_{i=2}^n \sum_{j=i-1}^n (\min(i + 1, j) - i + 1) \\
&= (n - 1) + \sum_{i=2}^n \left(\sum_{j=i-1}^i + \sum_{j=i+1}^n \right) (\min(i + 1, j) - i + 1) \\
&= (n - 1) + \sum_{i=2}^n \sum_{j=i-1}^i (j - i + 1) + \sum_{i=2}^n \sum_{j=i+1}^n ((i + 1) - i + 1) \\
&= (n - 1) + (n - 1) + 2 \cdot \frac{(n - 2)(n - 1)}{2} = n(n - 1).
\end{aligned}$$

□

Theorem 1.25. Let $A = (a_{ij})_{n \times n}$ be a lower Hessenberg matrix and $B = (b_{ij})_{n \times n}$ be an upper Hessenberg matrix. Write down the pseudocode to find $C = AB$.

Proof. By the given condition, $a_{ik} = 0$ if $i < k - 1$, $b_{kj} = 0$ if $k > j + 1$. Then $a_{ik} \neq 0$ only if $i \geq k - 1$, $b_{kj} \neq 0$ only if $k \leq j + 1$.

For non-zero entries in the sum $c_{ij} = \sum_{k=1}^n a_{ik}b_{kj}$, we need $a_{ik} \neq 0$ and $b_{kj} \neq 0$. Thus, $k \leq i + 1$ and $k \leq j + 1$, so $k \leq \min(i + 1, j + 1)$. We also need $k \leq n$, so $k \leq \min(n, i + 1, j + 1)$. Thus,

$$c_{ij} = \sum_{k=1}^n a_{ik}b_{kj} = \sum_{k=1}^{\min(n, i+1, j+1)} a_{ik}b_{kj}.$$

Multiplication of lower Hessenberg $A = (a_{ij})_{n \times n}$ and upper Hessenberg $B = (b_{ij})_{n \times n}$

```

1: for  $i = 1, \dots, n$  do
2:   for  $j = 1, \dots, n$  do
3:      $c_{ij} = a_{i1}b_{1j}$ 
4:     for  $k = 2, \dots, \min(n, i + 1, j + 1)$  do
5:        $c_{ij} \leftarrow c_{ij} + a_{ik}b_{kj}$ 
6: return  $C = (c_{ij})_{n \times n}$ 

```

□

1.5 Time complexity

Definition 1.26. We say $f(n) = O(g(n))$ if there exists a positive constant M and a real number N such that

$$|f(n)| \leq Mg(n) \quad \text{for all } n > N.$$

Example.

- $n(n+1)(n+2)/6 = O(n^3)$, since when $n > 2$,

$$\frac{n(n+1)(n+2)}{6} \leq \frac{n(2n)(2n)}{6} = \frac{2}{3}n^3.$$

- $(n-1)n(n+1)/6 = O(n^3)$ since

$$\frac{(n-1)n(n+1)}{6} = \frac{n(n^2-1)}{6} \leq \frac{1}{6}n^3.$$

Example. Let $\mathbf{u} = (u_1, \dots, u_n)^T$, $\mathbf{v} = (v_1, \dots, v_n)^T$, $\mathbf{w} = (w_1, \dots, w_n)^T$. Find $\mathbf{r} = \mathbf{u}\mathbf{v}^T\mathbf{w}$. (It turns out that the order of multiplication matters!)

Method 1: First compute $A = \mathbf{u}\mathbf{v}^T$:

$$a_{ij} = u_i v_j \quad (i, j = 1, \dots, n).$$

Then compute $\mathbf{r} = A\mathbf{w}$:

$$r_i = \sum_{j=1}^n a_{ij} w_j \quad (i = 1, \dots, n).$$

Computing $\mathbf{u}\mathbf{w}^T\mathbf{w}$

```

1: for  $i = 1, \dots, n$  do ▷ Compute  $A = \mathbf{u}\mathbf{v}^T$ 
2:   for  $j = 1, \dots, n$  do
3:      $a_{ij} \leftarrow u_i v_j$ 
4:   for  $i = 1, \dots, n$  do ▷ Compute  $\mathbf{r} = A\mathbf{w}$ 
5:      $r_i \leftarrow a_{i1}w_1$ 
6:     for  $j = 2, \dots, n$  do
7:        $r_i \leftarrow r_i + a_{ij}w_j$ 
8: return  $(r_1, \dots, r_n)^T$ 

```

The time complexity of this algorithm is $O(n^2)$.

Method 2: First compute $C = \mathbf{v}^T\mathbf{w} = \mathbf{v} \cdot \mathbf{w} = \sum_{i=1}^n v_i w_i$ (which is a 1×1 matrix, i.e., a real number). Then compute $\mathbf{r} = C\mathbf{u}$.

Computing $\mathbf{u}\mathbf{w}^T\mathbf{w}$

```

1:  $C \leftarrow v_1 w_1$   $\triangleright$  Compute  $C = \mathbf{v}^T \mathbf{w}$ 
2: for  $i = 2, \dots, n$  do
3:    $C \leftarrow C + v_i w_i$ 
4: for  $i = 1, \dots, n$  do  $\triangleright$  Compute  $\mathbf{r} = \mathbf{C}\mathbf{u}$ 
5:    $r_i \leftarrow C u_i$ 
6: return  $(r_1, \dots, r_n)^T$ 

```

The time complexity of this algorithm is $O(n)$.

1.6 Numerical stability

Consider an integration problem of finding the value of

$$I_n(\beta) = \int_0^1 \frac{x^n}{x + \beta} dx$$

for some fixed $\beta > 0$. We derive a **recurrence relation** for $I_n(\beta)$. If $n \geq 1$, the trick is to add a term:

$$\begin{aligned}
 I_n(\beta) &= \int_0^1 \frac{x^n + \beta x^{n-1}}{x + \beta} - \frac{\beta x^{n-1}}{x + \beta} dx \\
 &= \int_0^1 x^{n-1} dx - \beta \int_0^1 \frac{x^{n-1}}{x + \beta} dx \\
 &= \frac{1}{n} - \beta I_{n-1}(\beta).
 \end{aligned}$$

For the initial condition where $n = 0$,

$$I_0(\beta) = \int_0^1 \frac{1}{x + \beta} dx = [\ln(x + \beta)]_0^1 = \ln \frac{\beta + 1}{\beta}.$$

Computation of $I_n(\beta)$ for $\beta > 0$

```

1:  $I \leftarrow \ln \frac{\beta + 1}{\beta}$ 
2: for  $k = 1, \dots, n$  do
3:    $I \leftarrow \frac{1}{k} - \beta I$ 

```

Take a large $\beta = 10$. For large values of n , the numerical result produced by the above algorithm differs significantly from the exact result, due to **round-off error**.

Let $\tilde{I}_n(\beta)$ be the numerical solution.

- When $n = 0$,

$$\tilde{I}_0(\beta) = I_0(\beta) + \epsilon_0$$

where ϵ_0 is the round-off error.

- When $n = 1$,

$$\begin{aligned}
 \tilde{I}_1(\beta) &= 1 \oplus 1 \ominus \beta \otimes \tilde{I}_0(\beta) \\
 &= 1 - \beta \tilde{I}_0(\beta) + \epsilon_1 \\
 &= 1 - \beta I_0(\beta) - \beta \epsilon_0 + \epsilon_1 \\
 &= I_1(\beta) - \beta \epsilon_0 + \epsilon_1
 \end{aligned}$$

where ε_1 is the additional round-off error introduced at this step.

- When $n = 2$,

$$\begin{aligned}\tilde{I}_2(\beta) &= 1 \oplus 2 \ominus \beta \otimes \tilde{I}_1(\beta) \\ &= 1/2 - \beta \tilde{I}_1(\beta) + \varepsilon_2 \\ &= 1/2 - \beta I_1(\beta) + \beta^2 \varepsilon_0 - \beta \varepsilon_0 + \varepsilon_2 \\ &= I_2(\beta) + \beta^2 \varepsilon_0 - \beta \varepsilon_1 + \varepsilon_2\end{aligned}$$

- In general,

$$\tilde{I}_n(\beta) = I_n(\beta) + \sum_{k=0}^n (-\beta)^{n-k} \varepsilon_k.$$

When β and n are large, the error term becomes large even if every ε_k is small!

Our numerical method above using a recurrence relation is called an **iterative method**.

- In each iteration, the error introduced in the previous step is **amplified** by the **amplification factor** β .
- If $\beta \leq 1$, we say that the method is **numerically stable**.
- If $\beta > 1$, we say that the method is **numerically unstable**.

For unstable methods, the numerical error grows quickly, and the results turn unreliable after several iterations.

Hence, we should consider an alternative method. When $\beta > 1$, the **series expansion** of the integral of $I_n(\beta)$ is

$$\begin{aligned}I_n(\beta) &= \int_0^1 \frac{x^n}{x + \beta} dx = \int_0^1 x^n \cdot \frac{1}{x + \beta} = \int_0^1 x^n \cdot \frac{1}{\beta} \frac{1}{1 - \left(-\frac{x}{\beta}\right)} \\ &= \int_0^1 x^n \frac{1}{\beta} \sum_{k=0}^n \left(-\frac{x}{\beta}\right)^k dx = \sum_{k=0}^{\infty} \int_0^1 \frac{(-1)^k}{\beta} \frac{x^{n+k}}{\beta^k} dx \\ &= \sum_{k=0}^{\infty} \frac{(-1)^k}{\beta^{k+1}} \left[\frac{x^{n+k+1}}{n+k+1} \right]_0^1 = \sum_{k=0}^{\infty} \frac{(-1)^k}{\beta^{k+1}} \cdot \frac{1}{n+k+1} = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{n+k} \left(\frac{1}{\beta}\right)^k\end{aligned}$$

where we used the geometric series $\frac{1}{1-t} = 1 + t + t^2 + \dots$ ($|t| < 1$).

We approximate this infinite sum by truncating it to a finite sum:

$$I_n(\beta) \approx \sum_{k=1}^N \frac{(-1)^{k+1}}{n+k} \left(\frac{1}{\beta}\right)^k.$$

The truncation error can be bounded by

$$\begin{aligned}\varepsilon_N &:= \left| I_n(\beta) - \sum_{k=1}^N \frac{(-1)^{k+1}}{n+k} \left(\frac{1}{\beta}\right)^k \right| = \left| \sum_{k=N+1}^{\infty} \frac{(-1)^{k+1}}{n+k} \left(\frac{1}{\beta}\right)^k \right| \\ &\leq \sum_{k=N+1}^{\infty} \left| \frac{(-1)^{k+1}}{n+k} \left(\frac{1}{\beta}\right)^k \right| = \sum_{k=N+1}^{\infty} \frac{1}{n+k} \left(\frac{1}{\beta}\right)^k \\ &\leq \frac{1}{n+1} \sum_{k=N+1}^{\infty} \left(\frac{1}{\beta}\right)^k = \frac{1}{(n+1)(\beta-1)} \left(\frac{1}{\beta}\right)^N\end{aligned}$$

by computing the geometric series. Observe that as $N \rightarrow \infty$, the truncation error ε_N tends to 0.

In order that the truncation error does not exceed ε , we require that

$$\frac{1}{(n+1)(\beta-1)} \left(\frac{1}{\beta}\right)^N < \varepsilon$$

which is equivalent to

$$N > -\log_{\beta}[\varepsilon(n+1)(\beta-1)].$$

The approximation of $I_n(\beta)$ can be written as

$$I_n(\beta) \approx \sum_{k=1}^N \frac{(-1)^{k+1}}{n+k} \left(\frac{1}{\beta}\right)^k = \sum_{k=1}^N \frac{s_k}{n+k}$$

where $s_{k+1} = -s_k/\beta$ (replace power terms) and $s_0 = 1$.

Computation of $I_n(\beta)$ for $\beta > 1$ with error $< \varepsilon$

```

1:  $N \leftarrow \left\lceil -\log_{\beta}[\varepsilon(n+1)(\beta-1)] \right\rceil$ 
2:  $I \leftarrow 0, s \leftarrow -1$ 
3: for  $k = 1, \dots, N$  do
4:    $s \leftarrow -\frac{s}{\beta}$ 
5:    $I \leftarrow I + \frac{s}{n+k}$ 
6: return  $I$ 
```

2 Solution of Linear Systems of Equations

2.1 Linear systems & Gaussian elimination

Definition 2.1. A **linear system** with m equations and n unknowns has the form

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1, \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2, \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m. \end{aligned}$$

Let $A = (a_{ij}) \in \mathbb{R}^{m \times n}$, $\mathbf{x} = (x_1, \dots, x_n)^T$, and $\mathbf{b} = (b_1, \dots, b_m)^T$. Then the linear system can be written compactly as

$$A\mathbf{x} = \mathbf{b}.$$

For a linear system $A\mathbf{x} = \mathbf{b}$, the number of solutions is either 0, 1, or infinitely many.

Definition 2.2. Given $A\mathbf{x} = \mathbf{b}$ with $A \in \mathbb{R}^{m \times n}$, $\mathbf{b} \in \mathbb{R}^m$, the **augmented matrix** is

$$\tilde{A} = (A \mid \mathbf{b}) = \left(\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{array} \right).$$

Definition 2.3. Let $A = (a_{ij}) \in \mathbb{R}^{m \times n}$. The **rank** of A , denoted $\text{rank}(A)$, is the dimension of the column space of A .

Theorem 2.4 (Existence and uniqueness). *Let $\tilde{A} = (A \mid \mathbf{b})$ be the augmented matrix of $A\mathbf{x} = \mathbf{b}$, where $A \in \mathbb{R}^{m \times n}$ and $\mathbf{b} \in \mathbb{R}^m$.*

- (i) *A solution exists if and only if $\text{rank}(A) = \text{rank}(\tilde{A})$.*
- (ii) *The solution is unique if and only if $\text{rank}(A) = \text{rank}(\tilde{A}) = n$.*

Gaussian elimination is a two-stage procedure for solving $A\mathbf{x} = \mathbf{b}$:

- (1) **Elimination:** transform $\tilde{A}$ into an upper triangular matrix.
- (2) **Backward substitution:** solve the resulting triangular system by solving $x_n, x_{n-1}, \dots, x_2, x_1$.

Consider the augmented matrix

$$\tilde{A} = \left(\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & & \vdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} & b_n \end{array} \right).$$

If $a_{11} \neq 0$, we eliminate x_1 by performing

$$(j) \leftarrow (j) - \frac{a_{j1}}{a_{11}} \times (1) \quad \text{for } j = 2, \dots, n.$$

This produces the augmented matrix

$$\left(\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ 0 & \tilde{a}_{22} & \cdots & \tilde{a}_{2n} & \tilde{b}_2 \\ \vdots & \vdots & & \vdots & \vdots \\ 0 & \tilde{a}_{n2} & \cdots & \tilde{a}_{nn} & \tilde{b}_n \end{array} \right)$$

where for $j = 2, \dots, n$ and $k = 2, \dots, n$,

$$\tilde{a}_{jk} = a_{jk} - \frac{a_{j1}}{a_{11}}a_{1k}, \quad \tilde{b}_j = b_j - \frac{a_{j1}}{a_{11}}b_1.$$

In general, if $a_{ii} \neq 0$, to eliminate x_i , perform the operation

$$(j) \leftarrow (j) - \frac{a_{ji}}{a_{ii}} \times (i) \quad \text{for } j = i + 1, \dots, n.$$

Repeating this process yields an upper triangular augmented matrix

$$\left(\begin{array}{ccccc|c} a_{11} & a_{12} & \cdots & a_{1,n-1} & a_{1n} & b_1 \\ & a_{22} & \cdots & a_{2,n-1} & a_{2n} & b_2 \\ & & \ddots & \vdots & \vdots & \vdots \\ & & & a_{n-1,n-1} & a_{n-1,n} & b_{n-1} \\ 0 & & & & a_{nn} & b_n \end{array} \right).$$

Then the updated entries satisfy

$$\tilde{a}_{jk} = a_{jk} - \frac{a_{ji}}{a_{ii}}a_{ik}, \quad \tilde{b}_j = b_j - \frac{a_{ji}}{a_{ii}}b_i, \quad j, k = i + 1, \dots, n. \quad (2)$$

The new linear system is triangular,

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{nn}x_n &= b_n \end{aligned}$$

so **backward substitution** can be performed. Solving the n -th equation for x_n gives

$$x_n = \frac{b_n}{a_{nn}}.$$

Solving the $(n - 1)$ -th equation for x_{n-1} and using the known value for x_n yields

$$x_{n-1} = \frac{b_{n-1} - a_{n-1,n}x_n}{a_{n-1,n-1}}.$$

Continuing this process, we obtain

$$x_i = \frac{b_i - \sum_{j=i+1}^n a_{ij}x_j}{a_{ii}} \quad (3)$$

for each $i = n - 1, \dots, 1$.

For efficiency, we can make the elimination process more economic by avoid changing numbers to zero (since the lower triangular entries will eventually equal 0, there is no need to change them, leave them unchanged).

Algorithm 2 Solve the linear system with augmented matrix $\tilde{A} = (a_{ij})_{n \times (n+1)}$

```

1: for  $i = 1, \dots, n-1$  do  $\triangleright$  Elimination
2:   for  $j = i+1, \dots, n$  do
3:      $m_{ji} \leftarrow \frac{a_{ji}}{a_{ii}}$ 
4:     for  $k = i+1, \dots, n+1$  do
5:        $a_{jk} \leftarrow a_{jk} - m_{ji}a_{ik}$ 
6:  $x_n \leftarrow \frac{a_{n,n+1}}{a_{nn}}$   $\triangleright$  Backward substitution
7: for  $i = n-1, \dots, 1$  do
8:    $x_i \leftarrow a_{i,n+1}$ 
9:   for  $j = i+1, \dots, n$  do
10:     $x_i \leftarrow x_i - a_{ij}x_j$ 
11:    $x_i \leftarrow \frac{x_i}{a_{ii}}$ 
12: return  $(x_1, \dots, x_n)^T$ 

```

For **elimination**, the number of divisions is

$$\sum_{i=1}^{n-1} \sum_{j=i+1}^n 1 = \sum_{i=1}^{n-1} (n-i) = \frac{n(n-1)}{2}.$$

The number of subtractions equals the number of multiplications, which equals

$$\begin{aligned}
\sum_{i=1}^{n-1} \sum_{j=i+1}^n \sum_{k=i+1}^{n+1} 1 &= \sum_{i=1}^{n-1} \sum_{j=i+1}^n (n-i+1) \\
&= \sum_{i=1}^{n-1} (n-i)(n-i+1) \\
&= \sum_{i=1}^{n-1} (n-i)^2 + \sum_{i=1}^{n-1} (n-i) \\
&= \frac{(n-1)n(2n-1)}{6} + \frac{n(n-1)}{2} = \frac{n(n-1)(n+1)}{3}.
\end{aligned}$$

For **backward substitution**, the number of divisions is n . The number of multiplications equals the number of subtractions, which equals

$$\sum_{i=1}^{n-1} \sum_{j=i+1}^n 1 = \sum_{i=1}^{n-1} (n-i) = \frac{n(n-1)}{2}.$$

Theorem 2.5. The operation counts of Gaussian elimination are:

- Number of divisions: $\frac{n(n+1)}{2}$
- Number of multiplications (number of subtractions): $\frac{n(n-1)(2n+5)}{6}$

Hence, the time complexity is $O(n^3)$.

2.2 Gaussian elimination with row labels

In the case where the diagonal element = 0, we need to swap rows. Note that the cost of swapping rows is $O(n)$, since we need to copy over the n elements of the rows involved. Thus, we instead label all the rows, and **swap row labels**. Swapping real numbers incurs much less computational cost.

Example. Consider the augmented matrix

$$\begin{array}{l} 1 : \\ 2 : \\ 3 : \\ 4 : \end{array} \left(\begin{array}{cccc|c} 1 & -1 & 2 & -1 & -8 \\ 0 & 0 & -1 & -1 & -4 \\ 0 & 2 & -1 & 1 & 6 \\ 0 & -1 & 2 & 4 & 10 \end{array} \right)$$

where we add labels to each row. To eliminate x_2 , swap (2) $\leftrightarrow$ (3).

$$\begin{array}{l} 1 : \\ 3 : \\ 2 : \\ 4 : \end{array} \left(\begin{array}{cccc|c} 1 & -1 & 2 & -1 & -8 \\ 0 & 0 & -1 & -1 & -4 \\ 0 & 2 & -1 & 1 & 6 \\ 0 & -1 & 2 & 4 & 10 \end{array} \right)$$

Then perform Gaussian elimination as usual.

The row labels are stored in array of n elements $r_1, \dots, r_n$.

Definition 2.6. Let r_i be the index of the row with label i .

Read $r_i = j$ this as “label i is at the j -th row”.

Example.

$$\begin{array}{l} 1 : \\ 4 : \\ 2 : \\ 3 : \end{array} \left(\begin{array}{cccc|c} 1 & -1 & 2 & -1 & -8 \\ 0 & 0 & -1 & -1 & -4 \\ 0 & 2 & -1 & 1 & 6 \\ 0 & 0 & 3/2 & 9/2 & 13 \end{array} \right)$$

Then label 1 corresponds to row 1, label 2 corresponds to row 3, label 3 corresponds to row 4, label 4 corresponds to row 2:

$$r_1 = 1, \quad r_2 = 3, \quad r_3 = 4, \quad r_4 = 2.$$

To implement $(j) \leftarrow (j) - m_{ji} \times (i)$,

$$a_{r_j k} \leftarrow a_{r_j k} - m_{ji} \times a_{r_i k}, \quad k = i + 1, \dots, n + 1.$$

To implement $(i) \leftrightarrow (j)$,

$$r_i \leftrightarrow r_j.$$

How to determine m_{ji} in $(j) \leftarrow (j) - m_{ji} \times (i)$?

$$a_{r_j i} = m_{ji} a_{r_i i} \implies m_{ji} = \frac{a_{r_j i}}{a_{r_i i}}$$

We swap labels when the diagonal element is zero:

$$a_{r_i i} = 0 \quad \text{before computing } m_{ji}.$$

$a_{r_i i}$ is called the **pivot element**.

To swap labels, find the next non-zero element, i.e., find some $j > i$ and $a_{r_j i}$, then do $r_i \leftrightarrow r_j$.

If $a_{r_j i} = 0$ for all $j \geq i$, then number of equations $>$ unknowns, so there are zero or infinitely many solutions.

When eliminating the i -th variable, we do the following:

- Check if the pivot element $a_{r_i i}$ is zero.
- If $a_{r_i i} = 0$, try to find some $j > i$ such that $a_{r_j i} \neq 0$.
- If such a j cannot be found, return an error message.

- If found, do $(i) \leftrightarrow (j)$ or $r_i \leftrightarrow r_j$.
- Eliminate the i -th variable by

$$(j) \leftarrow (j) - m_{ji} \times (i), \quad j = i + 1, \dots, n$$

where $m_{ji} = \frac{a_{r_j i}}{a_{r_i i}}$.

Backward substitution:

$$x_i = \frac{1}{a_{r_i i}} \left(a_{r_i, n+1} - \sum_{j=i+1}^n a_{r_i j} x_j \right).$$

Searching for a nonzero pivot element:

Algorithm 3 Find $a_{r_j i} \neq 0$ for $j \geq i$ and swap the labels

```

1:  $j \leftarrow i$ 
2: while  $j \leq n$  and  $a_{r_j i} = 0$  do
3:    $j \leftarrow j + 1$ 
4: if  $j = n + 1$  then
5:   return "Error: Matrix is singular"
6: else if  $j \neq i$  then
7:   Swap  $r_i$  and  $r_j$ 
```

Algorithm 4 Elimination

```

1: for  $i = 1, \dots, n$  do
2:    $r_i \leftarrow i$ 
3: for  $i = 1, \dots, n - 1$  do
4:   Find  $a_{r_j i} \neq 0$  for  $j \geq i$  and swap row labels
5:   for  $j = i + 1, \dots, n$  do
6:      $m_{ji} \leftarrow \frac{a_{r_j i}}{a_{r_i i}}$ 
7:     for  $k = i + 1, \dots, n + 1$  do
8:        $a_{r_j k} \leftarrow a_{r_j k} - m_{ji} a_{r_i k}$ 
```

▷ Initialise row labels

Algorithm 5 Backward substitution

```

1: for  $i = n, \dots, 1$  do
2:    $x_i \leftarrow a_{r_i, n+1}$ 
3:   for  $j = i + 1, \dots, n$  do
4:      $x_i \leftarrow x_i - a_{r_i j} x_j$ 
5:    $x_i \leftarrow \frac{x_i}{a_{r_i i}}$ 
6: return  $(x_1, \dots, x_n)^T$ 
```

As an application, let A be a **diagonally dominant** tridiagonal matrix:

$$A = \begin{pmatrix} a_{11} & a_{12} & & & & \\ a_{21} & a_{22} & a_{23} & & & \\ & a_{32} & a_{33} & a_{34} & & \\ & & \ddots & \ddots & \ddots & \\ & & & a_{n-1, n-2} & a_{n-1, n-1} & a_{n-1, n} \\ & & & & a_{n, n-1} & a_{nn} \end{pmatrix}$$

and the elements of A satisfy $|a_{11}| > |a_{21}|$, $|a_{nn}| > |a_{n-1,n}|$, and

$$|a_{kk}| > |a_{k+1,k}| + |a_{k-1,k}| \quad k = 2, \dots, n-1.$$

Consider solving the linear system $A\mathbf{x} = \mathbf{b}$.

- First pivoting: No exchange since $|a_{11}| > |a_{21}|$.

Row operation to eliminate x_1 : $(2) \leftarrow (2) - \frac{a_{21}}{a_{11}} \times (1)$.

After the row operation, the submatrix

$$\begin{pmatrix} \tilde{a}_{22} & a_{23} & & & \\ a_{32} & a_{33} & a_{34} & & \\ & \ddots & \ddots & \ddots & \\ & & a_{n-1,n-2} & a_{n-1,n-1} & a_{n-1,n} \\ & & & a_{n,n-1} & a_{nn} \end{pmatrix}$$

is still a diagonally dominant tridiagonal matrix. To see this, note that

$$|\tilde{a}_{22}| = \left| a_{22} - \frac{a_{21}}{a_{11}} a_{12} \right| \geq |a_{22}| - \left| \frac{a_{21}}{a_{11}} \right| |a_{12}| \geq |a_{22}| - |a_{12}| > |a_{32}|$$

where we used the condition that $|a_{21}| \leq |a_{11}|$. More generally, for intermediate columns,

$$|a_{kk}| > |a_{k+1,k}| + |a_{k-1,k}|, \quad k = 2, \dots, n-1.$$

- Second pivoting: No exchange since $|\tilde{a}_{22}| > |a_{32}|$

Row operation: $(3) \leftarrow (3) - \frac{a_{32}}{\tilde{a}_{22}} \times (2)$

After the row operation, the submatrix is still a diagonally dominant tridiagonal matrix. To see this, note that

$$|\tilde{a}_{33}| = \left| a_{33} - \frac{a_{32}a_{23}}{\tilde{a}_{22}} \right| \geq |a_{33}| - \left| \frac{a_{32}a_{23}}{\tilde{a}_{22}} \right| \geq |a_{33}| - |a_{32}| > |a_{43}|$$

where we used the condition that $|\tilde{a}_{22}| \leq |a_{32}|$. More generally, for intermediate columns,

$$|a_{kk}| > |a_{k+1,k}| + |a_{k-1,k}|, \quad k = 2, \dots, n-1.$$

- Third pivoting: No exchange since $|\tilde{a}_{33}| > |a_{43}|$

Conclusion: Exchange of labels never happens during partial pivoting (labels are not need). Every row operation only changes one diagonal element:

$$a_{i+1,i+1} \leftarrow a_{i+1,i+1} - \frac{a_{i+1,i}a_{i,i+1}}{a_{ii}}.$$

After elimination,

$$\left(\begin{array}{ccccccccc|c} a_{11} & a_{12} & & & & & & & b_1 \\ a_{21} & a_{22} & a_{23} & & & & & & b_2 \\ & a_{32} & a_{33} & a_{34} & & & & & b_3 \\ & & \ddots & \ddots & \ddots & & & & \vdots \\ & & & a_{n-1,n-2} & a_{n-1,n-1} & a_{n-1,n} & & & b_{n-1} \\ & & & & a_{n,n-1} & a_{nn} & & & b_n \end{array} \right)$$

We do backward substitution:

$$x_n = \frac{b_n}{a_{nn}}$$

$$x_i = \frac{b_i - a_{i,i+1}x_{i+1}}{a_{ii}}, \quad i = n-1, \dots, 1.$$

Thomas algorithm

```

1: for  $i = 1, \dots, n-1$  do
2:    $m \leftarrow \frac{a_{i+1,i}}{a_{ii}}$ 
3:    $a_{i+1,i+1} \leftarrow a_{i+1,i+1} - m \times a_{i,i+1}$ 
4:    $b_{i+1} \leftarrow b_{i+1} - m \times b_i$ 
5:  $x_n \leftarrow \frac{b_n}{a_{nn}}$ 
6: for  $i = n-1, \dots, 1$  do
7:    $x_i \leftarrow \frac{b_i - a_{i,i+1}x_{i+1}}{a_{ii}}$ 
8: return  $(x_1, \dots, x_n)^T$ 

```

Operation counts: $2n-1$ divisions, $3(n-1)$ subtractions, $3(n-1)$ multiplications. Hence, time complexity is $O(n)$.

2.3 Partial pivoting

To reduce round-off error, it is often necessary to perform row interchanges even when the pivot elements are not zero.

Example. Consider

$$\begin{pmatrix} 0.003000 & 59.14 \\ 5.291 & -6.130 \end{pmatrix} \mathbf{x} = \begin{pmatrix} 59.17 \\ 46.78 \end{pmatrix}$$

using four-digit arithmetic with rounding. The exact solution is $x_1 = 10.00$, $x_2 = 1.000$. The first pivot element $a_{11} = 0.003000$ is small, and its associated multiplier

$$m_{21} = \frac{5.291}{0.003000} = 1763.6\bar{6}$$

rounds to the large number 1764. Elimination yields

$$\begin{pmatrix} 0.003000 & 59.14 \\ & -104300 \end{pmatrix} \mathbf{x} = \begin{pmatrix} 59.17 \\ -104400 \end{pmatrix}.$$

Backward substitution yields

$$x_2 \approx 1.001,$$

which is a close approximation to the actual value. However, because of the small pivot a_{11} ,

$$x_1 \approx \frac{59.17 - (59.14)(1.001)}{0.003000} = -10.00$$

which is very far off from the actual value. This is because $err_{abs}(x_1) = err_{abs}(x_2) \times \frac{b_1}{a_{11}}$, where $\frac{b_1}{a_{11}} \approx 2000$ is large.

To avoid this problem, we perform **partial pivoting**: select an element in the same column that is below the diagonal with the largest absolute value as the pivot element.

Algorithm 6 Find the largest $|a_{r_j i}|$ for $j \geq i$, and swap the row labels

```

1:  $j \leftarrow i$ 
2: for  $k = i + 1, \dots, n$  do
3:   if  $|a_{r_k i}| > |a_{r_j i}|$  then
4:      $j \leftarrow k$ 
5: if  $a_{r_j i} = 0$  then
6:   return "Error: Matrix is singular"
7: else if  $j \neq i$  then
8:   Swap  $r_i$  and  $r_j$ 

```

$\triangleright$ Find the largest $|a_{r_k i}|$
 $\triangleright$ Take that as the pivot
 $\triangleright$ All pivot candidates are 0

Number of comparisons is

$$\sum_{i=1}^{n-1} \sum_{k=i+1}^n 1 = \frac{n(n-1)}{2}.$$

Example. Assume that we obtain the following augmented matrix after the algorithm of partial pivoting:

$$\begin{array}{l} 3 : \\ 1 : \\ 4 : \\ 2 : \end{array} \left(\begin{array}{cccc|c} 2 & 4/3 & -3 & 14/3 & 32/3 \\ 3 & 1 & 9 & 2 & -6 \\ 1 & 2/3 & 0 & 7/3 & 7/3 \\ 0 & -2 & 3 & 9 & 7 \end{array} \right)$$

Find the original augmented matrix and the solution of the system.

Solution. Identify the pivot elements, and numbers that should be 0:

$$\begin{array}{l} 3 : \\ 1 : \\ 4 : \\ 2 : \end{array} \left(\begin{array}{cccc|c} 2 & 4/3 & -3 & 14/3 & 32/3 \\ 3 & 1 & 9 & 2 & -6 \\ 1 & 2/3 & 0 & 7/3 & 7/3 \\ 0 & -2 & 3 & 9 & 7 \end{array} \right)$$

First perform backward substitution, and solving the resulting linear system

$$\begin{aligned} -3x_3 + \frac{14}{3}x_4 &= \frac{32}{3} \\ 3x_1 + x_2 + 9x_3 + 2x_4 &= -6 \\ \frac{7}{3}x_4 &= \frac{7}{3} \\ -2x_2 + 3x_3 + 9x_4 &= 7 \end{aligned}$$

yields $x_4 = 1$, $x_3 = -2$, $x_2 = -2$, $x_1 = 4$. □

In the last step of elimination, row operation:

$$(4) \leftarrow (4) - \frac{0}{-3} \times (3)$$

Inverse operation:

$$(4) \leftarrow (4) + \frac{0}{-3} \times (3)$$

$$\begin{array}{l} 3 : \\ 1 : \\ 4 : \\ 2 : \end{array} \left(\begin{array}{cccc|c} 2 & 4/3 & -3 & 14/3 & 32/3 \\ 3 & 1 & 9 & 2 & -6 \\ 1 & 2/3 & 0 & 7/3 & 7/3 \\ 0 & -2 & 3 & 9 & 7 \end{array} \right)$$

Row operation:

$$\begin{aligned}(3) &\leftarrow (3) - \frac{4/3}{-2} \times (2) \\ (4) &\leftarrow (4) - \frac{2/3}{-2} \times (2)\end{aligned}$$

Inverse operation:

$$\begin{aligned}(3) &\leftarrow (3) + \frac{4/3}{-2} \times (2) \\ (4) &\leftarrow (4) + \frac{2/3}{-2} \times (2)\end{aligned}$$

$$\begin{array}{l} 3 : \\ 1 : \\ 4 : \\ 2 : \end{array} \left(\begin{array}{cccc|c} 2 & 4/3 & -5 & -4/3 & 6 \\ \textcolor{red}{3} & 1 & 9 & 2 & -6 \\ 1 & 2/3 & -1 & -2/3 & 0 \\ 0 & -2 & 3 & 9 & 7 \end{array} \right)$$

Row operation:

$$\begin{aligned}(2) &\leftarrow (2) - \frac{0}{3} \times (1) \\ (3) &\leftarrow (3) - \frac{2}{3} \times (1) \\ (4) &\leftarrow (4) - \frac{1}{3} \times (1)\end{aligned}$$

Inverse operation:

$$\begin{aligned}(2) &\leftarrow (2) + \frac{0}{3} \times (1) \\ (3) &\leftarrow (3) + \frac{2}{3} \times (1) \\ (4) &\leftarrow (4) + \frac{1}{3} \times (1)\end{aligned}$$

$$\begin{array}{l} 3 : \\ 1 : \\ 4 : \\ 2 : \end{array} \left(\begin{array}{cccc|c} 2 & 2 & 1 & 0 & 2 \\ 3 & 1 & 9 & 2 & -6 \\ 1 & 1 & 2 & 0 & -2 \\ 0 & -2 & 3 & 9 & 7 \end{array} \right)$$

Remark. To check answer, substitute values of x_1, x_2, x_3, x_4 back into the original linear system.

Example. Find the maximum possible value of β such that no label swapping is used when using partial pivoting to solve the linear system with coefficient matrix

$$\begin{pmatrix} 1 & 2 & 3 \\ \alpha & 3 & 1 \\ 1 & \beta & 5 \end{pmatrix}$$

Solution. First step: $|\alpha| \leq 1$

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & 3 - 2\alpha & 1 - 3\alpha \\ 0 & \beta - 2 & 2 \end{pmatrix}$$

Second step: $|3 - 2\alpha| \geq |\beta - 2|$

Expanding gives

$$2 - |3 - 2\alpha| \leq \beta \leq 2 + |3 - 2\alpha|.$$

From the first step, $-1 \leq \alpha \leq 1 \iff -2 \leq 2\alpha \leq 2 \iff 1 \leq 3 - 2\alpha \leq 5$. Hence,

$$\beta \leq 2 + |3 - 2\alpha| \leq 7.$$

□

2.4 Scaled partial pivoting

Although partial pivoting is sufficient for many linear systems, situations do arise when it is inadequate.

Example. The linear system

$$\begin{pmatrix} \textcolor{red}{30.00} & 591400 \\ 5.291 & -6.130 \end{pmatrix} \mathbf{x} = \begin{pmatrix} 591700 \\ 46.78 \end{pmatrix}$$

is the same as that in the previous subsection, except that all the entries in the first row have been multiplied by 10^4 . The partial pivoting procedure with four-digit arithmetic leads to the same inaccurate solutions $x_2 \approx 1.001$ and $x_1 \approx -10.00$.

The issue is the red entry should be regarded small because of its relative magnitude in the equations. To fix this, define the relative magnitude by dividing with the largest value in the row. Then use relative magnitude in the comparison when selecting the pivot element.

Scaled partial pivoting places the element in the pivot position that is largest relative to the entries in its row. Compare the relative magnitudes

$$\frac{|a_{ri}|}{s_{r_i}}, \quad \frac{|a_{r_{i+1}i}|}{s_{r_{i+1}}}, \quad \dots, \quad \frac{|a_{r_n i}|}{s_{r_n}}$$

where

$$s_{r_k} = \max \{ |a_{r_k i}|, |a_{r_{k,i+1}}|, \dots, |a_{r_k n}| \}.$$

Algorithm 7 Pivoting before eliminating the i -th variable

```

1:  $s_{r_k} \leftarrow \max \{ |a_{r_k i}|, |a_{r_{k,i+1}}|, \dots, |a_{r_k n}| \}$  for  $k = i, \dots, n$ 
2:  $j \leftarrow i, \max \leftarrow \frac{|a_{r_j i}|}{s_{r_j}}$ 
3: for  $k = i + 1, \dots, n$  do
4:    $r \leftarrow \frac{|a_{r_k i}|}{s_{r_k}}$ 
5:   if  $r > \max$  then
6:      $j \leftarrow k, \max \leftarrow r$ 
7: if  $a_{r_j i} = 0$  then
8:   return "Error: Matrix is singular"
9: else if  $j \neq i$  then
10:  Swap  $r_i$  and  $r_j$ 
```

Number of divisions is

$$\sum_{i=1}^{n-1} \left(1 + \sum_{k=i+1}^n 1 \right) = \frac{(n+2)(n-1)}{2}.$$

Number of comparisons is

$$\begin{aligned}
 \sum_{i=1}^{n-1} \left(\sum_{k=i}^n \sum_{j=i+1}^n 1 + \sum_{k=i+1}^n 1 \right) &= \sum_{i=1}^{n-1} \left[\left(\sum_{k=i}^n (n-i) \right) + (n-i) \right] \\
 &= \sum_{i=1}^{n-1} (n-i+2)(n-i) = \sum_{i=1}^{n-1} (n-i)^2 + 2 \sum_{i=1}^{n-1} (n-i) \\
 &= \frac{n(n-1)(2n-1)}{6} + 2 \times \frac{n(n-1)}{2} = \frac{n(n-1)(2n+5)}{6}.
 \end{aligned}$$

Since s_k 's are fixed, we compute s_k only once in the beginning of the algorithm:

$$s_k = \max \{ |a_{k1}|, |a_{k2}|, \dots, |a_{kn}| \}$$

and then use these values in all pivoting steps.

Algorithm 8 Solve $Ax = \mathbf{b}$ using scaled partial pivoting

```

1: Set  $s_k$  for  $k = 1, \dots, n$   $\triangleright$  Initialise  $s_k$ 
2: for  $i = 1, \dots, n$  do  $\triangleright$  Initialise row labels
3:    $r_i \leftarrow i$ 
4:   for  $i = 1, \dots, n-1$  do
5:     Pivoting
6:     Eliminate the  $i$ -th variable
7: Backward substitution
8: return  $(x_1, \dots, x_n)^T$ 

```

Pivoting

```

1:  $j \leftarrow i, \max \leftarrow \frac{a_{r_j i}}{s_{r_j}}$ 
2: for  $k = i+1, \dots, n$  do
3:    $r \leftarrow \frac{a_{r_k i}}{s_{r_k}}$ 
4:   if  $r > \max$  then
5:      $j \leftarrow k, \max \leftarrow r$ 
6: if  $a_{r_j i} = 0$  then
7:   return "Error: Matrix is singular"
8: else if  $j \neq i$  then
9:   Swap  $r_i$  and  $r_j$ 

```

$$a_{r_j k} \leftarrow a_{r_j k} - \frac{a_{r_j i}}{a_{r_i i}} a_{r_i k}, \quad j = i+1, \dots, n, \quad k = i+1, \dots, n+1.$$

Eliminate the i -th variable

```

1: for  $j = i+1, \dots, n$  do
2:    $m_{ji} \leftarrow \frac{a_{r_j i}}{a_{r_i i}}$ 
3:   for  $k = i+1, \dots, n+1$  do
4:      $a_{r_j k} \leftarrow a_{r_j k} - m_{ji} a_{r_i k}$ 

```

$$x_i = \frac{1}{a_{r_i i}} \left(a_{r_i, n+1} - \sum_{j=i+1}^n a_{r_i j} x_j \right)$$

Eliminate the i -th variable

```

1: for  $i = n, \dots, 1$  do
2:    $x_i \leftarrow a_{r_i, n+1}$ 
3:   for  $j = i + 1, \dots, n$  do
4:      $x_i \leftarrow x_i - a_{r_i j} x_j$ 
5:    $x_i \leftarrow \frac{x_i}{a_{r_i i}}$ 

```

Example. Consider a linear system with coefficient matrix

$$\begin{pmatrix} 2 & 4 & 5 \\ 4 & 6 & \alpha \\ 1 & \beta & 2 \end{pmatrix}.$$

Find all the possible values of α and β such that no label swapping is required when solving the system with scaled partial pivoting.

Solution. First compute the largest absolute value in each row:

$$s_1 = 5, \quad s_2 = \max(|\alpha|, 6), \quad s_3 = \max(|\beta|, 2).$$

First pivoting: compare

$$\frac{2}{5}, \quad \frac{4}{\max(|\alpha|, 6)}, \quad \frac{1}{\max(|\beta|, 2)}.$$

Since there is no label swapping, we require

$$\begin{aligned} \frac{2}{5} &\geq \frac{4}{\max(|\alpha|, 6)} \implies |\alpha| \geq 10, \\ \frac{2}{5} &\geq \frac{1}{\max(|\beta|, 2)} \implies |\beta| \geq \frac{5}{2}. \end{aligned}$$

Perform elimination

$$(2) \leftarrow (2) - 2 \times (1), \quad (3) \leftarrow (3) - \frac{1}{2} \times (1).$$

$$\begin{pmatrix} 2 & 4 & 5 \\ 0 & -2 & \alpha - 10 \\ 0 & \beta - 2 & -\frac{1}{2} \end{pmatrix}$$

Second pivoting: compare

$$\frac{2}{\max(|\alpha|, 6)}, \quad \frac{|\beta - 2|}{\max(|\beta|, 2)}.$$

Since there is no row swapping, we must have

$$\frac{2}{|\alpha|} \geq \frac{|\beta - 2|}{|\beta|}.$$

Solving the inequalities $|\alpha| = 10$ and $\beta = \frac{5}{2}$. □

2.5 LU factorization

Let $A \in \mathbb{R}^{n \times n}$ and $B \in \mathbb{R}^{n \times p}$. Consider solving $AX = B$. Writing

$$X = (\mathbf{x}_1, \dots, \mathbf{x}_p) \quad \text{and} \quad B = (\mathbf{b}_1, \dots, \mathbf{b}_p),$$

this is equivalent to solving $A\mathbf{x}_1 = \mathbf{b}_1, \dots, A\mathbf{x}_p = \mathbf{b}_p$.

If Gaussian elimination is performed independently for each right-hand side, the elimination of the coefficient matrix A is repeated. Since the elimination steps depend only on A , this redundancy can be removed by preprocessing A once.

Consider the linear system $Ax = \mathbf{b}$ for $A = (a_{ij})_{n \times n}$, $\mathbf{b} = (b_1, \dots, b_n)^T$. For the elimination step, for $i = 1, \dots, n-1$ and $j = i+1, \dots, n$, define the **multiplier**

$$m_{ji} = \frac{a_{ji}}{a_{ii}}.$$

Then update

$$a_{jk} \leftarrow a_{jk} - m_{ji}a_{ik} \quad (k = i+1, \dots, n).$$

After elimination, A becomes an upper triangular matrix U . The right-hand side transforms consistently with the same multipliers:

- Elimination of the first variable:

$$b_2 \leftarrow b_2 - m_{21}b_1, \quad \dots, \quad b_n \leftarrow b_n - m_{n1}b_1$$

- Elimination of the second variable:

$$b_3 \leftarrow b_3 - m_{32}b_2, \quad \dots, \quad b_n \leftarrow b_n - m_{n2}b_2$$

• ...

- Elimination of the $(n-1)$ -th variable:

$$b_n \leftarrow b_n - m_{n,n-1}b_{n-1}$$

We can rewrite the above more compactly as

$$\begin{aligned} b_2 &\leftarrow b_2 - m_{21}b_1, \\ b_3 &\leftarrow b_3 - m_{31}b_1 - m_{32}b_2, \\ &\vdots \\ b_n &\leftarrow b_n - m_{n1}b_1 - m_{n2}b_2 - \dots - m_{n,n-1}b_{n-1}. \end{aligned}$$

Equivalently, defining $\tilde{\mathbf{b}}$ after elimination,

$$\begin{aligned} \tilde{b}_1 &= b_1, \\ \tilde{b}_i &= b_i - \sum_{j=1}^{i-1} m_{ij}\tilde{b}_j, \quad i = 2, \dots, n. \end{aligned}$$

After that, we can do backward substitution using the upper triangular coefficient matrix to find $x_1, \dots, x_n$.

The multipliers m_{ji} and the resulting upper triangular matrix U are independent of the right-hand side. Hence, we **preprocess** A to store the upper triangular matrix U and the multipliers m_{ji} .

Elimination step for the coefficient matrix $(a_{ij})_{n \times n}$

<pre> 1: for $i = 1, \dots, n-1$ do 2: for $j = i+1, \dots, n$ do 3: $m_{ji} \leftarrow \frac{a_{ji}}{a_{ii}}$ 4: for $k = i+1, \dots, n$ do 5: $a_{jk} \leftarrow a_{jk} - m_{ji}a_{ik}$ 6: return a_{ij} for all $i \leq j$ 7: return m_{ji} for all $i < j$ </pre>	$\triangleright$ Using i-th row $\triangleright$ to eliminate $i+1, \dots, n$-th rows $\triangleright$ Compute each multiplier $\triangleright$ Update entries $\triangleright$ Upper triangular matrix A $\triangleright$ Lower triangular matrix of multipliers
---	---

Instead of storing L and U separately, combine them for memory-efficient storage:

$$\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ m_{21} & a_{22} & \cdots & a_{2n} \\ m_{31} & m_{32} & \cdots & a_{3n} \\ \vdots & \vdots & \ddots & \vdots \\ m_{n1} & m_{n2} & \cdots & a_{nn} \end{pmatrix}.$$

Elimination step for the coefficient matrix $(a_{ij})_{n \times n}$

```

1: for i = 1, ..., n - 1 do
2:   for j = i + 1, ..., n do
3:      $a_{ji} \leftarrow \frac{a_{ji}}{a_{ii}}$ 
4:     for k = i + 1, ..., n do
5:        $a_{jk} \leftarrow a_{jk} - a_{ji}a_{ik}$ 
return  $(a_{ij})_{n \times n}$ 

```

Elimination for the right-hand side:

$$b_i \leftarrow b_i - m_{i1}b_1 - \cdots - m_{i,i-1}b_{i-1} \quad (i = 2, \dots, n).$$

Elimination step for the right-hand side

```

1: for i = 2, ..., n do
2:   for j = 1, ..., i - 1 do
3:      $b_i \leftarrow b_i - a_{ij}b_j$ 

```

$\triangleright$ Elimination for the right-hand side

Backward substitution:

$$x_i \leftarrow \frac{b_i - a_{i,i+1}x_{i+1} - \cdots - a_{in}x_n}{a_{ii}} \quad (i = n, \dots, 1).$$

Backward substitution

```

1:  $x_n \leftarrow \frac{b_n}{a_{nn}}$ 
2: for i = n - 1, ..., 1 do
3:    $x_i \leftarrow b_i$ 
4:   for j = i + 1, ..., n do
5:      $x_i \leftarrow x_i - a_{ij}x_j$ 
6:    $x_i \leftarrow \frac{x_i}{a_{ii}}$ 
return  $(x_1, \dots, x_n)^T$ 

```

When eliminating the right-hand side, from above we have

$$\begin{aligned} \tilde{b}_1 &= b_1 \\ \tilde{b}_2 &= b_2 - m_{21}\tilde{b}_1 \\ \tilde{b}_3 &= b_3 - m_{31}\tilde{b}_1 - m_{32}\tilde{b}_2 \\ &\vdots \\ \tilde{b}_n &= b_n - m_{n1}\tilde{b}_1 - m_{n2}\tilde{b}_2 - \cdots - m_{n,n-1}\tilde{b}_{n-1} \end{aligned}$$

Rearrange terms such that LHS is $\tilde{b}$, RHS is b :

$$\begin{aligned} \tilde{b}_1 &= b_1 \\ m_{21}\tilde{b}_1 + \tilde{b}_2 &= b_2 \\ m_{31}\tilde{b}_1 + m_{32}\tilde{b}_2 + \tilde{b}_3 &= b_3 \\ m_{n1}\tilde{b}_1 + m_{n2}\tilde{b}_2 + \cdots + m_{n,n-1}\tilde{b}_{n-1} + \tilde{b}_n &= b_n \end{aligned}$$

This is equivalent to $L\tilde{\mathbf{b}} = \mathbf{b}$, where

$$L = \begin{pmatrix} 1 & & & & \\ m_{21} & 1 & & & \\ m_{31} & m_{32} & 1 & & \\ \vdots & \vdots & \ddots & \ddots & \\ m_{n1} & m_{n2} & \cdots & m_{n,n-1} & 1 \end{pmatrix}.$$

We have $A\mathbf{x} = \mathbf{b}$, $U\mathbf{x} = \tilde{\mathbf{b}}$, and $L\tilde{\mathbf{b}} = \mathbf{b}$. Hence, $A\mathbf{x} = LU\mathbf{x}$. Since this holds for any $\mathbf{x}$, we conclude that $A = LU$.

Definition 2.7. A **unit lower triangular matrix** is a lower triangular matrix with all diagonal entries equal to one.

Theorem 2.8. Let $A \in \mathbb{R}^{n \times n}$. If linear systems with coefficient matrix A can be solved by Gaussian elimination without pivoting, then there exist a unit lower triangular matrix L and an upper triangular matrix U such that

$$A = LU. \quad (4)$$

Given the LU factorisation $A = LU$, we can solve $A\mathbf{x} = \mathbf{b}$ as follows:

$$A\mathbf{x} = \mathbf{b} \iff LU\mathbf{x} = \mathbf{b}$$

- (1) Solve the lower triangular system $L\tilde{\mathbf{b}} = \mathbf{b}$ for $\tilde{\mathbf{b}}$ using **forward substitution**.

Since L is triangular, this requires $O(n^2)$ operations.

- (2) Solve the upper triangular system $U\mathbf{x} = \tilde{\mathbf{b}}$ for $\mathbf{x}$ using **backward substitution**.

Since U is triangular, this requires an additional $O(n^2)$ operations.

Hence, the overall time complexity is $O(n^2)$.

We now consider the modifications that must be made when row interchanges are required, i.e., we meet a zero pivot element. We first introduce a class of matrices that are used to rearrange, or permute, rows of a given matrix.

Definition 2.9. An $n \times n$ **permutation matrix** $P = (p_{ij})$ is obtained by rearranging the rows of identity matrix I_n .

This gives a matrix with precisely one non-zero entry in each row and in each column, and each non-zero entry is a 1.

Example. The matrix

$$P = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

is a 3×3 permutation matrix. For any 3×3 matrix A , multiplying on the left by P has the effect of interchanging the second and third rows of A :

$$PA = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{31} & a_{32} & a_{33} \\ a_{21} & a_{22} & a_{23} \end{pmatrix}.$$

Similarly, multiplying A on the right by P interchanges the second and third columns of A .

The idea is to rearrange the rows of A by pre-multiplying a permutation matrix P such that no row interchanges are required. Then apply LU factorisation to PA .

Theorem 2.10. *With partial pivoting, there exist a permutation matrix P , a unit lower triangular matrix L , and an upper triangular matrix U such that*

$$PA = LU. \quad (5)$$

Given the LU factorization $PA = LU$, we solve $A\mathbf{x} = \mathbf{b}$ as follows:

$$A\mathbf{x} = \mathbf{b} \iff PA\mathbf{x} = P\mathbf{b} \iff LU\mathbf{x} = P\mathbf{b}$$

- (1) Rearrange the rows of $\mathbf{b}$ to find $P\mathbf{b}$.
- (2) Solve $L\tilde{\mathbf{b}} = P\mathbf{b}$ using **forward substitution**.
- (3) Solve $U\mathbf{x} = \tilde{\mathbf{b}}$ using **backward substitution**.

Time complexity: $O(n^2)$

Two useful properties of permutation matrices relate to Gaussian elimination. Suppose $\{k_1, \dots, k_n\}$ is a permutation of $\{1, \dots, n\}$, and the permutation matrix $P = (p_{ij})$ is defined by

$$p_{ij} = \begin{cases} 1 & \text{if } j = k_i, \\ 0 & \text{otherwise.} \end{cases}$$

Then

- PA permutes the rows of A :

$$PA = \begin{pmatrix} a_{k_1 1} & \cdots & a_{k_1 n} \\ \vdots & & \vdots \\ a_{k_n 1} & \cdots & a_{k_n n} \end{pmatrix}.$$

- P^{-1} exists and $P^{-1} = P^T$.
- $\det P = (-1)^k$, where k is the number of row interchanges.

3 Interpolation and Least Squares Approximation

3.1 Lagrange interpolation

For an unknown function $f(x)$, given some data points on the graph of $f(x)$:

$$(x_0, y_0), (x_1, y_1), \dots, (x_n, y_n),$$

how can we recover the original function $f(x)$? Given x , how can we guess the value of $f(x)$?

The next theorem tells us that polynomials can approximate any continuous function defined on a finite closed interval up to any precision. (To get higher accuracy, we usually need a polynomial of a higher degree.)

Theorem 3.1 (Weierstrass approximation theorem). *Let f be a continuous function on $[a, b]$. For any $\varepsilon > 0$, there exists a polynomial $P(x)$ such that for any $x \in [a, b]$,*

$$|f(x) - P(x)| < \varepsilon.$$

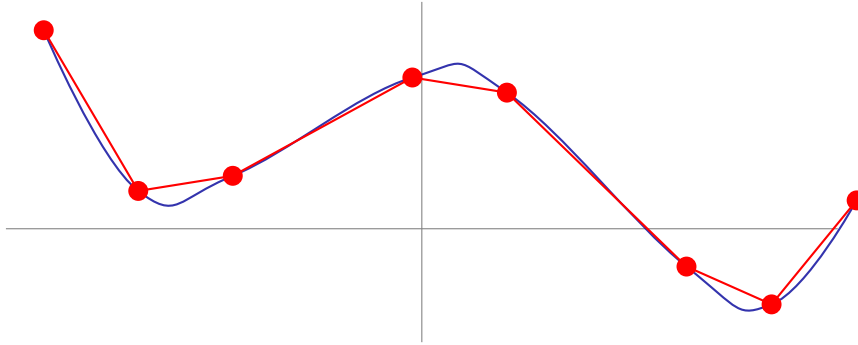
Remark. There is no guarantee whether $f(x)$ and $P(x)$ agree on any points.

Example. The Bernstein expansion

$$B_N(x) = \sum_{j=0}^N f\left(\frac{j}{N}\right) \binom{N}{j} x^j (1-x)^{N-j}$$

approximates any function $f(x)$ defined on $[0, 1]$.

Interpolation using **piecewise linear functions**: Given data points $\{(x_i, y_i)\}_{i=0}^n$, we want to find a continuous piecewise linear function $f(x)$ such that $f(x_k) = y_k$ for all $k = 1, \dots, n$, and $f(x)$ is a linear function on (x_{k-1}, x_k) .



It is natural to define

$$f(x) = \begin{cases} y_0 + \frac{y_1 - y_0}{x_1 - x_0}(x - x_0) & \text{if } x \in [x_0, x_1] \\ y_1 + \frac{y_2 - y_1}{x_2 - x_1}(x - x_1) & \text{if } x \in (x_1, x_2] \\ \dots & \\ y_{n-1} + \frac{y_n - y_{n-1}}{x_n - x_{n-1}}(x - x_{n-1}) & \text{if } x \in (x_{n-1}, x_n] \end{cases}$$

Alternatively, consider hat functions. Suppose the given data points satisfy $y_0 = 0, \dots, y_{k-1} = 0, y_k = 1, y_{k+1} = 0, \dots, y_n = 0$. The interpolating function $\phi_k(x)$ is

$$\phi_k(x) = \begin{cases} \frac{x - x_{k-1}}{x_k - x_{k-1}} & \text{if } x \in [x_{k-1}, x_k] \\ \frac{x - x_{k+1}}{x_k - x_{k+1}} & \text{if } x \in (x_k, x_{k+1}] \\ 0 & \text{otherwise} \end{cases}$$

More generally, let $y_k = f(x_k)$. Then the interpolating function $f(x)$ is

$$f(x) = y_k \phi_k(x) = \begin{cases} \frac{y_k}{x_k - x_{k-1}}(x - x_{k-1}) & \text{if } x \in [x_{k-1}, x_k] \\ \frac{y_k}{x_k - x_{k+1}}(x - x_{k+1}) & \text{if } x \in (x_k, x_{k+1}] \\ 0 & \text{otherwise} \end{cases}$$

Theorem 3.2. *Let*

$$\phi_0(x) = \begin{cases} \frac{x - x_1}{x_0 - x_1} & \text{if } x \in [x_0, x_1], \\ 0 & \text{otherwise} \end{cases}$$

$$\phi_n(x) = \begin{cases} \frac{x - x_{n-1}}{x_n - x_{n-1}} & \text{if } x \in [x_{n-1}, x_n], \\ 0 & \text{otherwise} \end{cases}$$

and $\phi_k(x)$, $k = 1, \dots, n-1$ be the hat functions defined above. Then the function

$$f(x) = \sum_{k=0}^n y_k \phi_k(x)$$

satisfies $f(x_k) = y_k$ for every $k = 0, 1, \dots, n$.

The functions $\phi_j(x)$ satisfies

$$\phi_j(x_k) = \delta_{jk}, \quad (j, k = 0, 1, \dots, n)$$

where δ_{jk} is the Kronecker delta symbol. The set of functions $\{\phi_0, \phi_1, \dots, \phi_n\}$ is called the **basis** of the linear interpolation problem; the functions $\phi_0, \phi_1, \dots, \phi_n$ are called the **basis functions** of the linear interpolation problem.

The function

$$I(x) = y_0 \phi_0(x) + y_1 \phi_1(x) + \dots + y_n \phi_n(x)$$

is the interpolating function for the data points $(x_0, y_0), (x_1, y_1), \dots, (x_n, y_n)$.

Proof. We just need to show that $I(x_0) = y_0, I(x_1) = y_1, \dots, I(x_n) = y_n$. For any $i = 0, 1, \dots, n$,

$$\begin{aligned} I(x_i) &= \sum_{k=0}^n y_k \phi_k(x_i) \\ &= 0 + \dots + 0 + y_i + 0 + \dots + 0 = y_i. \end{aligned}$$

Therefore $I(x)$ is an interpolating function for $f(x)$. □

Notation. Let Π_n denote the set of real polynomials of degree $\leq n$.

Let $f(x)$ be some unknown function, and $x_0, x_1, \dots, x_n$ be $n+1$ distinct real numbers. Can we find a polynomial P_n such that $P_n(x_i) = f(x_i)$? Such a polynomial is said to **interpolate** the data, and (as we shall see) can approximate f at other values of x if f is smooth enough. This is the most basic question in approximation theory.

Theorem 3.3. *The interpolating polynomial of degree $\leq n$ exists and is unique. That is, there exists a unique polynomial $P_n \in \Pi_n$ such that $P_n(x_i) = f(x_i)$ for $i = 0, 1, \dots, n$.*

Proof. For $k = 0, 1, \dots, n$, define the **k -th Lagrange basis polynomial** $L_k(x)$ by

$$L_k(x) = \prod_{j=0, j \neq k}^n \frac{x - x_j}{x_k - x_j} = \frac{(x - x_0) \cdots (x - x_{k-1})(x - x_{k+1}) \cdots (x - x_n)}{(x_k - x_0) \cdots (x_k - x_{k-1})(x_k - x_{k+1}) \cdots (x_k - x_n)} \in \Pi_n. \quad (6)$$

Then $L_k(x_i) = \delta_{ik}$, that is,

$$\begin{aligned} L_k(x_i) &= 0 \quad \text{for } i = 0, \dots, k-1, k+1, \dots, n \\ L_k(x_k) &= 1. \end{aligned}$$

Now define the **Lagrange interpolating polynomial**

$$P_n(x) = \sum_{k=0}^n f(x_k) L_k(x) \quad (7)$$

so that

$$P_n(x_i) = \sum_{k=0}^n f(x_k) L_k(x_i) = f(x_i)$$

for $i = 0, 1, \dots, n$.

To show uniqueness, consider two interpolating polynomials $P_n, Q_n \in \Pi_n$ for the same function $f(x)$ on the same nodes $x_0, x_1, \dots, x_n$. Then their difference $D_n = P_n - Q_n \in \Pi_n$ satisfies $D_n(x_k) = 0$ for $k = 0, 1, \dots, n$. That is, D_n is a polynomial of degree at most n and has at least $n+1$ distinct roots. By the fundamental theorem of algebra, D_n must be the zero polynomial, so that $P_n = Q_n$. $\square$

Example. Consider an interpolation of the gamma function

$$\Gamma(x) = \int_0^{+\infty} t^{x-1} e^{-t} dt.$$

Given the data points

$$\begin{aligned} x_0 &= 1, & x_1 &= 2, & x_2 &= 3, & x_3 &= 4 \\ \Gamma(x_0) &= 1, & \Gamma(x_1) &= 1, & \Gamma(x_2) &= 2, & \Gamma(x_3) &= 6 \end{aligned}$$

we can use a **cubic** polynomial to interpolate the data. The basis polynomials are

$$\begin{aligned} L_0(x) &= \frac{(x-2)(x-3)(x-4)}{(1-2)(1-3)(1-4)} = -\frac{1}{6}(x-2)(x-3)(x-4), \\ L_1(x) &= \frac{(x-1)(x-3)(x-4)}{(2-1)(2-3)(2-4)} = \frac{1}{2}(x-1)(x-3)(x-4), \\ L_2(x) &= \frac{(x-1)(x-2)(x-4)}{(3-1)(3-2)(3-4)} = -\frac{1}{2}(x-1)(x-2)(x-4), \\ L_3(x) &= \frac{(x-1)(x-2)(x-3)}{(4-1)(4-2)(4-3)} = \frac{1}{6}(x-1)(x-2)(x-3). \end{aligned}$$

Hence, the Lagrange interpolating polynomial is

$$P_3(x) = \sum_{k=0}^3 f(x_k) L_k(x) = L_0(x) + L_1(x) + 2L_2(x) + 6L_3(x).$$

Suppose the interpolation polynomial is $P_n(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$. The coefficients $a_0, a_1, \dots, a_n$ can also be found by solving the linear system

$$\begin{pmatrix} 1 & x_0 & x_0^2 & \cdots & x_0^n \\ 1 & x_1 & x_1^2 & \cdots & x_1^n \\ 1 & x_2 & x_2^2 & \cdots & x_2^n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^n \end{pmatrix} \begin{pmatrix} a_0 \\ a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix} = \begin{pmatrix} f(x_0) \\ f(x_1) \\ f(x_2) \\ \vdots \\ f(x_n) \end{pmatrix}.$$

3.2 Divided differences

Let $P_{n-1}(x)$ be the Lagrange interpolating polynomial of $f(x)$ with n nodes $x_0, x_1, \dots, x_{n-1}$. Suppose we get one more data point $(x_n, f(x_n))$. How can we obtain $P_n(x)$ efficiently? In other words, what is the relation between P_{n-1} and P_n ?

Note that $P_{n-1}(x)$ satisfies

$$P_{n-1}(x_0) = f(x_0), \quad \dots, \quad P_{n-1}(x_{n-1}) = f(x_{n-1})$$

and $P_n(x)$ satisfies

$$P_n(x_0) = f(x_0), \quad \dots, \quad P_n(x_{n-1}) = f(x_{n-1}), \quad P_n(x_n) = f(x_n).$$

Consider their difference: let $Q_n(x) = P_n(x) - P_{n-1}(x)$. Then $Q_n(x)$ satisfies

$$Q_n(x_0) = 0, \quad \dots, \quad Q_n(x_{n-1}) = 0, \quad Q_n(x_n) = f(x_n) - P_{n-1}(x_n).$$

Note that $Q_n(x)$ is a polynomial of degree $\leq n$. Since Q_n has n roots, it must be of the form

$$b_n(x - x_0)(x - x_1) \cdots (x - x_{n-1})$$

where b_n is some unknown constant to be determined. By Lagrange interpolation,

$$\begin{aligned} Q_n(x) &= \sum_{k=0}^n f(x_k) L_k(x) \\ &= \sum_{k=0}^n [P_n(x_k) - P_{n-1}(x_k)] \prod_{j=0, j \neq k}^n \frac{x - x_j}{x_k - x_j} \\ &= [f(x_n) - P_{n-1}(x_n)] \prod_{j=0}^{n-1} \frac{x - x_j}{x_n - x_j} \\ &= \left([f(x_n) - P_{n-1}(x_n)] \prod_{j=0}^{n-1} \frac{1}{x_n - x_j} \right) \prod_{j=0}^{n-1} (x - x_j) \end{aligned}$$

which matches the form of Q_n as discussed above. Thus, $P_n(x)$ adds a polynomial

$$Q_n(x) = b_n(x - x_0)(x - x_1) \cdots (x - x_{n-1})$$

to $P_{n-1}(x)$, which does not change the function values on the existing data points $x_0, x_1, \dots, x_{n-1}$. The coefficient b_n is chosen to match the new data point $f(x_n)$:

$$P_{n-1}(x) + b_n(x - x_0)(x - x_1) \cdots (x - x_{n-1}) = f(x_n).$$

That is, $P_{n-1}(x_n) + Q(x_n) = P_n(x_n)$ at $x = x_n$.

Example. Consider the Dirichlet eta function

$$\eta(x) = \frac{1}{\Gamma(x)} \int_0^{+\infty} \frac{t^{x-1}}{e^t + 1} dt.$$

We are given the data points

$$\eta(-4) = 0, \quad \eta(-3) = 1/8, \quad \eta(-1) = 1/4, \quad \eta(0) = 1/2.$$

The Lagrange interpolating polynomial on these data points is

$$P_3(x) = \frac{1}{2} + \frac{5}{24}x - \frac{1}{16}x^2 - \frac{1}{48}x^3.$$

Now add one more data point $\eta(1) = \log 2$. Thus, the new interpolating polynomial is $P_4(x) = P_3(x) + b_4(x + 4)(x + 3)(x + 1)x$, where b_4 is chosen such that $P_4(1) = \eta(1) = \log 2$. Solve to get $b_4 = \frac{1}{40} \left(\log 2 - \frac{5}{8} \right)$, so

$$P_4(x) = \frac{1}{2} + \frac{5}{24}x - \frac{1}{16}x^2 - \frac{1}{48}x^3 + \frac{1}{40} \left(\log 2 - \frac{5}{8} \right) (x + 4)(x + 3)(x + 1)x.$$

this is easier and faster to obtain the next interpolating polynomial, rather than calculating it from scratch

In practice, we are not directly given the previous interpolating polynomial; instead, we add data points successively and interpolate the nodes.

- Interpolation with only one node x_0 :

$$P_0(x) = f(x_0)$$

- Interpolation with two nodes x_0 and x_1 :

$$P_1(x) = P_0(x) + b_1(x - x_0)$$

- Interpolation with three nodes x_0, x_1, x_2 :

$$P_2(x) = P_1(x) + b_2(x - x_0)(x - x_1)$$

• ...

- Interpolation with $n + 1$ nodes $x_0, x_1, \dots, x_n$:

$$P_n(x) = P_{n-1}(x) + b_n(x - x_0)(x - x_1) \cdots (x - x_{n-1})$$

In general, we have

$$P_k(x) = P_{k-1}(x) + b_k(x - x_0)(x - x_1) \cdots (x - x_{k-1})$$

where the coefficient b_k is chosen such that $P_k(x)$ matches the function value at x_k :

$$P_k(x_k) = f(x_k).$$

Observe that b_k is the leading coefficient of the polynomial $P_k(x)$. We can determine the value of b_k by comparing coefficients of x^k with the Lagrange interpolating polynomial:

$$P_k(x) = \sum_{l=0}^k f(x_l) \prod_{j=0, j \neq l}^k \frac{x - x_j}{x_l - x_j} = \sum_{l=0}^k \left(f(x_l) \prod_{j=0, j \neq l}^k \frac{1}{x_l - x_j} \right) \prod_{j=0, j \neq l}^k (x - x_j).$$

Hence, for $k \geq 1$,

$$b_k = \sum_{l=0}^k f(x_l) \prod_{j=0, j \neq l}^k \frac{1}{x_l - x_j}$$

is called the k -th divided difference of $f(x)$.

Definition 3.4. For $k \geq 1$, the **k -th divided difference** of $f(x)$ is

$$f[x_0, x_1, \dots, x_k] := \sum_{l=0}^k f(x_l) \prod_{j=0, j \neq l}^k \frac{1}{x_l - x_j}.$$

Hence, the interpolating polynomial is given by

$$P_n(x) = f[x_0] + f[x_0, x_1](x - x_0) + f[x_0, x_1, x_2](x - x_0)(x - x_1) + \cdots + f[x_0, x_1, \dots, x_n](x - x_0)(x - x_1) \cdots (x - x_{n-1}) \quad (8)$$

where $f[x_0] = f(x_0)$ is the 0-th divided difference. To determine the interpolating polynomial, it suffices to compute the values of the divided differences, and plug them into the above expression.

Remark. The n -th divided difference $f[x_0, x_1, \dots, x_n]$ is the leading coefficient of the interpolating polynomial $P_n(x)$.

We frequently use the following recurrence relation to compute divided differences.

Theorem 3.5. Let $x_0, x_1, \dots, x_n$ be $n + 1$ distinct points. For any function f ,

$$f[x_0, x_1, \dots, x_{n-1}, x_n] = \frac{f[x_1, \dots, x_n] - f[x_0, \dots, x_{n-1}]}{x_n - x_0}. \quad (9)$$

To compute divided differences systematically, we use a table to tabulate the results.

x	0th d.d.	1st d.d.	2nd d.d.	...
x_0	$f[x_0] = f(x_0)$			
x_1	$f[x_1] = f(x_1)$	$f[x_0, x_1] = \frac{f[x_1] - f[x_0]}{x_1 - x_0}$	$f[x_0, x_1, x_2] = \frac{f[x_1, x_2] - f[x_0, x_1]}{x_2 - x_0}$	
x_2	$f[x_2] = f(x_2)$	$f[x_1, x_2] = \frac{f[x_2] - f[x_1]}{x_2 - x_1}$	$f[x_1, x_2, x_3] = \frac{f[x_2, x_3] - f[x_1, x_2]}{x_3 - x_1}$	...
x_3	$f[x_3] = f(x_3)$	$f[x_2, x_3] = \frac{f[x_3] - f[x_2]}{x_3 - x_2}$		...
	$\vdots$	$\vdots$	$\vdots$	

3.3 Hermite interpolation

Generalization of divided differences: We allow interpolating nodes to be equal, e.g. $x_0 = x_1$. Imagine $x_1 \neq x_0$, $x_1 \rightarrow x_0$. Then

$$f[x_0, x_1] = \frac{f[x_1] - f[x_0]}{x_1 - x_0} \rightarrow f'(x_0).$$

We define $f[x_0, x_0] = f'(x_0)$.

Example. Find a polynomial $f(x)$ such that $f(0) = 1$, $f'(0) = 2$, $f(1) = 2$.

Solution. We have $x_0 = 0$, $x_1 = 0$, $x_2 = 1$. Then

x	0th d.d.	1st d.d.	2nd d.d.
0	1		
0	1	$f[0, 0] = f'(0) = 2$	
1	2	$f[0, 1] = \frac{2 - 1}{1 - 0} = 1$	$f[0, 0, 1] = \frac{f[0, 1] - f[0, 0]}{1 - 0} = -1$

Hence, the interpolating polynomial is

$$\begin{aligned} f(x) &= f[x_0] + f[x_0, x_1](x - x_0) + f[x_0, x_1, x_2](x - x_0)(x - x_1) \\ &= f[0] + f[0, 0](x - 0) + f[0, 0, 1](x - 0)(x - 0) \\ &= 1 + 2x - x^2. \end{aligned}$$

□

Example. Given distinct nodes x_0, x_1, x_2 and a differentiable function f , find a cubic polynomial $P_3(x)$ such that

$$P_3(x_0) = f(x_0), \quad P_3(x_1) = f(x_1), \quad P'_3(x_1) = f'(x_1), \quad P_3(x_2) = f(x_2).$$

Solution. The Lagrange interpolating polynomial is

$$\begin{aligned} P_3(x) &= f[x_0] + f[x_0, x_1](x - x_0) + f[x_0, x_1, x_1](x - x_0)(x - x_1) \\ &\quad + f[x_0, x_1, x_1, x_2](x - x_0)(x - x_1)^2, \end{aligned}$$

where

$$\begin{aligned} f[x_0, x_1, x_1] &= \frac{f'(x_1) - f[x_0, x_1]}{x_1 - x_0}, \quad f[x_1, x_1, x_2] = \frac{f[x_1, x_2] - f'(x_1)}{x_2 - x_1}, \\ f[x_0, x_1, x_1, x_2] &= \frac{f[x_1, x_1, x_2] - f[x_0, x_1, x_1]}{x_2 - x_0}. \end{aligned}$$

□

3.4 Interpolation with Chebyshev nodes

Runge's phenomenon: Strong oscillation of the interpolating polynomial near the boundary of the interval (if using equally spaced interpolating nodes).

$$f(x) = \frac{1}{1 + 25x^2}, \quad x \in [-1, 1]$$

Consider the interpolation of $f(x)$ on the interval $[-1, 1]$. Assume that we can freely choose the location of the $n + 1$ interpolation nodes. We want to choose interpolation nodes $x_0, x_1, \dots, x_n$ to best control the fluctuation of

$$g_n(x) = |(x - x_0)(x - x_1) \cdots (x - x_n)|.$$

The idea is to choose $x_0, x_1, \dots, x_n$ such that all the **peak values** of $g_n(x)$ in $[-1, 1]$ are the same, so as to minimise the maximum fluctuation of error.

Distribution of interpolation nodes: closer near endpoints, further near center

Definition 3.6. The **Chebyshev nodes** are defined as

$$x_k = \cos\left(\frac{(k + 1/2)\pi}{n + 1}\right)$$

for $k = 0, 1, \dots, n$.

Definition 3.7. The **Chebyshev polynomial** of degree n is defined as

$$T_n(x) = \cos(n \arccos x).$$

Proposition 3.8. The Chebyshev polynomials satisfy the recurrence relation

$$T_n(x) = 2xT_{n-1}(x) - T_{n-2}(x)$$

for $n \geq 2$.

Proof. For $n \geq 2$, let $\theta = \arccos x$. Then

$$\begin{aligned}
 T_n(x) &= \cos(n\theta) = \cos((n-1)\theta + \theta) \\
 &= \cos((n-1)\theta) \cos \theta - \sin((n-1)\theta) \sin \theta \\
 &= xT_{n-1}(x) - \sin((n-2)\theta + \theta) \sin \theta \\
 &= xT_{n-1}(x) - [\cos((n-2)\theta) \sin \theta + \sin((n-2)\theta) \cos \theta] \sin \theta \\
 &= xT_{n-1}(x) - \cos((n-2)\theta) \sin^2 \theta - \sin((n-2)\theta) \cos \theta \sin \theta \\
 &= xT_{n-1}(x) - \cos((n-2)\theta) (1 - \cos^2 \theta) - \sin((n-2)\theta) \cos \theta \sin \theta \\
 &= xT_{n-1}(x) - (1 - x^2)T_{n-2}(x) - \sin((n-2)\theta) \cos \theta \sin \theta \\
 &= xT_{n-1}(x) - (1 - x^2)T_{n-2}(x) - [\cos((n-2)\theta) \cos \theta - \cos((n-1)\theta)] \cos \theta \\
 &= xT_{n-1}(x) - (1 - x^2)T_{n-2}(x) - [xT_{n-2}(x) - T_{n-1}(x)]x \\
 &= 2xT_{n-1}(x) - T_{n-2}(x).
 \end{aligned}$$

□

Corollary 3.9. *The Chebyshev polynomials are polynomials.*

Proof. Prove by induction. It is easy to see that $T_0(x) = 1$ and $T_1(x) = x$.

□

We show that the Chebyshev polynomials vanish at the Chebyshev nodes: for $x_k = \cos\left(\frac{(k+1/2)\pi}{n+1}\right)$, $k = 0, 1, \dots, n$, we have

$$\begin{aligned}
 T_{n+1}(x_k) &= \cos((n+1) \arccos x_k) \\
 &= \cos((k+1/2)\pi) = 0.
 \end{aligned}$$

Since these x_k are the roots of $T_{n+1}(x)$, the polynomial can be factorised as

$$T_{n+1}(x) = C(x - x_0)(x - x_1) \cdots (x - x_n)$$

where C is the leading coefficient. By induction, the leading coefficient of $T_m(x)$ for $m \geq 1$ is 2^{m-1} . Thus, the leading coefficient of $T_{n+1}(x)$ is $2^{(n+1)-1} = 2^n$. Hence, $C = 2^n$, so

$$T_{n+1}(x) = 2^n(x - x_0)(x - x_1) \cdots (x - x_n).$$

Therefore,

$$\begin{aligned}
 g_n(x) &= |(x - x_0)(x - x_1) \cdots (x - x_n)| \\
 &= |2^{-n} T_{n+1}(x)| \\
 &= |2^{-n} \cos((n+1) \arccos x)| \\
 &\leq 2^{-n}.
 \end{aligned}$$

The next result demonstrates the minimax property of Chebyshev polynomials: among all monic polynomials of degree n , the normalized Chebyshev polynomial $\tilde{T}_n(x) = \frac{1}{2^{n-1}} T_n(x)$ has the smallest possible maximum value on $[-1, 1]$.

Theorem 3.10. *Given any monic n -th order polynomial $p(x)$, there exist $x^* \in [-1, 1]$ such that*

$$|p(x^*)| \geq \frac{1}{2^{n-1}}.$$

Proof. Assume, for the sake of contradiction, that there exists a monic n -th order polynomial $p(x)$ such that for all $x \in [-1, 1]$,

$$|p(x)| < \frac{1}{2^{n-1}}.$$

Let $T_n(x)$ denote the n -th Chebyshev polynomial. Note that $T_n(x)$ attains its extreme values at the $n + 1$ points

$$y_k = \cos\left(\frac{k\pi}{n}\right), \quad k = 0, 1, \dots, n,$$

where $T_n(y_k) = (-1)^k$. We define the normalized (monic) Chebyshev polynomial as $\tilde{T}_n(x) = \frac{1}{2^{n-1}}T_n(x)$.

Consider the difference polynomial:

$$q(x) = \tilde{T}_n(x) - p(x).$$

Since both $\tilde{T}_n(x)$ and $p(x)$ are monic polynomials of degree n , $q(x)$ is a polynomial of degree at most $n - 1$. Evaluating $q(x)$ at the extreme points y_k , we have:

$$q(y_k) = \frac{(-1)^k}{2^{n-1}} - p(y_k).$$

Given our assumption that $|p(y_k)| < \frac{1}{2^{n-1}}$, it follows that:

$$\text{sgn}(q(y_k)) = \text{sgn}((-1)^k).$$

Thus, $q(x)$ changes sign n times over the interval $[-1, 1]$. By the Intermediate Value Theorem, $q(x)$ must have at least n distinct roots in the interval $[y_n, y_0]$.

However, $q(x)$ is a polynomial of degree at most $n - 1$. A non-zero polynomial of degree $n - 1$ cannot have n roots. This yields the desired contradiction. $\square$

3.5 Error of Lagrange interpolation

Let $P_n(x)$ be the interpolating polynomial for the nodes $x_0, x_1, \dots, x_n$. By our successive constructive of an interpolating polynomial, the interpolating polynomial for the nodes $x_0, x_1, \dots, x_n, x_{n+1}$ is

$$P_{n+1}(x) = P_n(x) + f[x_0, \dots, x_n, x_{n+1}](x - x_0) \cdots (x - x_n).$$

Since P_{n+1} matches f at x_{n+1} , we have $f(x_{n+1}) = P_{n+1}(x_{n+1})$, so

$$f(x_{n+1}) = P_n(x_{n+1}) + f[x_0, \dots, x_n, x_{n+1}](x_{n+1} - x_0) \cdots (x_{n+1} - x_n).$$

Since this holds for arbitrary x_{n+1} , we can replace x_{n+1} with x :

$$f(x) = P_n(x) + f[x_0, \dots, x_n, x](x - x_0) \cdots (x - x_n).$$

Theorem 3.11. Let $x_0, x_1, \dots, x_n$ be distinct real numbers. Then for any n th-order differentiable function $f(x)$, there exists

$$\xi \in (\min(x_0, \dots, x_n), \max(x_0, \dots, x_n))$$

such that

$$f[x_0, \dots, x_n] = \frac{f^{(n)}(\xi)}{n!}.$$

Proof. For $n = 1$, set $g(x) = f(x) - P_1(x)$. By Rolle's theorem, there exists ξ between x_0 and x_1 such that

$$f[x_0, x_1] = \frac{f(x_1) - f(x_0)}{x_1 - x_0} = f'(\xi).$$

For $n = 2$, similarly define $g(x) = f(x) - P_2(x)$, which satisfies $g(x_0) = g(x_1) = g(x_2) = 0$. WLOG assume $x_0 < x_1 < x_2$. Applying Rolle's theorem twice, there exists $\xi_0 \in (x_0, x_1)$ such that $g'(\xi_0) = 0$; there exists $\xi_1 \in (x_1, x_2)$ such that $g'(\xi_1) = 0$. By Rolle's theorem, there exists $\eta \in (\xi_0, \xi_1)$ such that $g''(\eta) = 0$.

Since $g(x) = f(x) - P_2(x)$, we have $f''(\eta) - P_2''(x) = 0$. Since $P_2''(x) = 2f[x_0, x_1, x_2]$, we conclude that $f[x_0, x_1, x_2] = \frac{1}{2}f''(\eta)$.

The proof can be generalized to any positive integer n . Note that Rolle's theorem is applied $n + (n - 1) + \cdots + 1$ times. $\square$

Theorem 3.12. Let $x_0, x_1, \dots, x_n$ be distinct interpolation nodes, and $P_n(x)$ be the Lagrange interpolating polynomial of $f(x)$. Suppose $f(x)$ is $(n + 1)$ th-order differentiable. Then for any x , there exists

$$\xi \in (\min(x_0, \dots, x_n, x), \max(x_0, \dots, x_n, x))$$

such that

$$f(x) - P_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!} (x - x_0)(x - x_1) \cdots (x - x_n). \quad (10)$$

A straightforward corollary is

$$|f(x) - P_n(x)| \leq C(x) |(x - x_0)(x - x_1) \cdots (x - x_n)|,$$

where

$$C(x) = \frac{1}{(n+1)!} \max_{\xi \in I_x} |f^{(n+1)}(\xi)|$$

with I_x being the interval

$$I_x = [\min(x_0, x_1, \dots, x_n, x), \max(x_0, x_1, \dots, x_n, x)].$$

Remark. Differentiate $f^{(n+1)}$ to find the maximum value.

Divided differences satisfy the following properties:

Proposition 3.13.

(1) *Linearity:*

$$\begin{aligned} f[x_0, \dots, x_n] + g[x_0, \dots, x_n] &= (f + g)[x_0, \dots, x_n] \\ \lambda f[x_0, \dots, x_n] &= (\lambda \cdot f)[x_0, \dots, x_n] \end{aligned}$$

(2) *Symmetry:* For an arbitrary permutation $\sigma: \{0, \dots, n\} \rightarrow \{0, \dots, n\}$,

$$f[x_0, \dots, x_n] = f[x_{\sigma(0)}, \dots, x_{\sigma(n)}].$$

That is, a divided difference is independent of the rearrangement of interpolation nodes.

(3) *Leibniz rule:*

$$(f \cdot g)[x_0, \dots, x_n] = \sum_{r=0}^n f[x_0, \dots, x_r] g[x_r, \dots, x_n].$$

Note that (3) is analogous to the product rule

$$(fg)^{(n)}(x) = \sum_{r=0}^n f^{(r)}(x) g^{(n-r)}(x).$$

3.6 Cubic spline interpolation

An alternative approach is to divide the approximation interval into a collection of sub-intervals and construct a different approximating polynomial on each sub-interval. This is called **piecewise-polynomial approximation**.

The simplest piecewise-polynomial approximation is **piecewise-linear interpolation**, which joins the data points by a series of straight lines. Then $S \in C^0$. A disadvantage is that there is likely no differentiability at the endpoints of the sub-intervals, i.e., $S \notin C^1$, so the interpolating function is not smooth.

The most common piecewise-polynomial approximation uses cubic polynomials between each successive pair of nodes, and is called **cubic spline interpolation**.

Definition 3.14. Let $S(x)$ be a cubic polynomial, denoted $p_k(x)$, on the sub-interval $[x_{k-1}, x_k]$ for each $k = 1, \dots, n$. Given a function f defined on $[a, b]$ and a set of nodes $a = x_0 < x_1 < \dots < x_n = b$, we say that S is **cubic spline interpolant** for f if:

- (1) $p_k(x_{k-1}) = f(x_{k-1})$ and $p_k(x_k) = f(x_k)$ for each $k = 1, \dots, n$.
- (2) $p'_k(x_k) = p'_{k+1}(x_k)$ for each $k = 1, \dots, n-1$.
- (3) $p''_k(x_k) = p''_{k+1}(x_k)$ for each $k = 1, \dots, n-1$.
- (4) One of the following sets of boundary conditions is satisfied:
 - (i) $p'_1(x_0) = f'(x_0)$ and $p'_n(x_n) = f'(x_n)$ (clamped boundary)
 - (ii) $S''(x_0) = S''(x_n) = 0$ (natural boundary)

In words, we require the cubic spline interpolant S for f to satisfy the following properties:

- (1) S matches f at all interpolating nodes.
- (2) derivatives on both sides of every interpolating node is equal (for interior points) - first-order differentiability
- (3) second-order differentiability
- (4) boundary conditions (for endpoints)

To determine p_k , let

$$p_k(x) = f[x_{k-1}] + f[x_{k-1}, x_k](x - x_{k-1}) + (\alpha_k x + \beta_k)(x - x_{k-1})(x - x_k),$$

such that p_k satisfies $p_k(x_{k-1}) = f(x_{k-1})$ and $p_k(x_k) = f(x_k)$, and is a cubic polynomial. Next, we determine the coefficients α_k and β_k by

$$p''_k(x_{k-1}) = S''(x_{k-1}), \quad p''_k(x_k) = S''(x_k).$$

The first-order derivative is

$$p'_k(x) = f[x_{k-1}, x_k] + (\alpha_k x + \beta_k)(x - x_k) + (\alpha_k x + \beta_k)(x - x_{k-1}) + \alpha_k(x - x_{k-1})(x - x_k),$$

so the second-order derivative is

$$p''_k(x) = \alpha_k(x - x_k) + (\alpha_k x + \beta_k) + \alpha_k(x - x_{k-1}) + (\alpha_k x + \beta_k) + \alpha_k(x - x_{k-1}) + \alpha_k(x - x_k)$$

and thus

$$\begin{aligned} p''_k(x_{k-1}) &= 2\alpha_k(x_{k-1} - x_k) + 2(\alpha_k x_{k-1} + \beta_k) \\ p''_k(x_k) &= 2\alpha_k(x - k - x_{k-1}) + 2(\alpha_k x + \beta_k) \end{aligned}$$

Solving for α_k and β_k gives

$$\begin{aligned} \alpha_k &= \frac{S''(x_k) - S''(x_{k-1})}{6(x_k - x_{k-1})} \\ \beta_k &= \frac{x_k - 2x_{k-1}}{6(x_k - x_{k-1})} S''(x_k) + \frac{2x_k - x_{k-1}}{6(x_k - x_{k-1})} S''(x_{k-1}) \end{aligned}$$

Therefore,

$$p_k(x) = f[x_{k-1}] + f[x_{k-1}, x_k](x - x_{k-1}) + \left(\frac{S''(x_k) - S''(x_{k-1})}{6(x_k - x_{k-1})}x + \frac{x_k - 2x_{k-1}}{6(x_k - x_{k-1})}S''(x_k) + \frac{2x_k - x_{k-1}}{6(x_k - x_{k-1})}S''(x_{k-1}) \right)(x - x_{k-1})(x - x_k). \quad (11)$$

So far, we have used conditions (1) and (3). To use condition (2), compute first order derivatives:

$$\begin{aligned} p'_k(x_{k-1}) &= f[x_{k-1}, x_k] + (\alpha_k x_{k-1} + \beta_k)(x_{k-1} - x_k), \\ p'_k(x_k) &= f[x_{k-1}, x_k] + (\alpha_k x_k + \beta_k)(x_k - x_{k-1}). \end{aligned}$$

Substituting the values of α_k and β_k above gives

$$\begin{aligned} \alpha_k x_{k-1} + \beta_k &= \frac{1}{3}S''(x_{k-1}) + \frac{1}{6}S''(x_k), \\ \alpha_k x_k + \beta_k &= \frac{1}{6}S''(x_{k-1}) + \frac{1}{3}S''(x_k). \end{aligned}$$

Therefore,

$$\begin{aligned} p'_k(x_{k-1}) &= f[x_{k-1}, x_k] + \frac{x_{k-1} - x_k}{6}[2S''(x_{k-1}) + S''(x_k)], \\ p'_k(x_k) &= f[x_{k-1}, x_k] + \frac{x_k - x_{k-1}}{6}[S''(x_{k-1}) + 2S''(x_k)], \end{aligned}$$

or by performing a linear shift $k \mapsto k + 1$,

$$\begin{aligned} p'_{k+1}(x_k) &= f[x_k, x_{k+1}] + \frac{x_k - x_{k+1}}{6}[2S''(x_k) + S''(x_{k+1})], \\ p'_k(x_k) &= f[x_{k-1}, x_k] + \frac{x_k - x_{k-1}}{6}[S''(x_{k-1}) + 2S''(x_k)]. \end{aligned}$$

For $k = 1, \dots, n-1$, equate the first-order derivatives $p'_{k+1}(x_k) = p'_k(x_k)$ and simplify to get

$$\frac{x_{k+1}x_k}{x_{k+1} - x_{k-1}}S''(x_{k+1}) + 2S''(x_k) + \frac{x_k - x_{k-1}}{x_{k+1} - x_{k-1}}S''(x_{k-1}) = 6f[x_{k-1}, x_k, x_{k+1}].$$

Assuming clamped boundary conditions, we have $p'_1(x_0) = f'(x_0)$, which solves to

$$2S''(x_0) + S''(x_1) = 6f[x_0, x_0, x_1]$$

and $p'_n(x_n) = f'(x_n)$, which solves to

$$S''(x_{n-1}) + 2S''(x_n) = 6f[x_{n-1}, x_n, x_n].$$

Therefore, for $k = 0, 1, \dots, n-1, n$,

$$\lambda_k S''(x_{k-1}) + 2S''(x_k) + (1 - \lambda_k)S''(x_{k+1}) = 6f[x_{k-1}, x_k, x_{k+1}],$$

where $x_{-1} = x_0$, $x_{n+1} = x_n$, and $\lambda_k = \frac{x_k - x_{k-1}}{x_{k+1} - x_{k-1}} \in [0, 1]$. This gives a linear system

$$\begin{pmatrix} 2 & 1 - \lambda_0 & & & \\ \lambda_1 & 2 & 1 - \lambda_1 & & \\ & \lambda_2 & 2 & 1 - \lambda_2 & \\ & & \ddots & \ddots & \ddots \\ & & & \lambda_{n-1} & 2 & 1 - \lambda_{n-1} \\ & & & & \lambda_n & 2 \end{pmatrix} \begin{pmatrix} S''(x_0) \\ S''(x_1) \\ S''(x_2) \\ \vdots \\ S''(x_{n-1}) \\ S''(x_n) \end{pmatrix} = \begin{pmatrix} 6f[x_0, x_0, x_1] \\ 6f[x_0, x_1, x_2] \\ 6f[x_1, x_2, x_3] \\ \vdots \\ 6f[x_{n-2}, x_{n-1}, x_n] \\ 6f[x_{n-1}, x_n, x_n] \end{pmatrix} \quad (12)$$

where we solve for the unknowns $S''(x_0), \dots, S''(x_n)$.

Now we check the existence and uniqueness of the solution to the above linear system. Let A be the transpose of the coefficient matrix:

$$A = \begin{pmatrix} 2 & \lambda_1 & & & \\ 1 - \lambda_0 & 2 & \lambda_2 & & \\ & 1 - \lambda_1 & 2 & \ddots & \\ & & 1 - \lambda_2 & \ddots & \lambda_{n-1} \\ & & & \ddots & 2 & \lambda_n \\ & & & & 1 - \lambda_{n-1} & 2 \end{pmatrix}$$

Then A is a tridiagonal diagonally dominant matrix. Since A is non-singular, A^T is non-singular. Hence, the solution exists and is unique.

On the other hand, for natural boundary conditions, i.e., $S''(x_0) = S''(x_n) = 0$, we have

$$\begin{pmatrix} 2 & 1 - \lambda_1 & & & \\ \lambda_2 & 2 & 1 - \lambda_2 & & \\ & \ddots & \ddots & \ddots & \\ & & \lambda_{n-2} & 2 & 1 - \lambda_{n-2} \\ & & & \lambda_{n-1} & 2 \end{pmatrix} \begin{pmatrix} S''(x_1) \\ S''(x_2) \\ \vdots \\ S''(x_{n-2}) \\ S''(x_{n-1}) \end{pmatrix} = \begin{pmatrix} 6f[x_0, x_1, x_2] \\ 6f[x_1, x_2, x_3] \\ \vdots \\ 6f[x_{n-3}, x_{n-2}, x_{n-1}] \\ 6f[x_{n-2}, x_{n-1}, x_n] \end{pmatrix}. \quad (13)$$

Procedure to perform the cubic spline interpolation:

- (1) Calculate second divided-differences
- (2) Solve second-order derivatives
- (3) Formulate the polynomials

To conclude this subsection, we state an error-bound formula for the cubic spline with clamped boundary conditions. The proof of this result can be found in [Schul], pp. 57-58.

Theorem 3.15. Let $f \in C^4[a, b]$ with $\max_{x \in [a, b]} |f^{(4)}(x)| = M$. If S is the unique clamped cubic spline interpolant to f with respect to the nodes $x_0, x_1, \dots, x_n$ satisfying $a = x_0 < x_1 < \dots < x_n = b$, then for all $x \in [a, b]$,

$$|f(x) - S(x)| \leq \frac{5M}{384} \max_{1 \leq k \leq n} (x_k - x_{k-1})^4.$$

3.7 Least squares approximation

Given data points $\{(x_i, y_i)\}_{i=1}^n$, find a polynomial of degree less than or equal to m :

$$P_m(x) = a_0 + a_1x + \dots + a_mx^m$$

which **best fit** all the data points. Here for $i \neq j$, the nodes x_i and x_j are allowed to be the same. We do not require P_m to pass through all data points.

The **least squares** approach involves determining the best approximating line, when the error is the sum of the squares of differences between y -values on the approximating line and the given y -values. Hence, constants a_0 and a_1 must be found that minimise the least squares error:

$$E(a_0, a_1) = \sum_{i=1}^n [y_i - (a_0 + a_1 x_i)]^2.$$

For a minimum to occur, we need both

$$\frac{\partial E}{\partial a_0} = 0 \quad \text{and} \quad \frac{\partial E}{\partial a_1} = 0,$$

that is,

$$\begin{aligned} 2 \sum_{i=1}^n [y_i - (a_0 + a_1 x_i)](-1) &= 0, \\ 2 \sum_{i=1}^n [y_i - (a_0 + a_1 x_i)](-x_i) &= 0. \end{aligned}$$

These equations simplify to the **normal equations**:

$$\begin{aligned} na_0 + \left(\sum_{i=1}^n x_i \right) a_1 &= \sum_{i=1}^n y_i, \\ \left(\sum_{i=1}^n x_i \right) a_0 + \left(\sum_{i=1}^n x_i^2 \right) a_1 &= \sum_{i=1}^n x_i y_i. \end{aligned}$$

The solution to this system of equations is

$$\begin{aligned} a_0 &= \frac{\sum_{i=1}^n x_i^2 \sum_{i=1}^n y_i - \sum_{i=1}^n x_i y_i \sum_{i=1}^n x_i}{n \left(\sum_{i=1}^n x_i^2 \right) - \left(\sum_{i=1}^n x_i \right)^2}, \\ a_1 &= \frac{n \sum_{i=1}^n x_i y_i - \sum_{i=1}^n x_i \sum_{i=1}^n y_i}{n \left(\sum_{i=1}^n x_i^2 \right) - \left(\sum_{i=1}^n x_i \right)^2}. \end{aligned}$$

Now consider the general problem of approximating $\{(x_i, y_i)\}_{i=1}^n$ with a polynomial

$$P_m(x) = a_0 + a_1 x + \cdots + a_m x^m$$

using the least squares procedure. We choose the constants $a_0, a_1, \dots, a_m$ to minimise the least squares error $E(a_0, a_1, \dots, a_m)$, where

$$E = \sum_{i=1}^n [y_i - P_m(x_i)]^2.$$

As in the linear case, for E to be minimised, it is necessary that $\frac{\partial E}{\partial a_j} = 0$ for each $j = 0, 1, \dots, m$. Thus, for each j , we must have

$$0 = \frac{\partial E}{\partial a_j} = -2 \sum_{i=1}^n y_i x_i^j + 2 \sum_{k=0}^m a_k \sum_{i=1}^n x_i^{j+k}.$$

This gives $m + 1$ **normal equations** in the $m + 1$ unknowns a_j . These are

$$\sum_{k=0}^m a_k \sum_{i=1}^n x_i^{j+k} = \sum_{i=1}^n y_i x_i^j.$$

Alternatively, we proceed by a linear algebra approach. Given data points $\{(x_i, y_i)\}_{i=0}^n$, define

$$\phi(a_0, a_2, \dots, a_m) = [y_0 - P_m(x_0)]^2 + [y_1 - P_m(x_1)]^2 + \dots + [y_n - P_m(x_n)]^2.$$

Assume $m < n$. Then $(y_0, y_1, \dots, y_n)^T$ is a point in $\mathbb{R}^{n+1}$, and $(P_m(x_0), P_m(x_1), \dots, P_m(x_n))^T$ forms a hyperplane as the coefficients $a_0, a_1, \dots, a_m$ vary. Hence, geometrically, the smallest value of ϕ is the distance from the point $(y_0, y_1, \dots, y_n)$ to the hyperplane.

Let $P(x) = a_0(1) + a_1(x) + a_2(x^2) + \dots + a_m(x^m)$. Evaluating the polynomial at $x_0, x_1, \dots, x_n$ gives a system of $n+1$ equations:

$$\begin{aligned} y_0 &\approx P(x_0) = a_0 + a_1 x_0 + a_2 x_0^2 + \dots + a_m x_0^m \\ y_1 &\approx P(x_1) = a_0 + a_1 x_1 + a_2 x_1^2 + \dots + a_m x_1^m \\ &\vdots \\ y_n &\approx P(x_n) = a_0 + a_1 x_n + a_2 x_n^2 + \dots + a_m x_n^m \end{aligned}$$

The right-hand side of this system can be decomposed into a linear combination of vectors

$$\mathbf{p}_0 = \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix}, \quad \mathbf{p}_1 = \begin{pmatrix} x_0 \\ x_1 \\ \vdots \\ x_n \end{pmatrix}, \quad \mathbf{p}_m = \begin{pmatrix} x_0^m \\ x_1^m \\ \vdots \\ x_n^m \end{pmatrix}.$$

These vectors define a hyperplane (a subspace) in $\mathbb{R}^{n+1}$. To find the minimum distance from $\mathbf{y}$ to the hyperplane, we find the orthogonal projection of $\mathbf{y}$ onto the subspace spanned by $\{\mathbf{p}_0, \dots, \mathbf{p}_m\}$.

Theorem 3.16 (Gram-Schmidt orthonormalization). *Given vectors $\mathbf{p}_0, \mathbf{p}_1, \dots, \mathbf{p}_m$, define*

$$\begin{aligned} \tilde{\mathbf{p}}_0 &= \mathbf{p}_0, & \tilde{\mathbf{p}}_0^* &= \tilde{\mathbf{p}}_0 / \|\tilde{\mathbf{p}}_0\|, \\ \tilde{\mathbf{p}}_1 &= \mathbf{p}_1 - (\mathbf{p}_1^T \tilde{\mathbf{p}}_0^*) \tilde{\mathbf{p}}_0^*, & \tilde{\mathbf{p}}_1^* &= \tilde{\mathbf{p}}_1 / \|\tilde{\mathbf{p}}_1\|, \\ \tilde{\mathbf{p}}_2 &= \mathbf{p}_2 - (\mathbf{p}_2^T \tilde{\mathbf{p}}_0^*) \tilde{\mathbf{p}}_0^* - (\mathbf{p}_2^T \tilde{\mathbf{p}}_1^*) \tilde{\mathbf{p}}_1^*, & \tilde{\mathbf{p}}_2^* &= \tilde{\mathbf{p}}_2 / \|\tilde{\mathbf{p}}_2\|, \\ \dots & \dots & \dots & \dots \\ \tilde{\mathbf{p}}_m &= \mathbf{p}_m - (\mathbf{p}_m^T \tilde{\mathbf{p}}_0^*) \tilde{\mathbf{p}}_0^* - (\mathbf{p}_m^T \tilde{\mathbf{p}}_1^*) \tilde{\mathbf{p}}_1^* - \dots - (\mathbf{p}_m^T \tilde{\mathbf{p}}_{m-1}^*) \tilde{\mathbf{p}}_{m-1}^*, & \tilde{\mathbf{p}}_m^* &= \tilde{\mathbf{p}}_m / \|\tilde{\mathbf{p}}_m\| \end{aligned}$$

Then $\tilde{\mathbf{p}}_0, \tilde{\mathbf{p}}_1, \dots, \tilde{\mathbf{p}}_m$ are orthonormal unit vectors.

We can rewrite the above as

$$\begin{aligned} \mathbf{p}_0 &= \|\tilde{\mathbf{p}}_0\| \tilde{\mathbf{p}}_0^*, \\ \mathbf{p}_1 &= (\mathbf{p}_1^T \tilde{\mathbf{p}}_0^*) \tilde{\mathbf{p}}_0^* + \|\tilde{\mathbf{p}}_1\| \tilde{\mathbf{p}}_1^*, \\ \mathbf{p}_2 &= (\mathbf{p}_2^T \tilde{\mathbf{p}}_0^*) \tilde{\mathbf{p}}_0^* + (\mathbf{p}_2^T \tilde{\mathbf{p}}_1^*) \tilde{\mathbf{p}}_1^* + \|\tilde{\mathbf{p}}_2\| \tilde{\mathbf{p}}_2^*, \\ &\dots \\ \mathbf{p}_m &= (\mathbf{p}_m^T \tilde{\mathbf{p}}_0^*) \tilde{\mathbf{p}}_0^* + (\mathbf{p}_m^T \tilde{\mathbf{p}}_1^*) \tilde{\mathbf{p}}_1^* + \dots + (\mathbf{p}_m^T \tilde{\mathbf{p}}_{m-1}^*) \tilde{\mathbf{p}}_{m-1}^* + \|\tilde{\mathbf{p}}_m\| \tilde{\mathbf{p}}_m^*. \end{aligned}$$

Then the old and new vectors are related by

$$\begin{pmatrix} \mathbf{p}_0 & \mathbf{p}_1 & \dots & \mathbf{p}_m \end{pmatrix} = \begin{pmatrix} \tilde{\mathbf{p}}_0^* & \tilde{\mathbf{p}}_1^* & \dots & \tilde{\mathbf{p}}_m^* \end{pmatrix} \begin{pmatrix} \|\tilde{\mathbf{p}}_0\| & \mathbf{p}_1^T \tilde{\mathbf{p}}_0^* & \mathbf{p}_2^T \tilde{\mathbf{p}}_0^* & \dots & \mathbf{p}_m^T \tilde{\mathbf{p}}_0^* \\ \|\tilde{\mathbf{p}}_1\| & \mathbf{p}_2^T \tilde{\mathbf{p}}_1^* & \dots & \dots & \mathbf{p}_m^T \tilde{\mathbf{p}}_1^* \\ & \ddots & \ddots & \ddots & \vdots \\ & & \|\tilde{\mathbf{p}}_{m-1}\| & \mathbf{p}_m^T \tilde{\mathbf{p}}_{m-1}^* & \vdots \\ & & & \|\tilde{\mathbf{p}}_m\| \end{pmatrix},$$

which is a **QR factorization** of $\begin{pmatrix} \mathbf{p}_0 & \mathbf{p}_1 & \cdots & \mathbf{p}_m \end{pmatrix}$, where Q satisfies $Q^T Q = I$.

Project $\mathbf{y}$ onto $\tilde{\mathbf{p}}_0^*, \tilde{\mathbf{p}}_1^*, \dots, \tilde{\mathbf{p}}_m^*$, and subtract $\mathbf{y}$ from the projection:

$$\mathbf{n} = \mathbf{y} - (\mathbf{y}^T \tilde{\mathbf{p}}_0^*) \tilde{\mathbf{p}}_0^* - (\mathbf{y}^T \tilde{\mathbf{p}}_1^*) \tilde{\mathbf{p}}_1^* - \cdots - (\mathbf{y}^T \tilde{\mathbf{p}}_m^*) \tilde{\mathbf{p}}_m^*.$$

Then $\mathbf{n}$ is perpendicular to $\mathbf{p}_0, \mathbf{p}_1, \dots, \mathbf{p}_m$. Therefore, the distance from $\mathbf{y}$ to the hyperplane is $\|\mathbf{n}\|$, and the point on the hyperplane that is closest to $\mathbf{y}$ is

$$\mathbf{y}^* = \mathbf{y} - \mathbf{n} = (\mathbf{y}^T \tilde{\mathbf{p}}_0^*) \tilde{\mathbf{p}}_0^* + (\mathbf{y}^T \tilde{\mathbf{p}}_1^*) \tilde{\mathbf{p}}_1^* + \cdots + (\mathbf{y}^T \tilde{\mathbf{p}}_m^*) \tilde{\mathbf{p}}_m^*.$$

To find the coefficients $a_0, a_1, \dots, a_m$, write the vector $\mathbf{y}^*$ as

$$\begin{aligned} \mathbf{y}^* &= \begin{pmatrix} P_m(x_0) \\ P_m(x_1) \\ \vdots \\ P_m(x_n) \end{pmatrix} = \begin{pmatrix} a_0 + a_1 x_0 + \cdots + a_m x_0^m \\ a_0 + a_1 x_1 + \cdots + a_m x_1^m \\ \vdots \\ a_0 + a_1 x_n + \cdots + a_m x_n^m \end{pmatrix} \\ &= a_0 \mathbf{p}_0 + a_1 \mathbf{p}_1 + \cdots + a_m \mathbf{p}_m. \end{aligned}$$

Since $\mathbf{y}^*$ is given by

$$\mathbf{y}^* (\mathbf{y}^T \tilde{\mathbf{p}}_0^*) \tilde{\mathbf{p}}_0^* + (\mathbf{y}^T \tilde{\mathbf{p}}_1^*) \tilde{\mathbf{p}}_1^* + \cdots + (\mathbf{y}^T \tilde{\mathbf{p}}_m^*) \tilde{\mathbf{p}}_m^*,$$

we just need to write the vectors $\tilde{\mathbf{p}}_0^*, \tilde{\mathbf{p}}_1^*, \dots, \tilde{\mathbf{p}}_m^*$ as linear combinations of $\mathbf{p}_0, \mathbf{p}_1, \dots, \mathbf{p}_m$.

Since $(\mathbf{p}_0, \mathbf{p}_1, \dots, \mathbf{p}_m) = (\tilde{\mathbf{p}}_0^*, \tilde{\mathbf{p}}_1^*, \dots, \tilde{\mathbf{p}}_m^*) R_1$, where

$$R = \begin{pmatrix} \|\tilde{\mathbf{p}}_0\| & \mathbf{p}_1^T \tilde{\mathbf{p}}_0^* & \mathbf{p}_2^T \tilde{\mathbf{p}}_0^* & \cdots & \mathbf{p}_m^T \tilde{\mathbf{p}}_0^* \\ & \|\tilde{\mathbf{p}}_1\| & \mathbf{p}_2^T \tilde{\mathbf{p}}_1^* & \cdots & \mathbf{p}_m^T \tilde{\mathbf{p}}_1^* \\ & & \ddots & \ddots & \vdots \\ & & & \|\tilde{\mathbf{p}}_{m-1}\| & \mathbf{p}_m^T \tilde{\mathbf{p}}_{m-1}^* \\ & & & & \|\tilde{\mathbf{p}}_m\| \end{pmatrix},$$

we have

$$(\tilde{\mathbf{p}}_0^*, \tilde{\mathbf{p}}_1^*, \dots, \tilde{\mathbf{p}}_m^*) = (\mathbf{p}_0, \mathbf{p}_1, \dots, \mathbf{p}_m) R^{-1}.$$

Therefore

$$\begin{aligned} \mathbf{y}^* &= (\mathbf{y}^T \tilde{\mathbf{p}}_0^*) \tilde{\mathbf{p}}_0^* + (\mathbf{y}^T \tilde{\mathbf{p}}_1^*) \tilde{\mathbf{p}}_1^* + \cdots + (\mathbf{y}^T \tilde{\mathbf{p}}_m^*) \tilde{\mathbf{p}}_m^* \\ &= (\tilde{\mathbf{p}}_0^*, \tilde{\mathbf{p}}_1^*, \dots, \tilde{\mathbf{p}}_m^*) (\tilde{\mathbf{p}}_0^*, \tilde{\mathbf{p}}_1^*, \dots, \tilde{\mathbf{p}}_m^*)^T \mathbf{y} \\ &= (\mathbf{p}_0, \mathbf{p}_1, \dots, \mathbf{p}_m) R^{-1} Q^T \mathbf{y} \end{aligned}$$

Therefore, we have $\mathbf{a} = R^{-1} Q^T \mathbf{y}$, or $R\mathbf{a} = Q^T \mathbf{y}$.

Procedure:

- (1) Compute the vectors $\mathbf{p}_k, k = 0, 1, \dots, m$ by

$$\mathbf{p}_k = (x_0^k, x_1^k, \dots, x_n^k)^T.$$

- (2) Orthonormalize these vectors by finding $Q = (\tilde{\mathbf{p}}_0^*, \tilde{\mathbf{p}}_1^*, \dots, \tilde{\mathbf{p}}_m^*)$ such that $Q^T Q = I$ and

$$(\mathbf{p}_0, \mathbf{p}_1, \dots, \mathbf{p}_m) = QR$$

with R being an upper triangular matrix.

- (3) Compute $\mathbf{b} = Q^T \mathbf{y}$ such that

$$\mathbf{y}^* = b_0 \tilde{\mathbf{p}}_0^* + b_1 \tilde{\mathbf{p}}_1^* + \cdots + b_m \tilde{\mathbf{p}}_m^*.$$

- (4) Solve the linear system $R\mathbf{a} = \mathbf{b}$ to find the optimal polynomial coefficients $\mathbf{a}$.

4 Numerical Integration

Given a function $f(x)$ defined on $[a, b]$, find the value of

$$\int_a^b f(x) \, dx.$$

The explicit form of the anti-derivative of $f(x)$ is usually unavailable, and we may not be able to express the integral in the series form. The expression of $f(x)$ may not be available, and we can only retrieve the function value $f(x)$ by a black box. We would like to develop an algorithm for any a, b and $f(x)$.

Given some function values $f(x_0), f(x_1), \dots, f(x_n)$, we can apply interpolation to approximate $f(x)$ by an interpolating function $P(x)$. Then the integral of $f(x)$ can be approximated by

$$\int_a^b f(x) \, dx \approx \int_a^b P(x) \, dx,$$

where the integral of $P(x)$ should be easily evaluated.

4.1 Trapezoidal rule

When $n = 1$, we know only two function values: $f(x_0) = f(a)$ and $f(x_1) = f(b)$. The Lagrange interpolating polynomial is

$$P(x) = f(a) \frac{x-b}{a-b} + f(b) \frac{x-a}{b-a}.$$

Then

$$\begin{aligned} \int_a^b f(x) \, dx &\approx \int_a^b P(x) \, dx = \int_a^b f(a) \frac{x-b}{a-b} + f(b) \frac{x-a}{b-a} \, dx \\ &= \left[\frac{1}{2} f(a) \frac{(x-b)^2}{a-b} + \frac{1}{2} f(b) \frac{(x-a)^2}{b-a} \right]_a^b \\ &= \frac{b-a}{2} [f(a) + f(b)]. \end{aligned}$$

This approximation is called the **trapezoidal rule**:

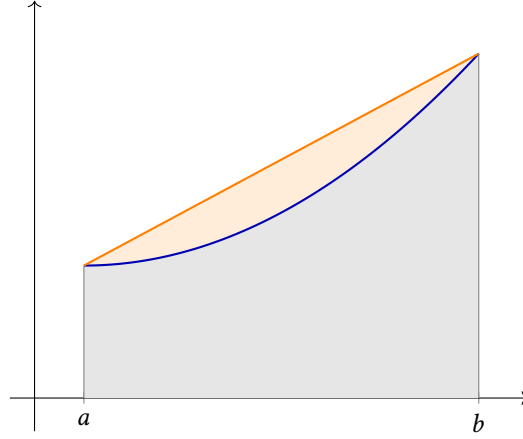
$$\int_a^b f(x) \, dx \approx \frac{b-a}{2} [f(a) + f(b)]. \quad (14)$$

Example. We would like to approximate

$$\int_0^2 \sqrt{1+x^2} \, dx.$$

Let $f(x) = \sqrt{1+x^2}$. Then $f(0) = 1$ and $f(2) = \sqrt{5}$. Thus,

$$\int_0^2 \sqrt{1+x^2} \, dx \approx \frac{b-a}{2} [f(a) + f(b)] = 1 + \sqrt{5}.$$



Theorem 4.1 (Error term). Suppose $f(x)$ is second-order differentiable on $[a, b]$. Then there exists $\xi \in (a, b)$ such that

$$\int_a^b f(x) dx = \frac{b-a}{2} [f(a) + f(b)] - \frac{1}{12} (b-a)^3 f''(\xi).$$

This theorem means that the absolute error is bounded by

$$\left| \int_a^b f(x) dx - \frac{b-a}{2} [f(a) + f(b)] \right| = \frac{1}{12} (b-a)^3 |f''(\xi)| \leq \frac{1}{12} (b-a)^3 \max_{\xi \in [a,b]} |f''(\xi)|.$$

Proof. Since the trapezoidal rule involves interpolating two points using a straight line, it gives exact results for polynomials no more than degree 1. Thus, we consider when $f(x)$ is quadratic. Let $f(x) = c_0 + c_1x + c_2x^2$. Then the exact integral is

$$\int_a^b f(x) dx = \left[c_0x + \frac{1}{2}c_1x^2 + \frac{1}{3}c_2x^3 \right]_a^b = (b-a) \left(c_0 + \frac{1}{2}c_1(b+a) + \frac{1}{3}c_2(a^2 + ab + b^2) \right).$$

The trapezoidal rule gives the approximation

$$\int_a^b f(x) dx \approx \frac{b-a}{2} [f(a) + f(b)] = (b-a) \left(c_0 + \frac{1}{2}c_1(b+a) + \frac{1}{2}c_2(a^2 + b^2) \right).$$

Hence, the error term is

$$\begin{aligned} \int_a^b f(x) dx - \frac{b-a}{2} [f(a) + f(b)] &= (b-a) \left(-\frac{1}{6}c_2(b-a)^2 \right) \\ &= -\frac{1}{12}(b-a)^3 \cdot (2c_2) \\ &= -\frac{1}{12}(b-a)^3 \cdot f''(\xi). \end{aligned}$$

For a general function $f(x)$, if we can find a quadratic function $Q(x)$ such that

$$Q(a) = f(a), \quad Q(b) = f(b), \quad \int_a^b Q(x) dx = \int_a^b f(x) dx,$$

then the error term is

$$\int_a^b f(x) dx - \frac{b-a}{2} [f(a) + f(b)] = \int_a^b f(x) dx - \frac{b-a}{2} [Q(a) + Q(b)] = -\frac{1}{12} (b-a)^3 Q''(\xi).$$

The equation for $Q(x)$ is of the form

$$Q(x) = f(a)\frac{x-b}{a-b} + f(b)\frac{x-a}{b-a} + C(x-a)(x-b),$$

where C is some constant to be determined by the condition $\int_a^b Q(x) dx = \int_a^b f(x) dx$.

Observe that $C = \frac{1}{2}Q''(x)$. If we choose C such that $Q''(x) = \min_{x \in [a,b]} f''(x)$, then the curve of $Q(x)$ lies entirely above $f(x)$, so that

$$\int_a^b Q(x) dx \geq \int_a^b f(x) dx.$$

On the other hand, if we choose C such that $Q''(x) = \max_{x \in [a,b]} f''(x)$, then the curve of $Q(x)$ lies entirely below $f(x)$, so that

$$\int_a^b Q(x) dx \leq \int_a^b f(x) dx.$$

It follows that our desired value of C lies in the interval

$$I = \left[\frac{1}{2} \min_{x \in [a,b]} f''(x), \frac{1}{2} \max_{x \in [a,b]} f''(x) \right].$$

Since f'' is continuous, by intermediate value theorem, there exists $\xi \in (a, b)$ such that $C = \frac{1}{2}f''(\xi)$. □

4.2 Simpson's rule

When $n = 2$, we have three nodes $x_0 = a$, $x_1 = \frac{a+b}{2}$, $x_2 = b$. We can find the Lagrange interpolating polynomial by divided differences:

$$P(x) = f(a) + \frac{f(b) - f(a)}{b - a}(x - a) + \frac{2}{(b - a)^2} \left[f(a) + f(b) - 2f\left(\frac{b + a}{2}\right) \right] (x - a)(x - b).$$

Hence, the numerical integration gives

$$\begin{aligned} \int_a^b f(x) dx &\approx \int_a^b P(x) dx \\ &= \frac{b-a}{2} [f(a) + f(b)] + \frac{2}{(b-a)^2} \left[f(a) + f(b) - 2f\left(\frac{b+a}{2}\right) \right] \cdot \frac{1}{6}(a-b)^3 \\ &= (b-a) \left[\frac{1}{6}f(a) + \frac{2}{3}f\left(\frac{a+b}{2}\right) + \frac{1}{6}f(b) \right]. \end{aligned}$$

This is known as **Simpson's rule**.

Example. Consider the integral

$$\int_0^2 \sqrt{1+x^2} dx.$$

Let $f(x) = \sqrt{1+x^2}$. Since $f(0) = 1$, $f(1) = \sqrt{2}$, $f(2) = \sqrt{5}$, Simpson's rule gives the following approximation:

$$\int_0^2 \sqrt{1+x^2} dx \approx \frac{2}{6}(1 + 4\sqrt{2} + \sqrt{5}).$$

Theorem 4.2 (Error term). *Suppose $f(x)$ is third-order differentiable on $[a, b]$. Then there exists $\xi \in (a, b)$ such that*

$$\int_a^b f(x) dx = \frac{b-a}{6} \left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right] - \frac{1}{90} \left(\frac{b-a}{2} \right)^5 f^{(4)}(\xi).$$

This theorem means that the absolute error is bounded by

$$\left| \int_a^b f(x) dx - \frac{b-a}{6} \left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right] \right| = \frac{1}{90} \left(\frac{b-a}{2} \right)^5 |f^{(4)}(\xi)| \leq \frac{1}{90} \left(\frac{b-a}{2} \right)^5 \max_{\xi \in [a,b]} |f^{(4)}(\xi)|.$$

In general, the error has the form

$$\int_a^b f(x) dx - (\text{numerical integration}) = C(b-a)^{k+1} f^{(k)}(\xi).$$

To determine the value of k , we use polynomials $f(x)$ to test.

- If $\deg f \leq k-1$, then $f^{(k)}(x) = 0$, so the numerical integration is exact.
- If $\deg f = k$, then the error is non-zero when $C \neq 0$. Also, the error is a constant term.

If a numerical integration formula is exact for all polynomials of degree $\leq m$ but is inexact for degree $m+1$, we say that the **degree of accuracy** of this formula is m .

Proof. To determine the error term, test over the domain $[-1, 1]$.

- When $f(x) = 1$,

$$\begin{aligned} \int_a^b f(x) dx &= b - a = 2, \\ \frac{b-a}{6} \left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right] &= \frac{1 - (-1)}{6} (1 + 4 + 1) = 2. \end{aligned}$$

The numerical integration is exact.

- When $f(x) = x$,

$$\begin{aligned} \int_a^b f(x) dx &= \frac{b^2 - a^2}{2} = 0, \\ \frac{b-a}{6} \left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right] &= \frac{2}{6} (-1 + 0 + 1) = 0. \end{aligned}$$

The numerical integration is exact.

- When $f(x) = x^2$,

$$\begin{aligned} \int_a^b f(x) dx &= \frac{b^3 - a^3}{3} = \frac{2}{3}, \\ \frac{b-a}{6} \left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right] &= \frac{2}{6} (1 + 0 + 1) = \frac{2}{3}. \end{aligned}$$

The numerical integration is exact.

- When $f(x) = x^3$,

$$\begin{aligned} \int_a^b f(x) dx &= \frac{b^4 - a^4}{4} = 0, \\ \frac{b-a}{6} \left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right] &= \frac{2}{6} (-1 + 0 + 1) = 0. \end{aligned}$$

The numerical integration is exact.

- When $f(x) = x^4$,

$$\int_a^b f(x) dx = \frac{b^5 - a^5}{5} = \frac{2}{5},$$

$$\frac{b-a}{6} \left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right] = \frac{2}{6}(1 + 0 + 1) = \frac{2}{3}.$$

The numerical integration is inexact.

The degree of accuracy of Simpson's rule is 3. Hence, the error term has the form

$$C(b-a)^5 f^{(4)}(\xi), \quad \xi \in (a, b).$$

That is,

$$\int_a^b f(x) dx - \frac{b-a}{6} \left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right] = C(b-a)^5 f^{(4)}(\xi).$$

To determine the constant C , again choose $a = -1$, $b = 1$ and $f(x) = x^4$. Then LHS is $\frac{2}{5} - \frac{2}{3} = -\frac{4}{15}$ and RHS is $C \cdot 2^5 \cdot 4! = 768C$. Solving gives $C = -\frac{1}{2880}$. $\square$

Example. The Trapezoidal rule applied to $\int_0^2 f(x) dx$ gives the value 4, and Simpson's rule gives the value 2. What is $f(1)$?

Solution. By Trapezoidal rule,

$$T = (b-a) \frac{f(a) + f(b)}{2} = 2 \cdot \frac{f(0) + f(2)}{2} = 4. \quad (1)$$

By Simpson's rule,

$$S = (b-a) \left[\frac{1}{6}f(a) + \frac{2}{3}f\left(\frac{a+b}{2}\right) + \frac{1}{6}f(b) \right] = 2 \cdot \frac{f(0) + 4f(1) + f(2)}{6} = 2. \quad (2)$$

Then (2) - (1) gives $f(1) = \frac{1}{2}$. $\square$

4.3 Newton-Cotes formulas

We are considering function values on

$$x_0 = a, \quad x_n = b, \quad x_k - x_{k-1} = \frac{b-a}{n} \quad \text{for } k = 1, \dots, n.$$

If $n = 1$, then this is the Trapezoidal rule; if $n = 2$, then this is Simpson's rule. In order to get a more accurate formula for the Lagrange interpolating polynomial, we now consider the general case for n .

$$\int_a^b f(x) dx \approx \int_a^b P_n(x) dx = \sum_{i=0}^n f(x_i) \int_a^b L_i(x) dx.$$

Hence, it suffices to compute the coefficients

$$a_i = \int_a^b L_i(x) dx = \int_a^b \prod_{j \neq i} \frac{x - x_j}{x_i - x_j} dx.$$

However, this requires tedious calculations to obtain. Moreover, for cases with different endpoints a and b , we also need to recalculate all these coefficients. Can we find some way to simplify it?

$$\sum_{i=0}^n a_i = \sum_{i=0}^n \int_a^b L_i(x) dx = \int_a^b \sum_{i=0}^n L_i(x) dx = \int_a^b 1 dx = b - a.$$

Let $c_i = a_i/(b - a)$. Then c_i is independent of a and b .

Then, consider the linear map from $[a, b]$ to $[0, 1]$. Let $x = a + (b - a)t$, where $t \in [0, 1]$, and define $g(t) = f(a + (b - a)t)$.

If we find an approximation $P(t)$ for $g(t)$, then $P\left(\frac{x-a}{b-a}\right)$ is an approximation for $f(x)$:

$$f(x) = f(a + (b - a)t) = g(t) \approx P(t) = P\left(\frac{x - a}{b - a}\right).$$

Therefore, for different problems on different domains $[a, b]$, the constant coefficients c_i are the same and determined by

$$c_i = \int_0^1 \prod_{j \neq i} \frac{t - t_j}{t_i - t_j} dt$$

where $t_j = j/n$ for $j = 0, 1, \dots, n$ (taking equally spaced t_j on $[0, 1]$).

Theorem 4.3. *The Newton-Cotes formula is*

$$\int_a^b f(x) dx \approx (b - a) \sum_{i=0}^n c_i f(x_i) = \frac{b - a}{n} \sum_{i=0}^n H_{n,i} f(x_i), \quad (15)$$

where

$$H_{n,i} = \frac{(-1)^{n-i}}{i!(n-i)!} \int_0^n s(s-1) \cdots (s-i+1)(s-i-1) \cdots (s-n) ds.$$

The case where $n = 3$ has a name:

Corollary 4.4 (Simpson's 3/8 rule).

$$\begin{aligned} \int_a^b f(x) dx &\approx \frac{b-a}{3} \left[\frac{3}{8} f(a) + \frac{9}{8} f\left(\frac{2a+b}{3}\right) + \frac{9}{8} f\left(\frac{a+2b}{3}\right) + \frac{3}{8} f(b) \right] \\ &= (b-a) \left[\frac{1}{8} f(a) + \frac{3}{8} f\left(\frac{2a+b}{3}\right) + \frac{3}{8} f\left(\frac{a+2b}{3}\right) + \frac{1}{8} f(b) \right]. \end{aligned}$$

Proof. When $n = 3$, compute each $H_{n,i}$:

$$\begin{aligned} H_{3,0} &= \frac{(-1)^3}{0!3!} \int_0^3 (s-1)(s-2)(s-3) ds = \frac{3}{8}, \\ H_{3,1} &= \frac{(-1)^2}{1!2!} \int_0^3 s(s-2)(s-3) ds = \frac{9}{8}, \\ H_{3,2} &= \frac{(-1)^1}{2!1!} \int_0^3 s(s-1)(s-3) ds = \frac{9}{8}, \\ H_{3,3} &= \frac{(-1)^0}{3!0!} \int_0^3 s(s-1)(s-2) ds = \frac{3}{8}. \end{aligned}$$

□

To determine the error term, let $a = -1$ and $b = 1$. One can show that when $f(x) = 1, x, x^2, x^3$, the numerical integration is exact; when $f(x) = x^4$, the numerical integration is inexact.

Thus, the degree of accuracy for Simpson's 3/8 formula is also 3, and the error term has the form

$$C(b-a)^5 f^{(4)}(\xi).$$

To determine the constant C , again choose $a = -1$, $b = 1$ and $f(x) = x^4$. Then $\frac{2}{5} - \frac{14}{27} = C \cdot 2^5 \cdot 4!$, so $C = -\frac{1}{6480}$. In general, the error term of the Newton-Cotes formula is as follows:

Theorem 4.5. Suppose that $\sum_{i=0}^n a_i f(x_i)$ denotes the $(n+1)$ -point (closed) Newton-Cotes formula with $x_0 = a$, $x_n = b$ and $h = (b-a)/n$. There exists $\xi \in (a, b)$ for which

$$\int_a^b f(x) dx = \sum_{i=0}^n a_i f(x_i) + \frac{h^{n+3} f^{(n+2)}(\xi)}{(n+2)!} \int_0^n t^2(t-1) \cdots (t-n) dt$$

if n is even and $f \in C^{n+2}[a, b]$, and

$$\int_a^b f(x) dx = \sum_{i=0}^n a_i f(x_i) + \frac{h^{n+2} f^{(n+1)}(\xi)}{(n+1)!} \int_0^n t(t-1) \cdots (t-n) dt$$

if n is odd and $f \in C^{n+1}[a, b]$.

For functions not defined on closed domains, we have to consider Newton-Cotes formulas on open domains (a, b) . The key idea is the same, and the difference is the distribution of points: the nodes are interior points

$$x_i = a + ih, \quad h = \frac{b-a}{n+2}.$$

For example,

- $n = 0$: Midpoint rule

$$\int_a^b f(x) dx = 2hf\left(\frac{a+b}{2}\right) + \frac{h^3}{3}f''(\xi)$$

- $n = 1$:

$$\int_a^b f(x) dx = \frac{3h}{2} \left[f\left(\frac{2a+b}{3}\right) + f\left(\frac{a+2b}{3}\right) \right] + \frac{3h^3}{4}f''(\xi)$$

- $n = 2$:

$$\int_a^b f(x) dx = \frac{4h}{3} \left[2f\left(\frac{3a+b}{4}\right) - f\left(\frac{a+b}{2}\right) + 2f\left(\frac{a+3b}{4}\right) \right] + \frac{14h^4}{45}f^{(4)}(\xi)$$

- $n = 3$:

$$\int_a^b f(x) dx = \frac{5h}{24} \left[11f\left(\frac{4a+b}{5}\right) + f\left(\frac{3a+2b}{5}\right) + f\left(\frac{2a+3b}{5}\right) + 11f\left(\frac{a+4b}{5}\right) \right] + \frac{95h^5}{144}f^{(4)}(\xi)$$

Theorem 4.6. Suppose that $\sum_{i=0}^n a_i f(x_i)$ denotes the $(n+1)$ -point open Newton-Cotes formula with $x_{-1} = a$, $x_{n+1} = b$ and $h = (b-a)/(n+2)$. There exists $\xi \in (a, b)$ for which

$$\int_a^b f(x) dx = \sum_{i=0}^n a_i f(x_i) + \frac{h^{n+3} f^{(n+2)}(\xi)}{(n+2)!} \int_{-1}^{n+1} t^2(t-1) \cdots (t-n) dt$$

if n is even and $f \in C^{n+2}[a, b]$, and

$$\int_a^b f(x) dx = \sum_{i=0}^n a_i f(x_i) + \frac{h^{n+2} f^{(n+1)}(\xi)}{(n+1)!} \int_{-1}^{n+1} t(t-1) \cdots (t-n) dt$$

if n is odd and $f \in C^{n+1}[a, b]$.

Example. Find constants c_0, c_1, x_1 so that

$$\int_{-1}^1 f(x) dx = c_0 f(-1) + c_1 f(x_1)$$

has the highest degree of precision.

Solution. To find the degree of precision, we test the formula with different polynomials. Since there are 3 unknowns, we need to write down 3 equations.

- If $f(x) = 1$, then $2 = c_0 + c_1$.
- If $f(x) = x$, then $0 = -c_0 + c_1x_1$.
- If $f(x) = x^2$, then $\frac{2}{3} = c_0 + c_1x_1^2$.

Solving for the unknowns gives $c_0 = \frac{1}{2}$, $c_1 = \frac{3}{2}$, $x_1 = \frac{1}{3}$.

Hence, $\int_{-1}^2 f(x) dx = \frac{1}{2}f(-1) + \frac{3}{2}\left(\frac{1}{3}\right)$ has highest degree of precision of 2, by testing with $f(x) = x^3$. $\square$

Example. Find constants c_1, x_0, x_1 so that

$$\int_{-1}^1 f(x) dx = f(x_0) + c_1 f(x_1)$$

has the highest degree of precision.

Solution. To determine degree of precision, again we test the formula with different polynomials.

- If $f(x) = 1$, then $c_1 = 1$.
- If $f(x) = x$, then $x_0 = -x_1$.
- If $f(x) = x^2$, then $\frac{2}{3} = x_0^2 + x_1^2$.

Solving gives $c_1 = 1$, $x_0 = -\frac{1}{\sqrt{3}}$, $x_1 = \frac{1}{\sqrt{3}}$. Hence, $\int_{-1}^1 f(x) dx = f\left(-\frac{1}{\sqrt{3}}\right) + f\left(\frac{1}{\sqrt{3}}\right)$ has highest degree of precision of 4, by testing up to $f(x) = x^4$. $\square$

Example. Find constants c_0, c_1, x_0, x_1 so that

$$\int_{-1}^1 f(x) dx = c_0 f(x_0) + c_1 f(x_1)$$

has the highest degree of precision.

Solution. By a similar method as above, we obtain

$$\int_{-1}^1 f(x) dx = f\left(-\frac{1}{\sqrt{3}}\right) + f\left(\frac{1}{\sqrt{3}}\right),$$

where the highest degree of precision is 3.

Here x_0, x_1 are called integration points/ Gaussian points, which are used to construct integration formulas with the highest possible degree. $\square$

4.4 Composite integration rules

The numerical result of Newton-Cotes formula does not always tend to the exact integral as $n \rightarrow \infty$, i.e., the numerical error may not tend to 0 (consider Runge's function).

One approach is to integrate using Chebyshev nodes, to get a better numerical result. Set $x_0, x_1, \dots, x_n$ to be the Chebyshev nodes. Write down the Lagrange basis polynomials

$$L_i(x) = \prod_{\substack{j=0 \\ j \neq i}}^n \frac{x - x_j}{x_i - x_j}.$$

Then integrate the basis polynomials:

$$w_i = \int_a^b L_i(x) dx.$$

Hence, the integration formula is

$$\int_a^b f(x) dx \approx \sum_{i=0}^n w_i f(x_i).$$

Composite trapezoidal rule

Another approach is to use **piecewise linear interpolation**. We use the area of the shaded region to approximate the integral.

[diagram]

In each subinterval $[x_{k-1}, x_k]$, $k = 1, \dots, n$, the shaded region is a trapezoid. The area of the trapezoid is $\frac{x_k - x_{k-1}}{2} [f(x_{k-1}) + f(x_k)]$. Thus, the total area of the shaded region is

$$\sum_{k=1}^n \frac{x_k - x_{k-1}}{2} [f(x_{k-1}) + f(x_k)].$$

Hence,

$$\int_a^b f(x) dx \approx \sum_{k=1}^n \frac{x_k - x_{k-1}}{2} [f(x_{k-1}) + f(x_k)].$$

In particular, if the interpolating nodes are equally spaced by $h = \frac{b-a}{n}$, then the above formula becomes

$$\begin{aligned} \int_a^b f(x) dx &\approx \frac{h}{2} \sum_{k=1}^n [f(x_{k-1}) + f(x_k)] \\ &= h \left[\frac{1}{2} f(x_0) + \sum_{k=1}^{n-1} f(x_k) + \frac{1}{2} f(x_n) \right]. \end{aligned}$$

This is known as the **composite trapezoidal rule**.

Another point of view is to write the integral on $[a, b]$ as the sum of integrals on subintervals:

$$\int_a^b f(x) dx = \sum_{k=1}^n \int_{x_{k-1}}^{x_k} f(x) dx.$$

For the integral on each subinterval, apply the trapezoidal rule:

$$\sum_{k=1}^n \int_{x_{k-1}}^{x_k} f(x) dx \approx \sum_{k=1}^n \frac{x_k - x_{k-1}}{2} [f(x_{k-1}) + f(x_k)].$$

Finally, sum up the approximations:

$$\int_a^b f(x) dx \approx \sum_{k=1}^n \frac{x_k - x_{k-1}}{2} [f(x_{k-1}) + f(x_k)].$$

Using error term of trapezoidal rule, the error term of composite trapezoidal rule is

$$\begin{aligned}
 E &= \int_a^b f(x) dx - \sum_{k=1}^n \frac{x_k - x_{k-1}}{2} [f(x_{k-1}) + f(x_k)] \\
 &= \sum_{k=1}^n \int_{x_{k-1}}^{x_k} f(x) dx - \sum_{k=1}^n \frac{x_k - x_{k-1}}{2} [f(x_{k-1}) + f(x_k)] \\
 &= \sum_{k=1}^n \left(\int_{x_{k-1}}^{x_k} f(x) dx - \frac{x_k - x_{k-1}}{2} [f(x_{k-1}) + f(x_k)] \right) \\
 &= -\frac{1}{12} \sum_{k=1}^n (x_k - x_{k-1})^3 f''(\xi_k).
 \end{aligned}$$

If the nodes are equally spaced, i.e., $x_k - x_{k-1} = h = \frac{b-a}{n}$ for all $k = 1, \dots, n$, then

$$E = -\frac{b-a}{12} h^2 \cdot \frac{1}{n} \sum_{k=1}^n f''(\xi_k) = -\frac{b-a}{12} h^2 f''(\xi)$$

by the intermediate value theorem. Hence, the absolute error is bounded by

$$|E| \leq \frac{b-a}{12} h^2 \max_{\xi \in [a,b]} |f''(\xi)| = \frac{(b-a)^3}{12n^2} \max_{\xi \in [a,b]} |f''(\xi)|.$$

Order of convergence: 2

As $n \rightarrow \infty$, the absolute error decreases with rate of decay n^{-2} , and thus tends to 0. For example, when h is reduced by a half, the right-hand side reduces to a quarter.

With an explicit upper bound for the error term, we can control the numerical error within a certain threshold.

Example. We would like to integrate $f(x) = \arctan x$ over $[0, 1]$ using the composite trapezoidal rule. How to choose n such that the numerical error is less than 10^{-8} ? Since the numerical error is bounded by

$$e_n \leq \frac{(1-0)^3}{12n^2} \max_{\xi \in [0,1]} |f''(\xi)| = \frac{1}{12n^2} \max_{\xi \in [0,1]} \frac{2\xi}{(1+\xi^2)^2},$$

it suffices to choose n such that

$$\frac{1}{12n^2} \max_{\xi \in [0,1]} \frac{2\xi}{(1+\xi^2)^2} < 10^{-8}.$$

Thus, we want to find an upper bound for

$$\max_{\xi \in [0,1]} \frac{2\xi}{(1+\xi^2)^2}.$$

Let $g(\xi) = \frac{\xi}{(1+\xi^2)^2}$. Then $g'(\xi) = \frac{1-3\xi^2}{(1+\xi^2)^3}$, so $g(\xi)$ has one critical point $\xi = \frac{1}{\sqrt{3}}$ in $[0, 1]$, which is a maximum

$g\left(\frac{1}{\sqrt{3}}\right) = \frac{3\sqrt{3}}{16}$. Hence, $n > \sqrt{\frac{10^8 \cdot 3\sqrt{3}}{6 \cdot 16}} \approx 2326.5$.

Alternatively, we can obtain a crude bound as follows:

$$\max_{\xi \in [0,1]} \left| \frac{\xi}{(1+\xi^2)^2} \right| \leq \max_{\xi \in [0,1]} |\xi| \cdot \max_{\xi \in [0,1]} \left| \frac{1}{(1+\xi^2)^2} \right| = 1 \cdot \frac{1}{(1+0^2)^2} = 1$$

where we used the fact that ξ and $\frac{1}{(1+\xi^2)^2}$ are monotone functions.

Composite Simpson's rule

The key idea is the same: Divide the interval into a number of subintervals. On each subinterval, apply Simpson's rule to approximate the integral. Sum up the approximations of integrals on all the subintervals.

Remark. The number of nodes must be odd (n must be even), since we need to use the midpoint of each subinterval.

The subintervals are $[x_0, x_2], [x_2, x_4], \dots, [x_{n-2}, x_n]$. Over each subinterval, approximate the integral using Simpson's rule:

$$\begin{aligned}\int_{x_0}^{x_2} f(x) dx &\approx \frac{x_2 - x_0}{6} [f(x_0) + 4f(x_1) + f(x_2)] \\ &= \frac{h}{3} [f(x_0) + 4f(x_1) + f(x_2)], \\ \int_{x_2}^{x_4} f(x) dx &\approx \frac{h}{3} [f(x_2) + 4f(x_3) + f(x_4)], \\ &\vdots \\ \int_{x_{n-2}}^{x_n} f(x) dx &\approx \frac{h}{3} [f(x_{n-2}) + 4f(x_{n-1}) + f(x_n)].\end{aligned}$$

Hence, the integral is approximated by

$$\begin{aligned}\int_a^b f(x) dx &= \int_{x_0}^{x_2} f(x) dx + \int_{x_2}^{x_4} f(x) dx + \dots + \int_{x_{n-2}}^{x_n} f(x) dx \\ &\approx \frac{h}{3} [f(x_0) + 4f(x_1) + f(x_2)] + \frac{h}{3} [f(x_2) + 4f(x_3) + f(x_4)] + \dots + \frac{h}{3} [f(x_{n-2}) + 4f(x_{n-1}) + f(x_n)] \\ &= \frac{h}{3} \left(f(a) + f(b) + 2 \sum_{k=1}^{n/2-1} f(x_{2k}) + 4 \sum_{k=1}^{n/2} f(x_{2k-1}) \right).\end{aligned}$$

Numerical error:

$$\int_a^b f(x) dx = \frac{h}{3} \left(f(a) + f(b) + 2 \sum_{k=1}^{n/2-1} f(x_{2k}) + 4 \sum_{k=1}^{n/2} f(x_{2k-1}) \right) - \frac{b-a}{180} h^4 f^{(4)}(\xi).$$

Order of convergence: 4

Example. Determine the smallest n required to approximate

$$I = \int_1^2 x \ln x dx$$

to within 10^{-4} . Use

- (1) composite trapezoidal rule;
- (2) composite Simpson's rule.

We first compute the required derivatives of f :

$$f(x) = x \ln x, \quad f'(x) = \ln x + 1, \quad f''(x) = \frac{1}{x}, \quad f^{(3)}(x) = -\frac{1}{x^2}, \quad f^{(4)}(x) = \frac{2}{x^3}.$$

- (1) For composite trapezoidal rule, we have

$$\text{error} \leq \frac{b-a}{12} h^2 \max_{\xi \in [a,b]} |f''(\xi)| = \frac{1}{12n^2}.$$

In order to have error $\leq 10^{-4}$, we only need $n \geq 29 > \sqrt{10^4/12}$.

- (2) For composite Simpson's rule, we have

$$\text{error} \leq \frac{b-a}{180n^4} \max_{\xi \in [a,b]} |f^{(4)}(\xi)| = \frac{1}{90n^4}.$$

In order to have error $\leq 10^{-4}$, we only need $n \geq 6 > \sqrt[4]{10^4/90}$.

But composite Simpson's rule requires n to be even! Hence, we can take $n = 6$.

A Tutorial Problems

A.1 Tutorial 1

Problem A.1.1. Convert the following binary numbers to decimal numbers:

- (1) $(110001)_2$
- (2) $(0.1001)_2$
- (3) $(110.0101)_2$
- (4) $(111111111111)_2$

Solution.

- (1) $(110001)_2 = 2^5 + 2^4 + 1 = 49$
- (2) $(0.1001)_2 = 2^{-1} + 2^{-4} = 0.5625$
- (3) $(110.0101)_2 = 2^2 + 2^1 + 2^{-2} + 2^{-4} = 6.3125$
- (4) $(111111111111)_2 = 2^{13} + \dots + 2^1 + 1 = 2^{14} - 1 = 16383$
 Alternatively, $(111111111111)_2 = (10000000000000)_2 - 1 = 2^{14} - 1 = 16383$

□

Problem A.1.2. True or False question: If true, provide a proof; if false, provide a counterexample.

- (1) $(x \otimes y) \otimes z = x \otimes (y \times z)$
- (2) $x \oplus x = 2 \otimes x$
- (3) $x \oplus y \ominus y \ominus x = 0$
- (4) $x \otimes x \oplus x \oplus x = 1$

Solution.

- (1) False. Consider the dangerous operation of multiplying two small numbers:

$$\frac{GmM}{r^2}$$

where G and m are very small.

- (2) True. By definition of float number operations,

$$\begin{aligned} x \oplus x &:= fl(fl(x) + fl(x)) \\ &= fl(2 \times fl(x)) \\ &= fl(fl(2) \times fl(x)) \\ &=: 2 \otimes x. \end{aligned}$$

- (3) False. Consider the dangerous operation of adding a small number and a large number, which may lead to round-off errors. From the lecture notes, a counterexample is

$$\begin{aligned} &7 \oplus 16777236 \ominus 16777236 \ominus 7 \\ &= 16777244 \ominus 16777236 \ominus 7 \\ &= 8 \ominus 7 = 1 \neq 0. \end{aligned}$$

- (4) False. Consider the dangerous operation of multiplying two small numbers, which might be smaller than the lower bound of float numbers. A counterexample is

$$\begin{aligned} & 10^{-23} \otimes 10^{-23} \oplus 10^{-23} \oplus 10^{-23} \\ &= 0 \oplus 10^{-23} \oplus 10^{-23} \\ &= 0 \oplus 10^{-23} = 0 \neq 1 \end{aligned}$$

□

Problem A.1.3. Compute $\mathbf{x}^T A \mathbf{x}$ for the symmetric matrix $A = (a_{ij})_{n \times n}$ and the vector $\mathbf{x} = (x_1, \dots, x_n)^T$. Solve the problem by pseudocodes and count the number of operations. Does symmetry of A help decrease time complexity? What if A is not symmetric?

Solution.

Method 1: Decompose the calculation into two steps: First compute $\mathbf{y} = A\mathbf{x}$, then compute $\mathbf{x}^T \mathbf{y}$.

- (1) Let $\mathbf{y} = (y_1, \dots, y_n)^T$. For each y_i ,

$$y_i = \sum_{j=1}^n a_{ij} x_j.$$

- (2) $\mathbf{x}^T \mathbf{y}$ is the dot product

$$\mathbf{x}^T \mathbf{y} = \mathbf{x} \cdot \mathbf{y} = \sum_{i=1}^n x_i y_i.$$

Compute $\mathbf{x}^T A \mathbf{x}$

```

1: for  $i = 1, \dots, n$  do
2:    $y_i \leftarrow a_{i1} x_1$ 
3:   for  $j = 2, \dots, n$  do
4:      $y_i \leftarrow y_i + a_{ij} x_j$ 
5:  $s \leftarrow x_1 y_1$ 
6: for  $j = 2, \dots, n$  do
7:    $s \leftarrow s + x_j y_j$ 
8: return  $s$ 

```

The number of additions is

$$\sum_{i=1}^n \sum_{j=2}^n 1 + \sum_{j=2}^n 1 = n^2 - 1.$$

The number of multiplications is

$$\sum_{i=1}^n \left(1 + \sum_{j=2}^n 1 \right) + 1 + \sum_{j=2}^n 1 = n^2 + n.$$

Since A is symmetric, $a_{ij} = a_{ji}$. Intuitively, this means we could half the number of operations. However, our algorithm above does not utilise this fact!

Method 2: Consider the quadratic form

$$\mathbf{x}^T A \mathbf{x} = \sum_{i=1}^n \sum_{j=1}^n x_i a_{ij} x_j.$$

Note that the matrix A consists of three regions: where $i < j$, $i = j$, and $i > j$. Accordingly, we split the above

summation according to index, and make use of the symmetric property of A :

$$\begin{aligned}
 \mathbf{x}^T A \mathbf{x} &= \sum_{i < j} x_i a_{ij} x_j + \sum_{i=j} x_i a_{ij} x_j + \sum_{i > j} x_i a_{ij} x_j \\
 &= \sum_{j < i} x_j a_{ji} x_i + \sum_{i=j} x_i a_{ij} x_j + \sum_{i > j} x_i a_{ij} x_j \\
 &= \sum_{i > j} x_i a_{ij} x_j + \sum_{i=j} x_i a_{ij} x_j + \sum_{i > j} x_i a_{ij} x_j \\
 &= \sum_{i=j} x_i a_{ij} x_j + 2 \sum_{i > j} x_i a_{ij} x_j \\
 &= \sum_{i=1}^n x_i a_{ii} x_i + 2 \sum_{i=1}^n \sum_{j=1}^{i-1} x_i a_{ij} x_j \\
 &= \sum_{i=1}^n x_i \left(a_{ii} x_i + 2 \sum_{j=1}^{i-1} a_{ij} x_j \right) \\
 &= 2 \sum_{i=1}^n x_i \left(\frac{1}{2} a_{ii} x_i + \sum_{j=1}^{i-1} a_{ij} x_j \right)
 \end{aligned}$$

Compute $\mathbf{x}^T A \mathbf{x}$

```

1: for  $i = 1, \dots, n$  do
2:    $y_i \leftarrow a_{ii} x_i / 2$ 
3:   for  $j = 1, \dots, i-1$  do
4:      $y_i \leftarrow y_i + a_{ij} x_j$ 
5:  $s \leftarrow x_1 y_1$ 
6: for  $j = 2, \dots, n$  do
7:    $s \leftarrow s + x_j y_j$ 
8: return  $2s$ 

```

$\triangleright$ Compute each $y_i = \frac{1}{2} a_{ii} x_i + \sum_{j=1}^{i-1} a_{ij} x_j$

$\triangleright$ Compute $\sum_{i=1}^n x_i y_i$

The number of additions is

$$\begin{aligned}
 \sum_{i=1}^n \sum_{j=1}^{i-1} 1 + \sum_{j=2}^n 1 &= \sum_{i=1}^n (i-1) + (n-1) \\
 &= \frac{n(n-1)}{2} + (n-1) \\
 &= \frac{(n-1)(n-2)}{2}.
 \end{aligned}$$

The number of multiplications is

$$\begin{aligned}
 \sum_{i=1}^n \left(2 + \sum_{j=1}^{i-1} 1 \right) + 1 + \sum_{j=2}^n 1 + 1 &= \sum_{i=1}^n (i+1) + (n+1) \\
 &= \frac{n^2 + 5n + 2}{2}.
 \end{aligned}$$

□

Problem A.1.4. Write down an $O(N)$ algorithm to compute

$$I_0(x) \approx \sum_{k=0}^N \frac{(x^2/4)^k}{(k!)^2}.$$

Could we minimize the number of divisions?

Solution. The idea is to relate the k -th term to the $(k + 1)$ -th term using a recursion relation, instead of calculating each term from scratch. Define

$$s_k = \frac{(x^2/4)^k}{(k!)^2}.$$

Then for $k \geq 0$,

$$\frac{s_{k+1}}{s_k} = \frac{x^2}{4(k+1)^2}.$$

Hence, the recursion formula for s_k is

$$s_{k+1} = \frac{x^2}{4(k+1)^2} s_k, \quad s_0 = 1.$$

Approximate $I_0(x)$

```

1:  $I_0 \leftarrow 0$ 
2:  $s \leftarrow 1$ 
3: for  $k = 0, \dots, N$  do
4:    $I_0 \leftarrow I_0 + s$ 
5:    $s \leftarrow \frac{x^2}{4(k+1)^2} s$ 
6: return  $I_0$ 

```

Some refinements: Note that we can compute and store the common ratio $r = \frac{x^2}{4}$. Also, we can set the first term as 1 before entering the for loop, and start the loop at $k = 1$.

Approximate $I_0(x)$

```

1:  $I_0 \leftarrow 1$ 
2:  $s \leftarrow 1, r \leftarrow \frac{x^2}{4}$ 
3: for  $k = 1, \dots, N$  do
4:    $s \leftarrow \frac{r s}{k^2}$ 
5:    $I_0 \leftarrow I_0 + s$ 
6: return  $I_0$ 

```

□

Problem A.1.5. Approximate

$$g(x) = \sqrt{\pi} - \sum_{k=0}^{\infty} \frac{(-1)^k x^{k+1/2}}{k!(k+1/2)}$$

such that the absolute truncation error is less than ε .

Solution. Since $g(x)$ is an infinite sum, we approximate $g(x)$ by truncating it to a finite sum:

$$g_N(x) = \sqrt{\pi} - \sum_{k=0}^N \frac{(-1)^k x^{k+1/2}}{k!(k+1/2)}.$$

Let $s_k = \frac{(-1)^k x^{k+1/2}}{k!}$. Thus,

$$g_N(x) = \sqrt{\pi} - \sum_{k=0}^N \frac{s_k}{k+1/2}.$$

Then s_k satisfies the recurrence relation

$$s_{k+1} = -\frac{x}{k+1}s_k, \quad s_0 = \sqrt{x}.$$

When $N+2 > x$, the absolute error can be bounded by

$$\begin{aligned} |g_N(x) - g(x)| &= \left| \sum_{k=N+1}^{\infty} \frac{(-1)^k x^{k+1/2}}{k!(k+1/2)} \right| \\ &\leq \sum_{k=N+1}^{\infty} \left| \frac{(-1)^k x^{k+1/2}}{k!(k+1/2)} \right| && \text{[triangle inequality]} \\ &= \sum_{k=N+1}^{\infty} \frac{x^{k+1/2}}{k!(k+1/2)} = x^{N+3/2} \sum_{k=N+1}^{\infty} \frac{x^{k-N-1}}{k!(k+1/2)} \\ &\leq \frac{x^{N+3/2}}{(N+1)!(N+3/2)} \sum_{k=N+1}^{\infty} x^{k-N-1} \\ &= \frac{x^{N+3/2}}{(N+1)!(N+3/2)} \left(1 - \frac{x}{N+2}\right)^{-1} \\ &= \frac{|s_{N+1}|}{N+3/2} \left(1 - \frac{x}{N+2}\right)^{-1}. \end{aligned}$$

The idea is as follows:

- (1) Add a term.
- (2) Compute the error.
- (3) If the error is still large, return to Step 1.
- (4) Else, stop and return the result.

Approximate $g(x)$

```

1:  $g \leftarrow \sqrt{\pi}, s \leftarrow \sqrt{x}$ 
2:  $k \leftarrow 0, \text{err} \leftarrow +\infty$ 
3: while  $k+1 \leq x$  or  $\text{err} \geq \varepsilon$  do
4:    $g \leftarrow g - \frac{s}{k+1/2}$ 
5:    $k \leftarrow k+1$ 
6:    $s \leftarrow -\frac{xs}{k}$ 
7:    $\text{err} \leftarrow \frac{|s|}{k+1/2} \left(1 - \frac{x}{k+1}\right)^{-1}$ 
8: return  $g$ 
```

□

Problem A.1.6. Multiply an upper triangular matrix and a lower Hessenberg matrix. Solve the problem by pseudocodes and count the number of operations.

Solution. Let $A \in \mathbb{R}^{n \times n}$ be upper triangular, $B \in \mathbb{R}^{n \times n}$ be lower Hessenberg. We first find the zero entries in the matrix product $C = AB$:

$$c_{ij} := \sum_{k=1}^n a_{ik}b_{kj} = 0,$$

i.e., for each $k = 1, \dots, n$, one has $a_{ik}b_{kj} = 0$. In the i -th row of A , the zero entries are $a_{i1}, \dots, a_{i,i-1}$; in the j -th row of B , the zero entries are $b_{1j}, \dots, b_{j-2,j}$. Hence, for each $k = 1, \dots, n$, we need that if $i \leq k$, then $k < j-1$. However, there does not exist i and j satisfying this condition, which means there are no trivially zero c_{ij} .

Since the first few a_{ik} and b_{kj} are zero, we can simplify the summation.

- Since $a_{ik} \neq 0$ only if $i \leq k$, the lower bound of k can be changed to i .
- Since $b_{kj} \neq 0$ only if $k \geq j - 1$, the lower bound of k can be changed to $\max(1, j - 1)$.

$$c_{ij} = \sum_{k=\max(i,1,j-1)}^n a_{ik} b_{kj} = \sum_{k=\max(i,j-1)}^n a_{ik} b_{kj}.$$

Multiply the upper triangular matrix $A = (a_{ij})_{n \times n}$ and the lower Hessenberg matrix $B = (b_{ij})_{n \times n}$

```

1: for  $i = 1, \dots, n$  do
2:   for  $j = 1, \dots, n$  do
3:      $l \leftarrow \max(i, j - 1)$ ,  $c_{ij} \leftarrow a_{il} b_{lj}$ 
4:     for  $k = l + 1, \dots, n$  do
5:        $c_{ij} \leftarrow c_{ij} + a_{ik} b_{kj}$ 
6: return  $C = (c_{ij})_{n \times n}$ 

```

The number of multiplications is

$$\sum_{i=1}^n \sum_{j=1}^n \left(1 + \sum_{k=\max(i,j-1)+1}^n 1 \right) = \sum_{i=1}^n \sum_{j=1}^n [n - \max(i, j - 1) + 1].$$

For $j = 1, \dots, i$, we have $i > j - 1$. For $j = i + 1, \dots, n$, we have $i \leq j - 1$. Thus, we split the summation into two:

$$\sum_{i=1}^n \left[\sum_{j=1}^i (n - i + 1) + \sum_{j=i+1}^n (n - j + 2) \right]$$

For the first term,

$$\begin{aligned}
\sum_{i=1}^n \sum_{j=1}^i (n - i + 1) &= \sum_{i=1}^n i(n - i + 1) \\
&= \sum_{i=1}^n i(n + 1) - \sum_{i=1}^n i^2 \\
&= \frac{n(n + 1)}{2}(n + 1) - \frac{n(n + 1)(2n + 1)}{6} \\
&= \frac{n(n + 1)(n + 2)}{6}.
\end{aligned}$$

For the second term,

$$\begin{aligned}
\sum_{i=1}^n \sum_{j=i+1}^n (n - j + 2) &= \sum_{i=1}^n [(n - i + 1) + (n - i) + \dots + 2 + 1] - n \\
&= \sum_{i=1}^n \frac{(n - i + 1)(n - i + 2)}{2} - n \\
&= \frac{1}{2} \sum_{i=1}^n (n - i + 1)^2 + \frac{1}{2} \sum_{i=1}^n (n - i + 1) - n \\
&= \frac{1}{2} [n^2 + (n - 1)^2 + \dots + 2^2 + 1^2] + \frac{1}{2} [n + (n - 1) + \dots + 2 + 1] - n \\
&= \frac{1}{2} \frac{n(n + 1)(2n + 1)}{6} + \frac{1}{2} \frac{n(n + 1)}{2} - n \\
&= \frac{n(n + 1)(n + 2)}{6} - n.
\end{aligned}$$

Hence, the total number of multiplications is

$$\frac{n(n+1)(n+2)}{6} + \left(\frac{n(n+1)(n+2)}{6} - n \right) = \frac{n(n^2 + 3n - 1)}{3}.$$

The number of additions is

$$\begin{aligned} \text{number of multiplications} - n^2 &= \frac{n(n^2 + 3n - 1)}{3} - n^2 \\ &= \frac{(n-1)n(n+1)}{3}. \end{aligned}$$

□

A.2 Tutorial 2

Problem A.2.1. Show that

$$\begin{aligned} & \left| H_1(x) - \frac{2}{\pi} \sum_{m=0}^N \frac{(-1)^m x^{2(m+1)}}{(2m+1)!!(2m+3)!!} \right| \\ & \leq \frac{2}{\pi} \frac{|x|^2(N+2)}{(2N+3)!!(2N+5)!!} \left(1 - \frac{|x|^2}{(2N+5)(2N+7)} \right)^{-1}, \end{aligned}$$

where

$$H_1(x) = \frac{2}{\pi} \sum_{m=0}^{\infty} \frac{(-1)^m x^{2(m+1)}}{(2m+1)!!(2m+3)!!}.$$

Solution. By triangle inequality,

$$\begin{aligned} \text{err} &:= \left| H_1(x) - \frac{2}{\pi} \sum_{m=0}^N \frac{(-1)^m x^{2(m+1)}}{(2m+1)!!(2m+3)!!} \right| \\ &= \frac{2}{\pi} \left| \sum_{m=N+1}^{\infty} \frac{(-1)^m x^{2(m+1)}}{(2m+1)!!(2m+3)!!} \right| \\ &\leq \frac{2}{\pi} \sum_{m=N+1}^{\infty} \frac{|(-1)^m x^{2(m+1)}|}{(2m+1)!!(2m+3)!!} \\ &= \frac{2}{\pi} \sum_{m=N+1}^{\infty} \frac{|x|^{2(m+1)}}{(2m+1)!!(2m+3)!!}. \end{aligned}$$

Note that when $m \geq N+1$,

$$\begin{aligned} (2m+1)!! &= (2m+1)(2m-1) \cdots (2N+5)(2N+3) \cdots 3 \cdot 1 \\ &\geq (2N+5)^{m-N-1} (2N+3)!!, \\ (2m+3)!! &= (2m+3)(2m+1) \cdots (2N+7)(2N+5) \cdots 3 \cdot 1 \\ &\geq (2N+7)^{m-N-1} (2N+5)!!. \end{aligned}$$

We then have

$$\begin{aligned} err &\leq \frac{2}{\pi} \frac{1}{(2N+3)!!(2N+5)!!} \sum_{m=N+1}^{\infty} \frac{|x|^{2(m+1)}}{[(2N+5)(2N+7)]^{m-N-1}} \\ &= \frac{2}{\pi} \frac{|x|^{2(N+2)}}{(2N+3)!!(2N+5)!!} \left(1 - \frac{|x|^2}{(2N+5)(2N+7)} \right)^{-1}. \end{aligned}$$

□

Problem A.2.2. Use Gaussian elimination without pivoting or row swapping to find solutions for

$$\left(\begin{array}{cccc|c} 2 & 0 & 1 & 2 & -3 \\ 1 & 1 & 0 & 2 & -9 \\ 2 & -1 & 3 & 1 & 9 \\ 3 & -1 & 4 & 3 & 5 \end{array} \right).$$

Solution. Perform Gaussian elimination. First eliminate the first variable:

$$\begin{aligned} (2) &\leftarrow (2) - \frac{1}{2} \times (1) \\ (3) &\leftarrow (3) - (1) \\ (4) &\leftarrow (4) - \frac{3}{2} \times (1) \end{aligned}$$

Then eliminate the second variable:

$$\begin{aligned} (3) &\leftarrow (3) + (2) \\ (4) &\leftarrow (4) + (2) \end{aligned}$$

Finally, eliminate the third variable:

$$(4) \leftarrow (4) - \frac{4}{3} \times (3)$$

Backward substitution gives

$$x_4 = -4, \quad x_3 = 3, \quad x_2 = -2, \quad x_1 = 1.$$

□

Problem A.2.3. The following augmented matrix is the result of Gaussian elimination without pivoting. Try to find out the solutions and the original augmented matrix.

$$\left(\begin{array}{cccc|c} 1 & 1 & 0 & 1 & 2 \\ 2 & -1 & -1 & -1 & -3 \\ 4 & -5 & 3 & 3 & 7 \\ 3 & -4 & 3 & 1 & 4 \end{array} \right)$$

Solution. Row operation:

$$(4) \leftarrow (4) - \frac{3}{3} \times (3)$$

Inverse operation:

$$\begin{aligned} (4) &\leftarrow (4) + \frac{3}{3} \times (3) \\ \left(\begin{array}{cccc|c} 1 & 1 & 0 & 1 & 2 \\ 2 & -1 & -1 & -1 & -3 \\ 4 & -5 & 3 & 3 & 7 \\ 3 & -4 & 3 & 4 & 11 \end{array} \right) \end{aligned}$$

Row operation:

$$(3) \leftarrow (3) - \frac{-5}{-1} \times (2), \quad (4) \leftarrow (4) - \frac{-4}{-1} \times (2)$$

Inverse operation:

$$(3) \leftarrow (3) + \frac{-5}{-1} \times (2), \quad (4) \leftarrow (4) + \frac{-4}{-1} \times (2)$$

$$\left(\begin{array}{cccc|c} 1 & 1 & 0 & 1 & 2 \\ 2 & -1 & -1 & -1 & -3 \\ 4 & -5 & -2 & -2 & -8 \\ 3 & -4 & -1 & 0 & -1 \end{array} \right)$$

Row operation:

$$(2) \leftarrow (2) - \frac{2}{1} \times (1), \quad (3) \leftarrow (3) - \frac{4}{1} \times (1), \quad (4) \leftarrow (4) - \frac{3}{1} \times (1)$$

Inverse operation:

$$(2) \leftarrow (2) + \frac{2}{1} \times (1), \quad (3) \leftarrow (3) + \frac{4}{1} \times (1), \quad (4) \leftarrow (4) + \frac{3}{1} \times (1)$$

$$\left(\begin{array}{cccc|c} 1 & 1 & 0 & 1 & 2 \\ 2 & 1 & -1 & 1 & 1 \\ 4 & -1 & -2 & 2 & 0 \\ 3 & -1 & -1 & 3 & 5 \end{array} \right)$$

□

Problem A.2.4. Use Gaussian elimination with pivoting to solve the following matrix. What problems do you encounter?

$$\left(\begin{array}{ccccc} 0 & 1 & 2 & 5 & 7 \\ 1 & 2 & 4 & 2 & 8 \\ 3 & 3 & 6 & 7 & 9 \\ 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 2 & 0 & 3 \end{array} \right)$$

Solution.

$$\begin{array}{l} 1: \\ 2: \\ 3: \\ 4: \\ 5: \end{array} \left(\begin{array}{ccccc} 0 & 1 & 2 & 5 & 7 \\ 1 & 2 & 4 & 2 & 8 \\ 3 & 3 & 6 & 7 & 9 \\ 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 2 & 0 & 3 \end{array} \right)$$

Since the first diagonal element is 0, we swap $(1) \leftrightarrow (2)$:

$$\begin{array}{l} 2: \\ 1: \\ 3: \\ 4: \\ 5: \end{array} \left(\begin{array}{ccccc} 0 & 1 & 2 & 5 & 7 \\ 1 & 2 & 4 & 2 & 8 \\ 3 & 3 & 6 & 7 & 9 \\ 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 2 & 0 & 3 \end{array} \right)$$

Then eliminate the first variable:

$$(2) \leftarrow (2) - 0 \times (1)$$

$$(3) \leftarrow (3) - 3 \times (1)$$

$$(4) \leftarrow (4) - (1)$$

$$(5) \leftarrow (5) - 0 \times (1)$$

$$\begin{array}{l} 2: \\ 1: \\ 3: \\ 4: \\ 5: \end{array} \begin{pmatrix} 0 & 1 & 2 & 5 & 7 \\ 1 & 2 & 4 & 2 & 8 \\ 3 & -3 & -6 & 1 & -15 \\ 1 & -1 & -3 & -1 & -7 \\ 0 & 0 & 2 & 0 & 3 \end{pmatrix}$$

Since the second diagonal element is non-zero, no row swapping is needed. Thus eliminate the second variable:

$$(3) \leftarrow (3) + 3 \times (2)$$

$$(4) \leftarrow (4) + (2)$$

$$(5) \leftarrow (5) - 0 \times (2)$$

$$\begin{array}{l} 2: \\ 1: \\ 3: \\ 4: \\ 5: \end{array} \begin{pmatrix} 0 & 1 & 2 & 5 & 7 \\ 1 & 2 & 4 & 2 & 8 \\ 3 & -3 & 0 & 16 & 6 \\ 1 & -1 & -1 & 4 & 0 \\ 0 & 0 & 2 & 0 & 3 \end{pmatrix}$$

Since the third diagonal element is 0, we swap (3) $\leftrightarrow$ (4):

$$\begin{array}{l} 2: \\ 1: \\ 4: \\ 3: \\ 5: \end{array} \begin{pmatrix} 0 & 1 & 2 & 5 & 7 \\ 1 & 2 & 4 & 2 & 8 \\ 3 & -3 & 0 & 16 & 6 \\ 1 & -1 & -1 & 4 & 0 \\ 0 & 0 & 2 & 0 & 3 \end{pmatrix}$$

Then eliminate the third variable:

$$(4) \leftarrow (4) - 0 \times (3)$$

$$(5) \leftarrow (5) + 2 \times (3)$$

$$\begin{array}{l} 2: \\ 1: \\ 4: \\ 3: \\ 5: \end{array} \begin{pmatrix} 0 & 1 & 2 & 5 & 7 \\ 1 & 2 & 4 & 2 & 8 \\ 3 & -3 & 0 & 16 & 6 \\ 1 & -1 & -1 & 4 & 0 \\ 0 & 0 & 2 & 8 & 3 \end{pmatrix}$$

Finally, eliminate the fourth variable:

$$(5) \leftarrow (5) - \frac{1}{2} \times (4)$$

$$\begin{array}{l} 2: \\ 1: \\ 4: \\ 3: \\ 5: \end{array} \begin{pmatrix} 0 & 1 & 2 & 5 & 7 \\ 1 & 2 & 4 & 2 & 8 \\ 3 & -3 & 0 & 16 & 6 \\ 1 & -1 & -1 & 4 & 0 \\ 0 & 0 & 2 & 8 & 0 \end{pmatrix}$$

Note that the matrix is singular! □

Problem A.2.5. When solving linear systems with a lower Hessenberg coefficient matrix, if no row/column swapping is needed, then the lower Hessenberg structure is always preserved after eliminating $x_1, x_2, \dots, x_{n-1}$, which helps much for backward substitution computations (how?). However, swapping may violate this property. In order to overcome this difficulty, we could actually do Gaussian elimination starting from eliminating x_n and

then x_{n-1} , and Think about what would happen in this situation.

$$\begin{pmatrix} a_{11} & a_{12} & & & \\ a_{21} & a_{22} & a_{23} & & \\ a_{31} & a_{32} & a_{33} & \ddots & \\ \vdots & \vdots & \vdots & \ddots & a_{n-1,n} \\ a_{n1} & a_{n2} & a_{n3} & \cdots & a_{nn} \end{pmatrix}$$

Solution. Elimination of the first variable:

$$a_{jk} \leftarrow a_{jk} - \frac{a_{j1}}{a_{11}} a_{1k}, \quad j = 2, \dots, n, \quad k = 2.$$

The subsystem still has a lower Hessenberg coefficient matrix! Elimination of the second variable:

$$a_{jk} \leftarrow a_{jk} - \frac{a_{j2}}{a_{22}} a_{2k}, \quad j = 3, \dots, n, \quad k = 3.$$

Elimination of the i -th variable:

$$a_{jk} \leftarrow a_{jk} - \frac{a_{ji}}{a_{ii}} a_{ik}, \quad j = i + 1, \dots, n, \quad k = i + 1.$$

After elimination,

$$\begin{pmatrix} a_{11} & a_{12} & & & \\ a_{21} & a_{22} & a_{23} & & \\ a_{31} & a_{32} & a_{33} & \ddots & \\ \vdots & \vdots & \vdots & \ddots & a_{n-1,n} \\ a_{n1} & a_{n2} & a_{n3} & \cdots & a_{nn} \end{pmatrix}$$

To do backward substitution, we solve

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 &= b_1 \\ a_{22}x_2 + a_{23}x_3 &= b_2 \\ &\vdots \\ a_{n-1,n-1}x_{n-1} + a_{n-1,n}x_n &= b_{n-1} \\ a_{nn}x_n &= b_n \end{aligned}$$

or

$$\begin{aligned} x_1 &= \frac{b_1 - a_{12}x_2}{a_{11}} \\ x_2 &= \frac{b_2 - a_{23}x_3}{a_{22}} \\ &\vdots \\ x_{n-1} &= \frac{b_{n-1} - a_{n-1,n}x_n}{a_{n-1,n-1}} \\ x_n &= \frac{b_n}{a_{nn}} \end{aligned}$$

In general, we have

$$x_n = \frac{b_n}{a_{nn}}$$

$$x_i = \frac{b_i - a_{i,i+1}x_{i+1}}{a_{ii}} \quad i = n-1, \dots, 1$$

Solve $Ax = b$ for the lower Hessenberg matrix A

```

1: for  $i = 1, \dots, n-1$  do
2:   for  $j = i+1, \dots, n$  do
3:      $m \leftarrow \frac{a_{ji}}{a_{ii}}$ 
4:      $a_{j,i+1} \leftarrow a_{j,i+1} - ma_{i,i+1}$ 
5:      $b_j \leftarrow b_j - mb_i$ 
6:  $x_n \leftarrow \frac{b_n}{a_{nn}}$ 
7: for  $i = n-1, \dots, 1$  do
8:    $x_i \leftarrow \frac{b_i - a_{i,i+1}x_{i+1}}{a_{ii}}$ 
9: return  $(x_1, \dots, x_n)^T$ 

```

▷ Elimination

▷ Backward substitution

What if the pivot element is zero? Row labels are needed. Can we preserve the lower Hessenberg structure? Can we keep the low computational cost?

$$\begin{array}{l}
 1 : \\
 2 : \\
 3 : \\
 \vdots \\
 n :
 \end{array}
 \left(
 \begin{array}{cccccc}
 a_{11} & a_{12} & & & & \\
 a_{21} & a_{22} & a_{23} & & & \\
 a_{31} & a_{32} & a_{33} & \ddots & & \\
 \vdots & \vdots & \vdots & \ddots & \ddots & \\
 a_{n1} & a_{n2} & a_{n3} & \cdots & a_{nn} &
 \end{array}
 \right)$$

New idea: Change the order of elimination

First step: If $a_{nn} \neq 0$, then eliminate x_n :

$$(n-1) \leftarrow (n-1) - \frac{a_{n-1,n}}{a_{nn}} \times (n)$$

If $a_{nn} = 0$, then $(n-1) \leftrightarrow (n)$.

Second step: If $a_{r_{n-1},n-1} \neq 0$, then eliminate x_{n-1} :

$$(n-2) \leftarrow (n-2) - \frac{a_{r_{n-2},n-1}}{a_{r_{n-1},n-1}} \times (n-1).$$

Otherwise, $(n-2) \leftrightarrow (n-1)$.

In general, to eliminate the i -th variable: If $a_{r_i,i} \neq 0$, then

$$a_{r_{i-1},k} \leftarrow a_{r_{i-1},k} - \frac{a_{r_{i-1},i}}{a_{r_i,i}} a_{r_i,k}, \quad k = 1, \dots, i-1.$$

If $a_{r_i,i} = 0$, then

$$(i) \leftrightarrow (i-1).$$

Solve $Ax = b$ for the lower Hessenberg matrix A

```

1: for  $i = 1, \dots, n$  do
2:    $r_i \leftarrow i$ 
3: for  $i = n, \dots, 1$  do
4:   if  $a_{r_i, i} \neq 0$  then
5:      $m \leftarrow \frac{a_{r_{i-1}, i}}{a_{r_i, i}}$ 
6:     for  $k = 1, \dots, i-1$  do
7:        $a_{r_{i-1}, k} \leftarrow a_{r_{i-1}, k} - m a_{r_i, k}$ 
8:        $b_{r_{i-1}} \leftarrow b_{r_{i-1}} - m b_{r_i}$ 
9:   else
10:     $\text{Swap } r_i \text{ and } r_{i-1}$ 

```

After elimination, we do forward substitution:

$$x_i = \frac{1}{a_{ii}} \left(b_i - \sum_{j=1}^{i-1} a_{ij} x_j \right).$$

Using rows labels,

$$x_i = \frac{1}{a_{r_i i}} \left(b_{r_i} - \sum_{j=1}^{i-1} a_{r_i j} x_j \right).$$

Solve $Ax = b$ for the lower Hessenberg matrix A

```

11:  $x_1 = \frac{b_{r_1}}{a_{r_1 1}}$ 
12: for  $i = 2, \dots, n$  do
13:    $x_i \leftarrow b_{r_i}$ 
14:   for  $j = 1, \dots, i-1$  do
15:      $x_i \leftarrow x_i - a_{r_i j} x_j$ 
16:    $x_i \leftarrow \frac{x_i}{a_{r_i i}}$ 
17: return  $(x_1, \dots, x_n)^T$ 

```

□

A.3 Tutorial 3

Problem A.3.1. Assume that the augmented matrix

$$\left(\begin{array}{cccc|c} 1 & 9/4 & -1 & 11/14 & -11/14 \\ -3 & 9/4 & 4 & 17/4 & -5/2 \\ 4 & -1 & 0 & 3 & 2 \\ 4 & 2 & -14/9 & -25/9 & 11/9 \end{array} \right)$$

is obtained by Gaussian elimination with partial pivoting which only makes necessary changes of matrix entries. Find the original augmented matrix and the solution of the linear system.

Solution. We proceed by the following steps:

- (1) Determine the row labels of the resulting augmented matrix.
- (2) Backward substitution to find the solution x_1, x_2, x_3, x_4 .
- (3) Restore the augmented matrix using inverse operations.

First, label the rows. For the first pivoting, swap rows 3 and 1.

$$\begin{array}{l} 3 : \\ 2 : \\ 1 : \\ 4 : \end{array} \left(\begin{array}{cccc|c} 1 & 9/4 & -1 & 11/14 & -11/14 \\ -3 & 9/4 & 4 & 17/4 & -5/2 \\ 4 & -1 & 0 & 3 & 2 \\ 4 & 2 & -14/9 & -25/9 & 11/9 \end{array} \right)$$

For the second pivoting, no row swaps are needed. For the third pivoting, swap rows 3 and 4.

$$\begin{array}{l} 4 : \\ 2 : \\ 1 : \\ 3 : \end{array} \left(\begin{array}{cccc|c} 1 & 9/4 & -1 & 11/14 & -11/14 \\ -3 & 9/4 & 4 & 17/4 & -5/2 \\ 4 & -1 & 0 & 3 & 2 \\ 4 & 2 & -14/9 & -25/9 & 11/9 \end{array} \right)$$

Next, perform backward substitution to find

$$x_4 = -1, \quad x_3 = 1, \quad x_2 = -1, \quad x_1 = 1.$$

Finally, restore the augmented matrix. Since (3) was used to eliminate (4), the inverse row operation is

$$(4) \leftarrow (4) + \frac{-1}{-14/9} \times (3).$$

$$\begin{array}{l} 4 : \\ 2 : \\ 1 : \\ 3 : \end{array} \left(\begin{array}{cccc|c} 1 & 9/4 & -1 & -1 & 0 \\ -3 & 9/4 & 4 & 17/4 & -5/2 \\ 4 & -1 & 0 & 3 & 2 \\ 4 & 2 & -14/9 & -25/9 & 11/9 \end{array} \right)$$

Since (2) was used to eliminate (3) and (4), the inverse row operations are

$$(3) \leftarrow (3) + \frac{2}{9/4} \times (2), \quad (4) \leftarrow (4) + \frac{9/4}{9/4} \times (2).$$

$$\begin{array}{l} 4 : \\ 2 : \\ 1 : \\ 3 : \end{array} \left(\begin{array}{cccc|c} 1 & 9/4 & 3 & 13/4 & -5/2 \\ -3 & 9/4 & 4 & 17/4 & -5/2 \\ 4 & -1 & 0 & 3 & 2 \\ 4 & 2 & 2 & 1 & -1 \end{array} \right)$$

Finally, since (1) was used to eliminate (2), (3) and (4), the inverse row operations are

$$(2) \leftarrow (2) + \frac{-3}{4} \times (1), \quad (3) \leftarrow (3) + \frac{4}{4} \times (1)$$

$$\begin{array}{l} 4 : \\ 2 : \\ 1 : \\ 3 : \end{array} \left(\begin{array}{cccc|c} 1 & 2 & 3 & 4 & -2 \\ -3 & 3 & 4 & 2 & -4 \\ 4 & -1 & 0 & 3 & 2 \\ 4 & 1 & 2 & 4 & 1 \end{array} \right)$$

□

Problem A.3.2. For a linear system with coefficient matrix

$$\begin{pmatrix} 2 & 2 & 4 \\ -1 & -2 & 0 \\ 1 & \alpha & 3 \end{pmatrix},$$

find the range of α such that no label swapping is required when solving the system with Gaussian elimination with scaled partial pivoting.

Solution. Label the rows:

$$\begin{array}{l} 1 : \begin{pmatrix} 2 & 2 & 4 \end{pmatrix} \\ 2 : \begin{pmatrix} -1 & -2 & 0 \end{pmatrix} \\ 3 : \begin{pmatrix} 1 & \alpha & 3 \end{pmatrix} \end{array}$$

First pre-compute each s_k , i.e., the largest absolute value in each row:

$$s_1 = 4, \quad s_2 = 2, \quad s_3 = \max(|\alpha|, 3).$$

When choosing the first pivot, compare

$$\frac{2}{4}, \quad \frac{1}{2}, \quad \frac{1}{\max(|\alpha|, 3)}.$$

Note that $\max(|\alpha|, 3) \leq 3$, so $\frac{1}{\max(|\alpha|, 3)} \leq \frac{1}{3} < \frac{1}{2}$. Thus, no row swapping is needed. Then perform elimination:

$$(2) \leftarrow (2) + \frac{1}{2} \times (1), \quad (3) \leftarrow (3) - \frac{1}{2} \times (1).$$

$$\begin{array}{l} 1 : \begin{pmatrix} 2 & 2 & 4 \end{pmatrix} \\ 2 : \begin{pmatrix} -1 & -2 & 0 \end{pmatrix} \\ 3 : \begin{pmatrix} \alpha - 1 & 1 & 1 \end{pmatrix} \end{array}$$

When choosing the second pivot, compare

$$\frac{1}{2}, \quad \frac{|\alpha - 1|}{\max(|\alpha|, 3)}.$$

Since no row swapping is required, we want to have

$$\frac{|\alpha - 1|}{\max(|\alpha|, 3)} \leq \frac{1}{2},$$

or

$$2|\alpha - 1| \leq \max(|\alpha|, 3).$$

To remove the absolute values and the maximum function, we notice that

- $|\alpha - 1| = \alpha - 1$ if $\alpha \geq 1$
- $|\alpha - 1| = 1 - \alpha$ if $\alpha < 1$
- $\max(|\alpha|, 3) = -\alpha$ if $\alpha < -3$
- $\max(|\alpha|, 3) = 3$ if $-3 \leq \alpha \leq 3$
- $\max(|\alpha|, 3) = \alpha$ if $\alpha \geq 3$

This gives us 4 cases:

Case 1: $\alpha < -3$. Then $2(1 - \alpha) \leq -\alpha$, so $\alpha \geq 2$. Thus, there is no solution, so we need to swap labels.

Case 2: $-3 \leq \alpha < 1$. Then $2(1 - \alpha) \leq 3$, so $\alpha \geq -\frac{1}{2}$.

Combining both conditions, when $-\frac{1}{2} \leq \alpha < 1$, no label swapping is needed.

Case 3: $1 \leq \alpha < 3$. Then $2(\alpha - 1) \leq 3$, so $\alpha \leq \frac{5}{2}$.

Combining both conditions, when $1 \leq \alpha \leq \frac{5}{2}$, no label swapping is needed.

Case 4: $\alpha \geq 3$. Then $2(\alpha - 1) \leq \alpha$, so $\alpha \leq 2$. Thus, there is no solution, so we need to swap labels.

Therefore, when $\alpha \in \left[\frac{1}{2}, \frac{5}{2}\right]$, no label swapping is needed. $\square$

Problem A.3.3. Use scaled partial pivoting to solve the linear system represented by the augmented matrix

$$\left(\begin{array}{cccc|c} 2 & 6 & 3 & 5 & -30 \\ 4 & 5 & 8 & -1 & 0 \\ 3 & 2 & -1 & -2 & 15 \\ -3 & 5 & -1 & 1 & -15 \end{array} \right)$$

Solution. Label the rows:

$$\begin{array}{l} 1 : \left(\begin{array}{cccc|c} 2 & 6 & 3 & 5 & -30 \\ 4 & 5 & 8 & -1 & 0 \\ 3 & 2 & -1 & -2 & 15 \\ -3 & 5 & -1 & 1 & -15 \end{array} \right) \\ 2 : \\ 3 : \\ 4 : \end{array}$$

We first pre-compute each s_k , i.e., the largest absolute value in each row:

$$s_1 = 6, \quad s_2 = 8, \quad s_3 = 3, \quad s_4 = 5.$$

When choosing the first pivot element, compare

$$\frac{2}{6}, \quad \frac{4}{8}, \quad \frac{3}{3}, \quad \frac{5}{5},$$

so swap rows 1 and 3.

$$\begin{array}{l} 3 : \left(\begin{array}{cccc|c} 2 & 6 & 3 & 5 & -30 \\ 4 & 5 & 8 & -1 & 0 \\ 1 & 3 & 2 & -1 & 15 \\ -3 & 5 & -1 & 1 & -15 \end{array} \right) \\ 2 : \\ 1 : \\ 4 : \end{array}$$

Perform row operations:

$$(2) \leftarrow (2) - \frac{4}{3} \times (1), \quad (3) \leftarrow (3) - \frac{2}{3} \times (1), \quad (4) \leftarrow (4) + (1)$$

$$\begin{array}{l} 3 : \left(\begin{array}{cccc|c} 2 & 14/3 & 11/3 & 19/3 & -40 \\ 4 & 7/3 & 28/3 & 5/3 & -20 \\ 1 & 3 & 2 & -1 & 15 \\ -3 & 7 & -2 & -1 & 0 \end{array} \right) \\ 2 : \\ 1 : \\ 4 : \end{array}$$

When choosing the second pivot, compare

$$\frac{14/3}{6}, \quad \frac{7/3}{8}, \quad \frac{7}{5}$$

so swap rows 2 and 4. Note that (3) uses s_1 , because s_k is fixed.

$$\begin{array}{l} 3 : \left(\begin{array}{cccc|c} 2 & 14/3 & 11/3 & 19/3 & -40 \\ 4 & 7/3 & 28/3 & 5/3 & -20 \\ 1 & 3 & 2 & -1 & 15 \\ 2 & -3 & 7 & -2 & 0 \end{array} \right) \\ 4 : \\ 1 : \\ 2 : \end{array}$$

Next, perform row operations:

$$(3) \leftarrow (3) - \frac{14/3}{7} \times (2), \quad (4) \leftarrow (4) - \frac{7/3}{7} \times (2).$$

$$\begin{array}{l} 3 : \\ 4 : \\ 1 : \\ 2 : \end{array} \left(\begin{array}{cccc|c} 2 & 14/3 & 5 & 7 & -40 \\ 4 & 7/3 & 10 & 2 & -20 \\ 3 & 2 & -1 & -2 & 15 \\ -3 & 7 & -2 & -1 & 0 \end{array} \right)$$

When choosing the third pivot, compare

$$\frac{5}{6}, \quad \frac{10}{8}$$

so swap rows 3 and 4.

$$\begin{array}{l} 4 : \\ 3 : \\ 1 : \\ 2 : \end{array} \left(\begin{array}{cccc|c} 2 & 14/3 & 5 & 7 & -40 \\ 4 & 7/3 & 10 & 2 & -20 \\ 3 & 2 & -1 & -2 & 15 \\ -3 & 7 & -2 & -1 & 0 \end{array} \right)$$

Then perform row operations:

$$(4) \leftarrow (4) - \frac{5}{10} \times (3).$$

$$\begin{array}{l} 4 : \\ 3 : \\ 1 : \\ 2 : \end{array} \left(\begin{array}{cccc|c} 2 & 14/3 & 5 & 6 & -30 \\ 4 & 7/3 & 10 & 2 & -20 \\ 3 & 2 & -1 & -2 & 15 \\ -3 & 7 & -2 & -1 & 0 \end{array} \right)$$

Finally, having the resulting augmented matrix, we can perform backward substitution to find

$$x_4 = -5, \quad x_3 = -1, \quad x_2 = -1, \quad x_1 = 2.$$

□

Problem A.3.4. Given the LU factorization of A stored in $(a_{ij})_{n \times n}$,

- (a) solve $A^T \mathbf{x} = \mathbf{b}$ for $\mathbf{b} = (b_1, \dots, b_n)^T$ and think about what the pseudocode would be like.
- (b) compute A^{-1} .

Solution.

- (a) Since $A = LU$, the linear system $A^T \mathbf{x} = \mathbf{b}$ is equivalent to

$$U^T L^T \mathbf{x} = \mathbf{b}$$

where U^T is a lower triangular matrix and L^T is a unit upper triangular matrix.

Since $L^T \mathbf{x}$ is a vector, let $\mathbf{y} = L^T \mathbf{x}$. Then we solve in the following steps:

1. Solve $U^T \mathbf{y} = \mathbf{b}$.
2. Solve $L^T \mathbf{x} = \mathbf{y}$.

From the memory-efficient storage of L and U , recall that

$$U = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ & a_{22} & \cdots & a_{2n} \\ & & \ddots & \vdots \\ & & & a_{nn} \end{pmatrix},$$

so that

$$U^T = \begin{pmatrix} a_{11} & & & \\ a_{12} & a_{22} & & \\ \vdots & \vdots & \ddots & \\ a_{1n} & a_{2n} & \cdots & a_{nn} \end{pmatrix}.$$

Thus, U^T is a lower triangular matrix, so we solve $U^T \mathbf{y} = \mathbf{b}$ using forward substitution.

Next, from the memory-efficient storage of L and U , recall that

$$L = \begin{pmatrix} 1 & & & & \\ a_{21} & 1 & & & \\ a_{31} & a_{32} & 1 & & \\ \vdots & \vdots & \vdots & \ddots & \\ a_{n-1,1} & a_{n-1,2} & a_{n-1,3} & \cdots & 1 \\ a_{n1} & a_{n2} & a_{n3} & \cdots & a_{n,n-1} & 1 \end{pmatrix}$$

so that

$$L^T = \begin{pmatrix} 1 & a_{21} & a_{31} & \cdots & a_{n-1,1} & a_{n1} \\ & 1 & a_{32} & \cdots & a_{n-1,2} & a_{n2} \\ & & 1 & \cdots & a_{n-1,3} & a_{n3} \\ & & & \ddots & \vdots & \vdots \\ & & & & 1 & a_{n,n-1} \\ & & & & & 1 \end{pmatrix}$$

Thus, L^T is a lower triangular matrix, so we solve $L^T \mathbf{x} = \mathbf{y}$ using backward substitution.

(b) We want to find $C = A^{-1}$, which satisfies $AC = I$.

Write $C = (\mathbf{c}_1 \quad \cdots \quad \mathbf{c}_n)$, and compare the columns

$$(A\mathbf{c}_1 \quad \cdots \quad A\mathbf{c}_n) = \begin{pmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{pmatrix}$$

to get $A\mathbf{c}_k = \mathbf{e}_k$, for $k = 1, 2, \dots, n$. Since $A = LU$, we need to solve $LU\mathbf{c}_k = \mathbf{e}_k$, for $k = 1, 2, \dots, n$.

1. Solve $L\mathbf{y}_k = \mathbf{e}_k$.

2. Solve $U\mathbf{c}_k = \mathbf{y}_k$.

□

Problem A.3.5. Find a_0 , a_3 , x_0 and x_3 :

$$P_3(x) = a_0 - 8x - 2x^2 + a_3x^3$$

$$x_1 = -2, \quad x_2 = 1, \quad x_3 = 3$$

$$f(x_0) = -11, \quad f(x_1) = 10, \quad f(x_2) = 1$$

Solution. We proceed by the following steps.

(1) Since $P_3(x)$ has two unknowns, we can use two values to determine the two unknowns. Find a_0 and a_3 by $P_3(-2) = 10$ and $P_3(1) = 1$, so

$$a_0 - 8a_3 = 2, \quad a_0 + a_3 = 11.$$

This yields $a_0 = 10$, $a_3 = 1$.

(2) Since $a_0 = 10$ and $a_3 = 1$, the interpolating polynomial is

$$P_3(x) = 10 - 8x - 2x^2 + x^3.$$

Therefore, by direct substitution of $x_3 = 3$,

$$f(x_3) = P_3(x_3) = P_3(3) = 10 - 24 - 18 + 27 = -5.$$

(3) To find x_0 , we use $P_3(x_0) = f(x_0)$, so

$$10 - 8x_0 + 2x_0^2 + x_0^3 = -11.$$

Rearranging and factorising gives

$$(x_0 + 3)(x_0^2 - 5x_0 + 7) = 0,$$

so $x_0 = -3$.

□

Problem A.3.6. Assume that the Lagrange basis polynomials for the interpolation nodes x_0, x_1, x_2, x_3 are

$$L_0(x) = \ell_{00} + \ell_{01}x + \frac{1}{4}x^2 - \frac{1}{24}x^3,$$

$$L_1(x) = \ell_{10} + \ell_{11}x - x^2 + \frac{1}{4}x^3,$$

$$L_2(x) = \ell_{20} + \ell_{21}x + x^2 - \frac{1}{3}x^3,$$

$$L_3(x) = \ell_{30} + \ell_{31}x - \frac{1}{4}x^2 + \frac{1}{8}x^3.$$

Find x_0, x_1, x_2, x_3 .

Solution. The idea is to expand each Lagrange basis polynomial, and compare the coefficients of x^2 and x^3 . Since

$$\begin{aligned} L_0(x) &= \frac{(x - x_1)(x - x_2)(x - x_3)}{(x_0 - x_1)(x_0 - x_2)(x_0 - x_3)} \\ &= \frac{1}{(x_0 - x_1)(x_0 - x_2)(x_0 - x_3)}x^3 - \frac{x_1 + x_2 + x_3}{(x_0 - x_1)(x_0 - x_2)(x_0 - x_3)}x^2 + \dots, \end{aligned}$$

we can compare it with

$$L_0(x) = \ell_{00} + \ell_{01}x + \frac{1}{4}x^2 - \frac{1}{24}x^3$$

to see that the ratio of coefficient of x^2 to x^3 is $\frac{1/4}{-1/24} = -6$, so that

$$x_1 + x_2 + x_3 = 6.$$

Similarly, we have

$$x_0 + x_2 + x_3 = 4,$$

$$x_0 + x_1 + x_3 = 3,$$

$$x_0 + x_1 + x_2 = 2.$$

Summing up the four equations, we get

$$3(x_0 + x_1 + x_2 + x_3) = 15,$$

which yields $x_0 + x_1 + x_2 + x_3 = 5$. By subtracting this equation by each of the four equations, we get $x_0 = -1, x_1 = 1, x_2 = 2, x_3 = 3$. $\square$

A.4 Tutorial 4

Problem A.4.1. Let $x_0, x_1, \dots, x_n$ be $n + 1$ Chebyshev nodes on the interval $[-3, 1]$. Define

$$g(x) = |(x - x_0)(x - x_1) \cdots (x - x_n)|, \quad x \in [-3, 1].$$

Find the maximum value of $g(x)$ for $x \in [-3, 1]$.

Solution. If the given interval is $[-1, 1]$, then we are done, since the maximum value of $g(x)$ would be 2^{-n} . But the given interval is $[-3, 1]$, so we need to perform a linear transformation from $[-1, 1]$ to $[-3, 1]$.

The original Chebyshev nodes in $[-1, 1]$ are

$$\tilde{x}_k = \cos\left(\frac{2k+1}{2(n+1)}\right), \quad k = 0, 1, \dots, n.$$

They satisfy the following properties:

- All the local maximum values of the function

$$\tilde{g}(x) = |(x - \tilde{x}_0)(x - \tilde{x}_1) \cdots (x - \tilde{x}_n)|, \quad x \in [-1, 1]$$

are the same.

- The maximum value of $\tilde{g}(x)$ is 2^{-n} .

To find the nodes with similar properties in $[-3, 1]$, we translate and scale the Chebyshev nodes by a linear transformation. Thus, we need to find a linear function $\ell(x)$ that maps endpoint to endpoint:

$$\ell(-1) = -3, \quad \ell(1) = 1.$$

Let $\ell(x) = ax + b$. Solving gives $a = 2, b = -1$. Thus, the new Chebyshev nodes in $[-3, 1]$ are

$$x_k = \ell(\tilde{x}_k) = 2\tilde{x}_k - 1.$$

Then substitute these values into $g(x)$, and rewrite $g(x)$ to match the form of $\tilde{g}(x)$, i.e., $\tilde{x}_k$ are the zeros.

$$\begin{aligned} g(x) &= |(x - x_0)(x - x_1) \cdots (x - x_n)| \\ &= |(x - 2\tilde{x}_0 + 1)(x - 2\tilde{x}_1 + 1) \cdots (x - 2\tilde{x}_n + 1)| \\ &= \left| 2\left(\frac{x+1}{2} - \tilde{x}_0\right) \times 2\left(\frac{x+1}{2} - \tilde{x}_1\right) \times \cdots \times 2\left(\frac{x+1}{2} - \tilde{x}_n\right) \right| \\ &= 2^{n+1} \tilde{g}\left(\frac{x+1}{2}\right). \end{aligned}$$

Let $y = \frac{x+1}{2}$. Then $g(x) = 2^{n+1} \tilde{g}(y)$, where $y \in [-1, 1]$. We already know that on $[-1, 1]$, the maximum value of Chebyshev polynomial $\tilde{g}$ is 2^{-n} . Thus, the maximum value of g is 2. $\square$

More generally, to scale the Chebyshev nodes from $[-1, 1]$ to $[a, b]$ for general a and b , use a linear transformation $\ell(x)$ to map $[-1, 1]$ to $[a, b]$. Let $\ell(x) = cx + d$. Then solve $\ell(-1) = a, \ell(1) = b$ for c and d .

Problem A.4.2. Prove that

$$U_n(x) = \frac{\sin((n+1) \arccos x)}{\sin(\arccos x)}$$

is a polynomial of degree n with leading coefficient 2^n .

Solution. We proceed by induction. For the base case when $n = 0$,

$$U_0(x) = \frac{\sin(\arccos x)}{\sin(\arccos x)} = 1,$$

which is a polynomial of degree 0 with leading coefficient 2^0 .

Suppose $U_n(x)$ is a polynomial of degree n with leading coefficient 2^n . Then we want to relate U_{n+1} to U_n .

$$\begin{aligned} U_{n+1}(x) &= \frac{\sin((n+2)\arccos x)}{\sin(\arccos x)} \\ &= \frac{\sin((n+1)\arccos x) \cos(\arccos x)}{\sin(\arccos x)} + \frac{\cos((n+1)\arccos x) \sin(\arccos x)}{\sin(\arccos x)} \\ &= xU_n(x) + T_{n+1}(x) \end{aligned}$$

where we used the trigonometric formula $\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$. By inductive assumption, $xU_n(x)$ is a polynomial of degree $n + 1$ with leading coefficient 2^n . We know that $T_{n+1}(x)$ is a polynomial of degree $n + 1$ with leading coefficient 2^n . Hence, $U_{n+1}(x)$ is a polynomial of degree $n + 1$ with leading coefficient $2^n + 2^n = 2^{n+1}$. $\square$

Alternative solution. Let $\theta = \arccos x$. For $n \geq 1$, we have

$$\begin{aligned} T'_n(x) &= \frac{d}{dx} \cos(n\theta) = -n \sin(n\theta) \frac{d\theta}{dx} \\ &= -n \sin(n\theta) \left(\frac{dx}{d\theta} \right)^{-1} = -n \sin(n\theta) (-\sin \theta)^{-1} \\ &= n \frac{\sin(n\theta)}{\sin \theta} = nU_{n-1}(x). \end{aligned}$$

Therefore, for any $n \geq 0$,

$$\begin{aligned} U_n(x) &= \frac{1}{n+1} T'_{n+1}(x) \\ &= \frac{1}{n+1} (2^n x^{n+1} + \dots)' \\ &= 2^n x^n + \dots. \end{aligned}$$

$\square$

Problem A.4.3. Fill in the values.

0th d.d.	1st d.d.	2nd d.d.	3rd d.d.	4th d.d.
$f[-2] =$	$f[-2, -1] = 4$			
$f[-1] =$	$f[-1, 1] =$	$f[-2, -1, 1] =$	$f[-2, -1, 1, 3] = 2$	
$f[1] =$	$f[1, 3] =$	$f[-1, 1, 3] = 3$	$f[-1, 1, 3, 4] =$	$f[-2, -1, 1, 3, 4] = 1$
$f[3] =$	$f[3, 4] =$	$f[1, 3, 4] =$		
$f[4] = 5$				

Solution. Simply use the recurrence relation.

0th d.d.	1st d.d.	2nd d.d.	3rd d.d.	4th d.d.
$f[-2] = -79$				
	$f[-2, -1] = 4$			
$f[-1] = -75$		$f[-2, -1, 1] = -7$		
	$f[-1, 1] = -17$		$f[-2, -1, 1, 3] = 2$	
$f[1] = -109$		$f[-1, 1, 3] = 3$		$f[-2, -1, 1, 3, 4] = 1$
	$f[1, 3] = -5$		$f[-1, 1, 3, 4] = 8$	
$f[3] = -119$		$f[1, 3, 4] = 43$		
	$f[3, 4] = 124$			
$f[4] = 5$				

□

Problem A.4.4. Suppose the Lagrange interpolating polynomial of $f(x)$ with nodes $x_0, x_1, \dots, x_{100}$ is

$$1 + 5(x - x_0)(x - x_1) \cdots (x - x_{99}).$$

Find the value of $f[x_0, x_2, x_4, \dots, x_{98}]$.

Solution. By the above definition of the interpolating polynomial, the 0th divided differences are given by

$$f(x_0) = f(x_2) = \cdots = f(x_{98}) = 1.$$

Hence, the 1st divided differences all vanish:

$$f[x_0, x_2] = f[x_2, x_4] = \cdots = f[x_{96}, x_{98}] = 0.$$

Since all subsequent divided differences are 0, we conclude that $f[x_0, x_2, x_4, \dots, x_{98}] = 0$.

□

Problem A.4.5. Show that the polynomial

$$P_2(x) = f[a] + f[a, b](x - a) + f[a, b, b](x - a)(x - b)$$

satisfies

$$P_2(a) = f(a), \quad P_2(b) = f(b), \quad P_2'(b) = f'(b).$$

Solution. Compute

$$\begin{aligned} P_2(a) &= f[a] + 0 + 0 = f(a), \\ P_2(b) &= f[a] + f[a, b](b - a) + 0 = f(b), \\ P_2'(b) &= f[a, b] + f[a, b, b](2b - (a + b)) \\ &= f[a, b] + \frac{f[b, b] - f[a, b]}{b - a} = f[b, b] = f'(b). \end{aligned}$$

□

Problem A.4.6 (Extra). When $f(x)$ is second-order differentiable, show that there exists ξ between a and b such that

$$f[a, b, b] = \frac{1}{2}f''(\xi).$$

Solution. Let

$$g(x) = f(x) - P_2(x).$$

Since P_2 and f match on the interpolating nodes, we have $g(a) = g(b) = 0$ and $g'(b) = 0$.

By Rolle's theorem, there exists $\eta \in (\min(a, b), \max(a, b))$ such that $g'(\eta) = 0$.

Since $g'(\eta) = g'(b) = 0$, there exists ξ between η and b such that $g''(\xi) = 0$.

Finally, note that

$$g''(\xi) = f''(\xi) - 2f[a, b, b].$$

Rearranging gives the desired expression. $\square$

Problem A.4.7 (Extra). When $f(x)$ is third-order differentiable, show that for any x , there exists $\eta \in (\min(a, b, x), \max(a, b, x))$ such that

$$f(x) - P_2(x) = \frac{f'''(\eta)}{6}(x-a)(x-b)^2.$$

Solution. First simplify the left-hand side:

$$\begin{aligned} f(x) - P_2(x) &= f(x) - f[a] - f[a, b](x-a) - f[a, b, b](x-a)(x-b) \\ &= f[x, a, b](x-a)(x-b) - f[a, b, b](x-a)(x-b) \\ &= (f[x, a, b] - f[a, b, b])(x-a)(x-b) \\ &= f[x, a, b, b](x-a)(x-b)^2. \end{aligned}$$

Similar to the previous question, the proof can be completed by showing $f[x, a, b, b] = \frac{1}{6}f'''(\eta)$ for some η . Let $g(x) = f[x, a, b, b]$. Then $g(a) = g(b) = g'(b) = g(x) = 0$.

- By Rolle's theorem, there exist α_1, α_2 such that $g'(\alpha_1) = g'(\alpha_2) = g'(\beta) = 0$.
- There exist β_1, β_2 such that $g''(\beta_1) = g''(\beta_2) = 0$.
- There exists η such that $g'''(\eta) = 0$.

$\square$

A.5 Tutorial 5

Problem A.5.1. Assume that $f(x)$ is third-order differentiable. Show that for any distinct $a, b \in \mathbb{R}$, there exists ξ between a and b such that

$$f[a, b, b, b] = \frac{1}{6}f'''(\xi).$$

Solution. Consider the interpolating polynomial using the nodes a, b, b, b :

$$P_3(x) = f[a] + f[a, b](x-a) + f[a, b, b](x-a)(x-b) + f[a, b, b, b](x-a)(x-b)^2.$$

It satisfies $P_3(a) = f(a)$ and $P_3(b) = f(b)$. The first- and second-order derivatives of $P_3(x)$ are

$$\begin{aligned} P'_3(x) &= f[a, b] + f[a, b, b] \cdot [(x-a) + (x-b)] + f[a, b, b, b] \cdot [2(x-a) + (x-b)](x-b), \\ P''_3(x) &= 2f[a, b, b] + f[a, b, b, b] \cdot [2(x-a) + (x-b) + 3(x-b)]. \end{aligned}$$

Setting $x = b$ yields

$$\begin{aligned} P'_3(b) &= f[a, b] + f[a, b, b] \cdot (b-a) = f[b, b] = f'(b), \\ P''_3(b) &= 2f[a, b, b] + 2f[a, b, b, b](b-a) = 2f[b, b, b] = f''(b). \end{aligned}$$

Therefore, if we define $g(x) = f(x) - P_3(x)$, we have $g(a) = g(b) = 0$, $g'(b) = 0$, $g''(b) = 0$.

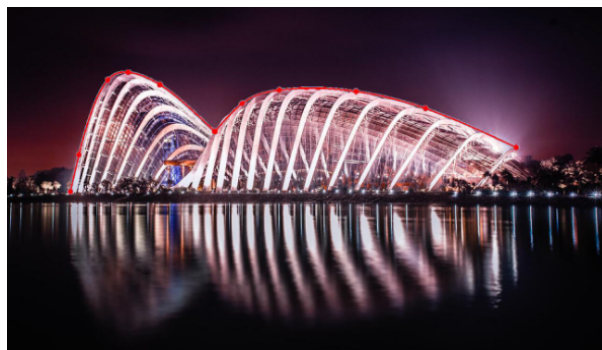
- By Rolle's theorem, there exists $\eta \in (a, b)$ such that $g'(\eta) = 0$.
- Using $g'(b) = 0$ and applying Rolle's theorem again, we can find $\zeta \in (\eta, b)$ such that $g''(\zeta) = 0$.
- Finally, we use $g''(b) = 0$ and applying Rolle's theorem one more time, we find $\xi \in (\zeta, b)$ such that $g'''(\xi) = 0$.

Since $P_3'''(\xi) = 6f[a, b, b, b]$, we obtain

$$g'''(\xi) = f'''(\xi) - P_3'''(\xi) = f'''(\xi) - 6f[a, b, b, b] = 0.$$

The proof is completed by equating the equation above to zero. $\square$

Problem A.5.2. Let $f(x)$ be the function of the red curve.



We have the following table:

i	0	1	2	3	4	5	6	7	8	9	10
x_i	0	14	62	95	147	237	282	343	469	583	732
$f(x_i)$	60	0	183	195	175	98	44	166	164	136	72

In the picture, x_5 is a non-smooth point. Therefore we consider two cubic spline interpolations: one with nodes $x_0, x_1, \dots, x_5$, the other with nodes $x_5, x_6, \dots, x_{10}$. The natural boundary conditions are used.

Solution. For the first cubic spline interpolation, calculate divided differences:

x	0th d.d.	1st d.d.	2nd d.d.
$x_0 = 0$	$f[x_0] = 0$		
$x_1 = 14$	$f[x_1] = 60$	$f[x_0, x_1] = 4.286$	
$x_2 = 62$	$f[x_2] = 183$	$f[x_1, x_2] = 2.563$	$f[x_0, x_1, x_2] = -0.02779$
$x_3 = 95$	$f[x_3] = 195$	$f[x_2, x_3] = 0.3636$	$f[x_1, x_2, x_3] = -0.02715$
$x_4 = 147$	$f[x_4] = 175$	$f[x_3, x_4] = -0.3846$	$f[x_2, x_3, x_4] = -0.008803$
$x_5 = 237$	$f[x_5] = 98$	$f[x_4, x_5] = -0.8556$	$f[x_3, x_4, x_5] = -0.003316$

Next, solve for the second-order derivatives:

$$\begin{aligned} \lambda_1 &= \frac{x_1 - x_0}{x_2 - x_0} = 0.2258, & \lambda_2 &= \frac{x_2 - x_1}{x_3 - x_1} = 0.5926, \\ \lambda_3 &= \frac{x_3 - x_2}{x_4 - x_2} = 0.3882, & \lambda_4 &= \frac{x_4 - x_3}{x_5 - x_3} = 0.3662. \end{aligned}$$

$$\begin{pmatrix} 2 & \lambda_1 & 0 & 0 \\ 1 - \lambda_2 & 2 & \lambda_2 & 0 \\ 0 & 1 - \lambda_3 & 2 & \lambda_3 \\ 0 & 0 & 1 - \lambda_4 & 2 \end{pmatrix} \begin{pmatrix} S''(x_1) \\ S''(x_2) \\ S''(x_3) \\ S''(x_4) \end{pmatrix} = \begin{pmatrix} 6f[x_0, x_1, x_2] \\ 6f[x_1, x_2, x_3] \\ 6f[x_2, x_3, x_4] \\ 6f[x_3, x_4, x_5] \end{pmatrix}$$

$$\begin{pmatrix} 2 & 0.7742 & 0 & 0 \\ 0.5926 & 2 & 0.4074 & 0 \\ 0 & 0.3882 & 2 & 0.6118 \\ 0 & 0 & 0.3662 & 2 \end{pmatrix} \begin{pmatrix} S''(x_1) \\ S''(x_2) \\ S''(x_3) \\ S''(x_4) \end{pmatrix} = \begin{pmatrix} -0.1668 \\ -0.1629 \\ -0.05282 \\ -0.01990 \end{pmatrix}$$

Solving gives $S''(x_1) = -0.05966$, $S''(x_2) = -0.06129$, $S''(x_3) = -0.01215$, $S''(x_4) = -0.007725$.

Finally, construct the polynomials:

$$\begin{aligned} p^{(1)}(x) &= 4.286x + x(x-14)(-0.0007102t - 0.009943) \\ p^{(2)}(x) &= 60 + 2.563(x-14) + (x-14)(x-62)(-5.666 \times 10^{-6}x - 0.03002) \\ p^{(3)}(x) &= 183 + 0.3636(x-62) + (x-62)(x-95)(0.0002482x - 0.03784) \\ p^{(4)}(x) &= 195 - 0.3846(x-95) + (x-95)(x-147)(0.00001418x - 0.006684) \\ p^{(5)}(x) &= 175 - 0.8556(x-147) + (x-147)(x-237)(0.00001431x - 0.004678) \end{aligned}$$

For the second cubic spline interpolation, calculate divided differences:

x	0th d.d.	1st d.d.	2nd d.d.
$x_5 = 237$	$f[x_5] = 98$		
		$f[x_5, x_6] = 1.022$	
$x_6 = 282$	$f[x_6] = 144$		$f[x_5, x_6, x_7] = -0.006241$
		$f[x_6, x_7] = 0.3607$	
$x_7 = 343$	$f[x_7] = 166$		$f[x_6, x_7, x_8] = -0.002014$
		$f[x_7, x_8] = -0.01587$	
$x_8 = 469$	$f[x_8] = 164$		$f[x_7, x_8, x_9] = -0.0009573$
		$f[x_8, x_9] = -0.2456$	
$x_9 = 583$	$f[x_9] = 136$		$f[x_8, x_9, x_{10}] = -0.0006993$
		$f[x_9, x_{10}] = -0.4295$	
$x_{10} = 732$	$f[x_{10}] = 72$		

Next, solve for the second-order derivatives:

$$\begin{aligned} \lambda_6 &= \frac{x_6 - x_5}{x_7 - x_5} = 0.4245, & \lambda_7 &= \frac{x_7 - x_6}{x_8 - x_6} = 0.3262, \\ \lambda_8 &= \frac{x_8 - x_7}{x_9 - x_7} = 0.5250, & \lambda_9 &= \frac{x_9 - x_8}{x_{10} - x_8} = 0.4335. \end{aligned}$$

$$\begin{pmatrix} 2 & \lambda_6 & 0 & 0 \\ 1 - \lambda_7 & 2 & \lambda_7 & 0 \\ 0 & 1 - \lambda_8 & 2 & \lambda_8 \\ 0 & 0 & 1 - \lambda_9 & 2 \end{pmatrix} \begin{pmatrix} S''(x_6) \\ S''(x_7) \\ S''(x_8) \\ S''(x_9) \end{pmatrix} = \begin{pmatrix} 6f[x_5, x_6, x_7] \\ 6f[x_6, x_7, x_8] \\ 6f[x_7, x_8, x_9] \\ 6f[x_8, x_9, x_{10}] \end{pmatrix}$$

$$\begin{pmatrix} 2 & 0.5755 & 0 & 0 \\ 0.3262 & 2 & 0.6738 & 0 \\ 0 & 0.5250 & 2 & 0.4750 \\ 0 & 0 & 0.4335 & 2 \end{pmatrix} \begin{pmatrix} S''(x_6) \\ S''(x_7) \\ S''(x_8) \\ S''(x_9) \end{pmatrix} = \begin{pmatrix} -0.03745 \\ -0.01208 \\ -0.005744 \\ -0.004196 \end{pmatrix}$$

Solving gives $S''(x_6) = -0.01801$, $S''(x_7) = -0.002493$, $S''(x_8) = -0.001812$, $S''(x_9) = -0.001705$.

Finally, construct the cubic spline polynomials:

$$\begin{aligned}
 p^{(6)}(x) &= 98 + 1.022(x - 237) + (x - 237)(x - 282)(-0.00006669x + 0.01280) \\
 p^{(7)}(x) &= 144 + 0.3607(x - 282) + (x - 282)(x - 343)(0.00004239x - 0.01837) \\
 p^{(8)}(x) &= 166 - 0.01587(x - 343) + (x - 343)(x - 469)(9.005 \times 10^{-7}x - 0.001442) \\
 p^{(9)}(x) &= 164 - 0.2456(x - 469) + (x - 469)(x - 583)(1.568 \times 10^{-7}x - 0.0009618) \\
 p^{(10)}(x) &= 136 - 0.4295(x - 583) + (x - 583)(x - 732)(1.907 \times 10^{-6}x - 0.001680)
 \end{aligned}$$

□

Problem A.5.3. Consider the spring that satisfies Hooke's law:

$$F = k(x - L),$$

where F is the force needed to stretch the spring, x is the length of the spring, k is the spring constant, and L is the length of the spring when not compressed or stretched. Given the following experimental data:

x	7.0	8.3	9.4	11.3	12.3	14.4	15.9
F	2	3	4	5	6	8	10

find the approximation of k and L up to four significant figures.

Solution. Let

$$\begin{aligned}
 \mathbf{p}_0 &= (1, 1, 1, 1, 1, 1, 1)^T \\
 \mathbf{p}_1 &= (7.0, 8.3, 9.4, 11.3, 12.3, 14.4, 15.9)^T
 \end{aligned}$$

Then apply Gram-Schmidt:

$$\begin{aligned}
 \tilde{\mathbf{p}}_0 &= \mathbf{p}_0 = (1, 1, 1, 1, 1, 1, 1)^T \\
 \tilde{\mathbf{p}}_0^* &= \frac{\tilde{\mathbf{p}}_0}{\|\tilde{\mathbf{p}}_0\|} = 0.377964(1, 1, 1, 1, 1, 1, 1)^T \\
 \tilde{\mathbf{p}}_1 &= \mathbf{p}_1 - (\mathbf{p}_1 \cdot \tilde{\mathbf{p}}_0^*)\tilde{\mathbf{p}}_0^* = 1/7(-29.6, -20.5, -12.8, 0.5, 7.5, 22.2, 32.7)^T \\
 \tilde{\mathbf{p}}_1^* &= \frac{\tilde{\mathbf{p}}_1}{\|\tilde{\mathbf{p}}_1\|} = (-0.533452, -0.369452, -0.230682, 0.00901101, 0.135165, 0.400089, 0.58932)^T
 \end{aligned}$$

This gives the QR factorization

$$(\mathbf{p}_0 \quad \mathbf{p}_1) = \begin{pmatrix} 0.377964 & -0.533452 \\ 0.377964 & -0.369452 \\ 0.377964 & -0.230682 \\ 0.377964 & 0.00901101 \\ 0.377964 & 0.135165 \\ 0.377964 & 0.400089 \\ 0.377964 & 0.58932 \end{pmatrix} \begin{pmatrix} 2.64575 & 29.7080 \\ 0 & 7.92681 \end{pmatrix}.$$

Let $\mathbf{y} = (2, 3, 4, 5, 6, 8, 10)^T$. The coefficients in the linear least squares approximation can be solved by

$$R\mathbf{a} = Q^T\mathbf{y}.$$

The linear equations are

$$\begin{pmatrix} 2.64575 & 29.7080 \\ 0 & 7.92681 \end{pmatrix} \begin{pmatrix} a_0 \\ a_1 \end{pmatrix} = \begin{pmatrix} 14.3626 \\ 6.85197 \end{pmatrix}.$$

Solving gives $a_0 = -4.27746$, $a_1 = 0.864405$. Comparing the polynomial

$$P_1(x) = -4.27746 + 0.864405x$$

with Hooke's law

$$F = k(x - L),$$

we can see that

$$k = 0.864405, \quad L = \frac{4.27746}{0.864405} = 4.94845.$$

□

Problem A.5.4. Given the following data points:

x_i	1.00	1.25	1.50	1.75	2.00
y_i	5.10	5.79	6.53	7.45	8.46

find the approximation using the exponential function $y = Ce^{ax}$ with coefficients up to five significant figures.

Solution. The idea is to transform the exponential function into a polynomial relationship between x and y . Since $y = Ce^{ax}$, we have $\log y = \log C + ax$. This gives a linear relationship between x and $\log y$.

x_i	1.00	1.25	1.50	1.75	2.00
y_i	1.6292	1.7561	1.8764	2.0082	2.1353

Least squares approximation gives

$$\log y \approx 1.1225 + 0.50572x,$$

which yields

$$y \approx e^{1.1225+0.50572x} = 3.0725e^{0.50572x}.$$

□

Problem A.5.5. Find w_0 , w_1 , x_0 and x_1 so that the quadrature formula attains highest degree of accuracy:

$$\int_{-1}^1 f(x) dx \approx w_0 f(x_0) + w_1 f(x_1).$$

Solution. Since there are 4 unknowns, we guess that the numerical integration formula is exact for $f(x) = 1, x, x^2, x^3$, which gives us the following equations:

$$w_0 + w_1 = 2, \tag{1}$$

$$w_0 x_0 + w_1 x_1 = 0, \tag{2}$$

$$w_0 x_0^2 + w_1 x_1^2 = \frac{2}{3}, \tag{3}$$

$$w_0 x_0^3 + w_1 x_1^3 = 0. \tag{4}$$

Solving gives $x_0 = -\frac{1}{\sqrt{3}}$, $x_1 = \frac{1}{\sqrt{3}}$, $w_0 = w_1 = 1$.

□

Problem A.5.6. Determine the constants C and k in the formula

$$\int_a^b f(x) dx - \frac{b-a}{4} \left[f(a) + 2f\left(\frac{a+b}{2}\right) + f(b) \right] = C(b-a)^{k+1} f^{(k)}(\xi),$$

and prove the equality.

Solution. Let $a = -1$, $b = 1$. Test using different polynomials $f(x)$:

- If $f(x) = 1$, then

$$\int_a^b f(x) dx = 2, \quad \frac{b-a}{4} \left[f(a) + 2f\left(\frac{a+b}{2}\right) + f(b) \right] = 2.$$

The numerical integration is exact.

- If $f(x) = x$, then

$$\int_a^b f(x) dx = 0, \quad \frac{b-a}{4} \left[f(a) + 2f\left(\frac{a+b}{2}\right) + f(b) \right] = 0.$$

The numerical integration is exact.

- If $f(x) = x^2$, then

$$\int_a^b f(x) dx = \frac{2}{3}, \quad \frac{b-a}{4} \left[f(a) + 2f\left(\frac{a+b}{2}\right) + f(b) \right] = 1.$$

The numerical integration is inexact.

Hence, the lowest degree of inexact polynomial is 2, so $k = 2$. From above, the error term is

$$C(b-a)^3 f^{(2)}(\xi) = -\frac{1}{3},$$

so $C = -\frac{1}{48}$. Hence, the formula is: there exists $\delta \in (a, b)$ such that

$$\int_a^b f(x) dx - \frac{b-a}{4} \left[f(a) + 2f\left(\frac{a+b}{2}\right) + f(b) \right] = -\frac{1}{48} (b-a)^3 f^{(2)}(\xi).$$

□

A.6 Sample questions

Problem A.6.1. Find the LU factorization of the matrix

$$\begin{pmatrix} 2 & -1 & 3 & -1 \\ -4 & 0 & -8 & 7 \\ 2 & 1 & 9 & 4 \\ 6 & -5 & 9 & 4 \end{pmatrix}.$$

Problem A.6.2. Find the value of $\sum_{i=0}^n \sum_{j=0}^n \sum_{k=0}^n L_i(x_k) L_j(x_k)$.

Solution. Recall the property of Lagrange basis polynomials $L_i(x_k) = \delta_{ik}$. Then we can rewrite the triple summation by substituting the Kronecker deltas, then swap the order of summation:

$$\begin{aligned} \sum_{i=0}^n \sum_{j=0}^n \sum_{k=0}^n L_i(x_k) L_j(x_k) &= \sum_{i=0}^n \sum_{j=0}^n \sum_{k=0}^n \delta_{ik} \delta_{jk} \\ &= \sum_{k=0}^n \left(\sum_{i=0}^n \delta_{ik} \right) \left(\sum_{j=0}^n \delta_{jk} \right) \\ &= \sum_{k=0}^n (1) \cdot (1) = \sum_{k=0}^n 1 = n+1. \end{aligned}$$

□

Problem A.6.3. Find the value of $\sum_{i=0}^n \sum_{j=0}^n \sum_{k=0}^n x_i x_j L_i(x_k) L_j(x_k)$.

Solution. Using the property $L_i(x_k) = \delta_{ik}$, we substitute these into the expression:

$$\sum_{i=0}^n \sum_{j=0}^n \sum_{k=0}^n x_i x_j \delta_{ik} \delta_{jk}$$

We can rearrange the order of summation to group terms related to each Kronecker delta:

$$= \sum_{k=0}^n \left(\sum_{i=0}^n x_i \delta_{ik} \right) \left(\sum_{j=0}^n x_j \delta_{jk} \right)$$

For a fixed k , the inner sums collapse because the Kronecker delta is non-zero only when the index matches k . Specifically, $\sum_{i=0}^n x_i \delta_{ik} = x_k$ and $\sum_{j=0}^n x_j \delta_{jk} = x_k$. Substituting these in:

$$= \sum_{k=0}^n (x_k) \cdot (x_k) = \sum_{k=0}^n x_k^2.$$

□

Problem A.6.4. Let $L_0(x), L_1(x), L_2(x), L_3(x)$ be the Lagrange basis polynomials with four interpolation nodes. Suppose

$$\begin{aligned} L_0(x) &= 4 + \ell_{01}x + \ell_{02}x^2 - \frac{4}{3}x^3, \\ L_1(x) &= \ell_{10} + \ell_{11}x - 16x^2 + 4x^3, \\ L_2(x) &= \ell_{20} + \ell_{21}x + 14x^2 - 4x^3, \\ L_3(x) &= -1 + \ell_{31}x + \ell_{32}x^2 + \frac{4}{3}x^3. \end{aligned}$$

Find the values of $\ell_{01}, \ell_{10}, \ell_{20}, \ell_{31}$ and $\ell_{02}, \ell_{11}, \ell_{21}, \ell_{32}$.

Solution. The idea is to first find the interpolation nodes, so that we can directly compute the Lagrange basis polynomials. Let $x_0 < x_1 < x_2 < x_3$ be the interpolation nodes. We have

$$\begin{aligned} x_1 x_2 x_3 &= 3, \\ x_0 + x_2 + x_3 &= 4, \\ x_0 + x_1 + x_3 &= 7/2, \\ x_0 x_1 x_2 &= 3/4. \end{aligned}$$

The first and the last equations yield $x_0 = x_3/4$. Therefore

$$x_2 = 4 - x_0 - x_3 = 4 - \frac{5}{4}x_3, \quad x_1 = \frac{7}{2} - x_0 - x_3 = \frac{7}{2} - \frac{5}{4}x_3.$$

Thus we obtain the following equation of x_3 :

$$\left(\frac{7}{2} - \frac{5}{4}x_3 \right) \left(4 - \frac{5}{4}x_3 \right) x_3 = 3.$$

The interpolation nodes are

$$\begin{aligned}x_3 &= 2, \\x_0 &= \frac{1}{4}x_3 = \frac{1}{2}, \\x_2 &= 4 - x_0 - x_3 = 4 - 2 - \frac{1}{2} = \frac{3}{2}, \\x_1 &= \frac{7}{2} - x_0 - x_3 = \frac{7}{2} - 2 - \frac{1}{2} = 1.\end{aligned}$$

This allows us to get the explicit expressions of the Lagrange basis polynomials. The coefficients are

$$\begin{aligned}\ell_{01} &= -\frac{26}{3}, & \ell_{10} &= -1, & \ell_{20} &= 4, & \ell_{31} &= \frac{11}{3}, \\ \ell_{02} &= 6, & \ell_{11} &= 19, & \ell_{21} &= -14, & \ell_{32} &= -4.\end{aligned}$$

□

Problem A.6.5. The polynomial

$$P_3(x) = -3 + a_1x + a_2x^2 - \frac{1}{2}x^3$$

is the Lagrange interpolating polynomial of $f(x)$ with nodes

$$x_0 = -3, \quad x_1 = -2, \quad x_2 = -1, \quad x_3.$$

Given that $f(x_1) = -1$, $f(x_2) = -4$ and $f(x_3) = -6$, find the values of x_3 and $f(x_0)$.

Problem A.6.6. Suppose $x_0 \neq x_1$. Show that for any smooth function $f(x)$, there exists a unique quadratic function $P_2(x)$ such that

$$P_2(x_0) = f(x_0), \quad P_2'(x_0) = f'(x_0), \quad P_2(x_1) = f(x_1).$$

Solution. Assume $P_2(x) = a_0 + a_1x + a_2x^2$. Then

$$\begin{pmatrix} 1 & x_0 & x_0^2 \\ 0 & 1 & 2x_0 \\ 1 & x_1 & x_1^2 \end{pmatrix} \begin{pmatrix} a_0 \\ a_1 \\ a_2 \end{pmatrix} = \begin{pmatrix} f(x_0) \\ f'(x_0) \\ f(x_1) \end{pmatrix}.$$

We show that the coefficient matrix is nonsingular by Gaussian elimination:

$$\begin{pmatrix} 1 & x_0 & x_0^2 \\ 0 & 1 & 2x_0 \\ 1 & x_1 & x_1^2 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & x_0 & x_0^2 \\ 0 & 1 & 2x_0 \\ 0 & x_1 - x_0 & x_1^2 - x_0^2 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & x_0 & x_0^2 \\ 0 & 1 & 2x_0 \\ 0 & 0 & (x_1 - x_0)^2 \end{pmatrix}$$

□

Problem A.6.7. Find the Lagrange interpolating polynomial with the following data points:

$$f(1) = 1, \quad f(3) = 5, \quad f(5) = 65.$$

Given two more data points:

$$f(2) = 2, \quad f(4) = 16,$$

find the new Lagrange interpolating polynomial.

Problem A.6.8. Complete the following table of divided differences:

$f[-3] = -2$			
	$f[-3, -1] =$		
$f[-1] =$		$f[-3, -1, 1] = -1/2$	
	$f[-1, 1] =$		$f[-3, -1, 1, 4] = 1/10$
$f[1] =$		$f[-1, 1, 4] =$	
	$f[1, 4] =$		
$f[4] = 5$			

Given $f(2) = 5$, find the Lagrange interpolating polynomial for $f(x)$ with interpolation nodes $-3, -1, 1, 2, 4$.

Problem A.6.9. Given a third-order differentiable function $f(x)$, show that for any $x_0 < x_1$, there exists $\xi \in (x_0, x_1)$ such that

$$f[x_0, x_1, x_1, x_1] = \frac{1}{6}f'''(\xi).$$

Problem A.6.10. Let $x_0 = 1, x_1 = 2, x_2 = 3$ and $x_3 = 4$. Find a function $S(x)$ satisfying the following conditions:

- (1) $S(x_0) = 1, S(x_1) = 3, S(x_2) = 0$.
- (2) $S(x)$ is a quadratic polynomial on $[x_0, x_1]$.
- (3) $S(x)$ is a quadratic polynomial on $[x_1, x_2]$.
- (4) $S(x)$ is a quadratic polynomial on $[x_2, x_3]$.
- (5) $S(x)$ is a first-order differentiable periodic function satisfying $S(x+3) = S(x)$ for any x .

Solution. Assume that

$$S(x) = \begin{cases} 1 + 2(x-1) + C_1(x-1)(x-2), & \text{if } x \in [1, 2], \\ 3 - 3(x-2) + C_2(x-2)(x-3), & \text{if } x \in [2, 3], \\ 0 + (x-3) + C_3(x-3)(x-4), & \text{if } x \in [3, 4]. \end{cases}$$

According to the first-order differentiability of $S(x)$, we have

$$\begin{aligned} S'(1) &= 2 + (1-2)C_1 = 1 + (4-3)C_3 = S'(4), \\ S'(2) &= 2 + (2-1)C_1 = -3 + (2-3)C_2, \\ S'(3) &= -3 + (3-2)C_2 = 1 + (3-4)C_3. \end{aligned}$$

It can be solved that

$$C_1 = -4, \quad C_2 = -1, \quad C_3 = 5.$$

The quadratic spline is

$$S(x) = \begin{cases} 1 + 2(x-1) - 4(x-1)(x-2), & \text{if } x \in [1, 2], \\ 3 - 3(x-2) - (x-2)(x-3), & \text{if } x \in [2, 3], \\ 0 + (x-3) + 5(x-3)(x-4), & \text{if } x \in [3, 4]. \end{cases}$$

□

Problem A.6.11. Find the QR factorization of the matrix

$$\begin{pmatrix} 1 & 3 & 2 & 0 \\ 1 & 1 & 4 & -1 \\ -1 & 2 & -1 & 3 \\ 1 & -2 & 1 & 0 \end{pmatrix}.$$

Solution.

$$\begin{pmatrix} 1/2 & 1/\sqrt{2} & -1/2 & 0 \\ 1/2 & 1/(3\sqrt{2}) & 5/6 & 0 \\ -1/2 & 2/(3\sqrt{2}) & 1/6 & 1/\sqrt{2} \\ 1/2 & -2/(3\sqrt{2}) & -1/6 & 1/\sqrt{2} \end{pmatrix} \begin{pmatrix} 2 & 0 & 4 & -2 \\ 0 & 3\sqrt{2} & \sqrt{2} & 5/(3\sqrt{2}) \\ 0 & 0 & 2 & -1/3 \\ 0 & 0 & 0 & 3/\sqrt{2} \end{pmatrix}$$

□

Problem A.6.12. Use Simpson's rule to find the approximation of the following integral up to four significant figures:

$$\int_1^2 x^{1/3} e^{-x} dx.$$

Provide an upper bound for the absolute error (up to four significant figures).

Solution. Let $f(x) = x^{1/3} e^{-x}$. By Simpson's rule,

$$\begin{aligned} \int_1^2 x^{1/3} e^{-x} dx &\approx \frac{2-1}{6} [f(1) + 4f(3/2) + f(2)] \\ &\approx \frac{1}{6} (0.3679 + 4 \times 0.2554 + 0.1705) = 0.2600. \end{aligned}$$

To find an upper bound of the error, we first compute the derivatives of f :

$$\begin{aligned} f'(x) &= \frac{1}{3} x^{-2/3} e^{-x} - x^{1/3} e^{-x} = \left(\frac{1}{3} x^{-2/3} - x^{1/3} \right) e^{-x}, \\ f''(x) &= \left(-\frac{2}{9} x^{-5/3} - \frac{1}{3} x^{-2/3} \right) e^{-x} - \left(\frac{1}{3} x^{-2/3} - x^{1/3} \right) e^{-x} \\ &= \left(x^{1/3} - \frac{2}{3} x^{-2/3} - \frac{2}{9} x^{-5/3} \right) e^{-x}, \\ f'''(x) &= \left(\frac{10}{27} x^{-8/3} + \frac{2}{3} x^{-5/3} + x^{-2/3} - x^{1/3} \right) e^{-x}, \\ f^{(4)}(x) &= \left(x^{1/3} - \frac{4}{3} x^{-2/3} - \frac{4}{3} x^{-5/3} - \frac{40}{27} x^{-8/3} - \frac{80}{81} x^{-11/3} \right) e^{-x} \\ &= \left(x^4 - \frac{4}{3} x^3 - \frac{4}{3} x^2 - \frac{40}{27} x - \frac{80}{81} \right) x^{-11/3} e^{-x}. \end{aligned}$$

To find the maximum of $|f^{(4)}(x)|$ for $x \in [1, 2]$, we take one more derivative to get

$$f^{(5)}(x) = \left(\frac{880}{243} + \frac{400}{81} x + \frac{100}{27} x^2 + \frac{20}{9} x^3 + \frac{5}{3} x^4 - x^5 \right) x^{-14/3} e^{-x}.$$

This function is positive for every $x \in [1, 2]$ because

$$\frac{20}{9} x^3 + \frac{5}{3} x^4 - x^5 = x^3 \left(\frac{20}{9} - \frac{1}{6} x^2 \right) + x^4 \left(\frac{5}{3} - \frac{5}{6} x \right) > 0.$$

Therefore

$$\max_{\xi \in [1, 2]} |f^{(4)}(\xi)| = \max\{|f^{(4)}(1)|, |f^{(4)}(2)|\} = \frac{335}{81e}.$$

Using

$$\max_{\xi \in [1, 2]} |f^{(4)}(\xi)| = \frac{335}{81e},$$

we can find that the upper bound of the absolute error is

$$\frac{1}{90} \left(\frac{2-1}{2} \right)^5 \max_{\xi \in [1, 2]} |f^{(4)}(\xi)| = \frac{67}{46656e} \approx 0.0005283.$$

□

Problem A.6.13. Consider the numerical integration formula:

$$\int_{-1}^1 f(x) dx \approx w_0 f(x_0) + w_1 f(0) + w_2 f(x_2),$$

where $x_0 < 0 < x_2$. Find the values of w_0, w_1, w_2, x_0, x_2 such that the numerical integration formula attains the highest degree of accuracy.

Solution. We let $f(x)$ be $1, x, x^2, x^3, x^4$ and equate both sides of the numerical integration formula:

$$w_0 + w_1 + w_2 = 2, \quad (1)$$

$$w_0 x_0 + w_2 x_2 = 0, \quad (2)$$

$$w_0 x_0^2 + w_2 x_2^2 = \frac{2}{3}, \quad (3)$$

$$w_0 x_0^3 + w_2 x_2^3 = 0, \quad (4)$$

$$w_0 x_0^4 + w_2 x_2^4 = \frac{2}{5}. \quad (5)$$

Solving gives $x_0 = -\sqrt{\frac{3}{5}}, x_2 = \sqrt{\frac{3}{5}}, w_0 = \frac{5}{9}, w_1 = \frac{8}{9}, w_2 = \frac{5}{9}$.

□

Problem A.6.14. Consider the numerical integration formula:

$$\int_a^b f(x) dx \approx \frac{b-a}{4} \left[f(a) + 2f\left(\frac{a+b}{2}\right) + f(b) \right].$$

For any smooth function $f(x)$, the error term is given by

$$\int_a^b f(x) dx - \frac{b-a}{4} \left[f(a) + 2f\left(\frac{a+b}{2}\right) + f(b) \right] = C(b-a)^{k+1} f^{(k)}(\xi)$$

for some $\xi \in (a, b)$. Find the values of C and k , and then prove the equality.

Solution. We first determine the degree of accuracy. Let $a = -1, b = 1$, and test $f(x)$ using polynomials $1, x, x^2$.

• $f(x) = 1$:

$$\text{LHS} = 2, \quad \text{RHS} = \frac{1}{2}(1 + 2 + 1) = 2.$$

• $f(x) = x$:

$$\text{LHS} = 0, \quad \text{RHS} = \frac{1}{2}(-1 + 0 + 1) = 0.$$

• $f(x) = x^2$:

$$\text{LHS} = \frac{2}{3}, \quad \text{RHS} = \frac{1}{2}(1 + 0 + 1) = 1.$$

□

Problem A.6.15. Use the composite Simpson's rule to approximate the integral

$$\int_1^3 \ln x dx$$

with absolute error less than 0.01. Write the result up to four significant figures.

Solution. Suppose we use $n + 1$ equally spaced nodes. Let $f(x) = \ln x$. The required derivatives are

$$f'(x) = \frac{1}{x}, \quad f''(x) = -\frac{1}{x^2}, \quad f'''(x) = \frac{2}{x^3}, \quad f^{(4)}(x) = -\frac{6}{x^4}.$$

It is clear that the maximum value of $|f^{(4)}(x)|$ for $x \in [1, 3]$ is 6. Thus, the upper bound for the error of the composite Simpson's rule is

$$\frac{2^5}{180n^4} \times 6 = \frac{16}{15n^4}.$$

We bound

$$\frac{16}{15n^4} < 0.01.$$

We get that the minimum value of the positive even integer n is 4.

The numerical integral is

$$\frac{1}{6} \left(\ln 1 + 4 \ln \frac{3}{2} + 2 \ln 2 + 4 \ln \frac{5}{2} + \ln 3 \right) = 1.295.$$

□

B Preliminaries

Theorem B.1 (Intermediate value theorem). *Let f be continuous on $[a, b]$. Then for any u between $f(a)$ and $f(b)$, that is,*

$$\min(f(a), f(b)) < u < \max(f(a), f(b)),$$

there exists $c \in (a, b)$ such that $f(c) = u$.

Theorem B.2 (Rolle's theorem). *Let f be continuous on $[a, b]$. If f is differentiable on (a, b) and $f(a) = f(b)$, then there exists $c \in (a, b)$ such that*

$$f'(c) = 0.$$

Theorem B.3 (Mean value theorem). *Let f be continuous on $[a, b]$. If f is differentiable on (a, b) , then there exists $c \in (a, b)$ such that*

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Proof. Apply Rolle's theorem on the function

$$g(x) = f(x) - \left[\frac{f(b) - f(a)}{b - a}(x - a) + f(a) \right].$$

□